supplementary file

A Hybrid Quantum–Classical Computing Method for Electromechanical Oscillatory Stability Analysis

This supplementary document provides comprehensive datasets and extended diagnostic metrics to support the analyses and conclusions presented in the main paper. Specifically, it details: **1)** the complete numerical right eigenpairs (eigenvalues and complex eigenvectors) extracted by the proposed hybrid quantum-classical eigensolver across multiple operating scenarios (supplementing Tables I and II); **2)** comparative spectral results from the RVVQE baseline algorithm; **3)** full computational outputs from the component ablation experiments to validate the efficacy of the dual-matrix seeding and two-step optimization strategies (supplementing Tables III and IV); and **4)** the exact classical reference spectra computed via MATLAB eig function, which serve as the rigorous benchmark for evaluating all quantum algorithms. To facilitate clear cross-referencing, each table herein explicitly indicates its corresponding section and table in the main paper.

1 Scope and reading conventions

Each subsequent table presents the calculated complex eigenvalues alongside their corresponding right eigenvectors. Vector entries are denoted as [01], [02], etc., corresponding to the state variables in the physical network. Complex quantities are formatted as $a + bj$, where j represents the imaginary unit. Eigenvalues and eigenvector components are reported with thirteen and nine significant digits, respectively. Right eigenvectors satisfy the state equation $\boldsymbol{A}_{sys}\hat{v} = \hat{\lambda}\hat{v}$, with classical reference solutions obtained via the standard eigensolver in MATLAB. All results reflect the complete quantum-classical hybrid pipeline, including post-optimization polishing and conjugate completion.

All results reflect the complete quantum-classical hybrid pipeline, including post-optimization polishing and conjugate completion. For performance validation, the **residual norm**, defined as $residual = \left\|\boldsymbol{A}_{sys}\hat{v} - \hat{\lambda}\hat{v}\right\|_2$, is reported for each computed eigenpair to quantify the precision with which the pair satisfies the eigenvalue equation.

2 Table set index

Tables S1–S8 detail the results of the proposed method across all operating scenarios; Tables S9–S10 present the RVVQE baseline comparisons; Tables S11–S18 provide the complete computational outputs from the component ablation experiments; and Tables S19–S26 list the classical reference spectra used as ground-truth benchmarks. The detailed mappings to the main paper are summarized in the index below.

Table ID	System	Method or strategy	Count	Main paper scope
Table S1	9-bus system	(Ours) Base	6/6	Table I
Table S2	9-bus system	(Ours) Light load	6/6	Table I
Table S3	9-bus system	(Ours) Heavy load	6/6	Table I
Table S4	9-bus system	(Ours) Reactive stress	6/6	Table I
Table S5	39-bus system	(Ours) Base	20/20	Table II
Table S6	39-bus system	(Ours) Light load	20/20	Table II
Table S7	39-bus system	(Ours) Heavy load	20/20	Table II
Table S8	39-bus system	(Ours) Reactive stress	20/20	Table II
Table S9	9-bus system	RVVQE Base	6/6	Table I
Table S10	39-bus system	RVVQE Base	18/20	Table II
Table S11	9-bus system	Full pipeline	6/6	Table III
Table S12	9-bus system	Without Step-1	6/6	Table III
Table S13	9-bus system	H only	6/6	Table III
Table S14	9-bus system	K only	6/6	Table III
Table S15	39-bus system	Full pipeline	20/20	Table IV
Table S16	39-bus system	Without Step-1	16/20	Table IV
Table S17	39-bus system	H only	6/20	Table IV
Table S18	39-bus system	K only	17/20	Table IV
Table S19	9-bus system	Reference Base	6	Reference for matching
Table S20	9-bus system	Reference Light load	6	Reference for matching
Table S21	9-bus system	Reference Heavy load	6	Reference for matching
Table S22	9-bus system	Reference Reactive stress	6	Reference for matching
Table S23	39-bus system	Reference Base	20	Reference for matching
Table S24	39-bus system	Reference Light load	20	Reference for matching
Table S25	39-bus system	Reference Heavy load	20	Reference for matching
Table S26	39-bus system	Reference Reactive stress	20	Reference for matching

3 Proposed method and RVVQE comparison

The proposed method is listed for all four operating conditions of both matrices. The RVVQE comparison contains only 9-bus system Base and 39-bus system Base, corresponding to the comparison rows in main paper Tables I and II. Numerical values and vector phases are retained from each individual run.

Table S1 9-bus system (Ours) Base

Returned 6/6 eigenpairs; Eigenvalue MAE 3.268396e-07; RMSE 5.710774e-07; mean residual 3.623158e-13; mean MAC 1.0000000000.

Table S1. Computed eigenpairs by the proposed method for the 9-bus system under the Base scenario (Supplementary to Table I and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -9.049113375636e-15 Im -8.689799888785e+00	1.26e-14	[01] -1.48156394e-16-3.60131621e-02j [02] -5.38284587e-17+9.41468493e-02j [03] -2.51307019e-17+5.39366471e-02j [04] -3.12947172e-01-1.89960548e-16j [05] 8.18117281e-01+2.78735747e-35j [06] 4.68698670e-01-1.07683353e-16j
2	Re -4.664608679178e-17 Im -9.658385477926e-07	9.58e-13	[01] 5.77350269e-01+2.05616948e-18j [02] 5.77350269e-01-1.77008577e-16j [03] 5.77350269e-01+2.08270950e-33j [04] -7.42202599e-17-5.57627146e-07j [05] -6.85767024e-17-5.57627145e-07j [06] 6.20019100e-17-5.57627146e-07j
3	Re 6.282010963885e-16 Im 8.689799888785e+00	2.67e-13	[01] 7.90554384e-16+3.60131621e-02j [02] -9.28083374e-15-9.41468493e-02j [03] -2.06048098e-14-5.39366471e-02j [04] -3.12947172e-01+4.06225573e-14j [05] 8.18117281e-01+6.71040727e-35j [06] 4.68698670e-01-3.04101982e-13j
4	Re 5.178718390205e-14 Im -1.336021110593e+01	4.98e-14	[01] 4.52684804e-16-2.97384465e-03j [02] 3.42024000e-16-2.21449322e-02j [03] 4.95002509e-17+7.12175300e-02j [04] -3.97311924e-02+3.87081745e-17j [05] -2.95860969e-01-1.10671382e-16j [06] 9.51481235e-01+3.24907636e-34j
5	Re 5.178718390205e-14 Im 1.336021110593e+01	4.98e-14	[01] 4.52684804e-16+2.97384465e-03j [02] 3.42024000e-16+2.21449322e-02j [03] 4.95002509e-17-7.12175300e-02j [04] -3.97311924e-02-3.87081745e-17j [05] -2.95860969e-01+1.10671382e-16j [06] 9.51481235e-01-8.08277946e-51j
6	Re 3.971587095095e-07 Im 8.366362606539e-07	8.37e-13	[01] 5.77350269e-01+4.96350612e-35j [02] 5.77350269e-01-5.21378896e-15j [03] 5.77350269e-01-4.63878596e-15j [04] 2.29299688e-07+4.83032170e-07j [05] 2.29299688e-07+4.83032171e-07j [06] 2.29299688e-07+4.83032170e-07j

Table S2 9-bus system (Ours) Light load

Returned 6/6 eigenpairs; Eigenvalue MAE 2.076241e-07; RMSE 3.596153e-07; mean residual 2.643967e-13; mean MAC 1.0000000000.

Table S2. Computed eigenpairs by the proposed method for the 9-bus system under the Light load scenario (Supplementary to Table I and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -7.502705472468e-17 Im -1.036749849783e-06	7.08e-13	[01] 5.77350269e-01+1.57098110e-17j [02] 5.77350269e-01-1.07473876e-16j [03] 5.77350269e-01+1.65829991e-50j [04] -1.34408685e-16-5.98567805e-07j [05] -8.13373022e-17-5.98567805e-07j [06] 8.57959837e-17-5.98567805e-07j
2	Re -7.502705472468e-17 Im 1.036749849783e-06	7.08e-13	[01] 5.77350269e-01-1.57098110e-17j [02] 5.77350269e-01+1.07473876e-16j [03] 5.77350269e-01+1.17173809e-33j [04] -1.34408685e-16+5.98567805e-07j [05] -8.13373022e-17+5.98567805e-07j [06] 8.57959837e-17+5.98567805e-07j
3	Re 1.952175552535e-16 Im -8.763042094030e+00	2.39e-14	[01] -1.40704018e-16-3.49476907e-02j [02] -1.28336934e-16+9.35685399e-02j [03] -1.82013458e-16+5.36522151e-02j [04] -3.06248085e-01-3.07245404e-16j [05] 8.19945054e-01-3.11754735e-50j [06] 4.70156619e-01+9.66677437e-17j
4	Re 1.952175552535e-16 Im 8.763042094030e+00	2.39e-14	[01] -1.40704018e-16+3.49476907e-02j [02] -1.28336934e-16-9.35685399e-02j [03] -1.82013458e-16-5.36522151e-02j [04] -3.06248085e-01+3.07245404e-16j [05] 8.19945054e-01+5.81990036e-34j [06] 4.70156619e-01-9.66677437e-17j
5	Re 1.345378203143e-15 Im -1.337096006609e+01	1.14e-13	[01] -9.13686724e-18-2.94927630e-03j [02] 6.19478708e-17-2.22528967e-02j [03] 2.71101486e-17+7.11223224e-02j [04] -3.94346556e-02+2.61885227e-18j [05] -2.97542593e-01+1.00222723e-16j [06] 9.50973732e-01+4.19017179e-35j
6	Re 6.207191048921e-15 Im 1.337096006609e+01	8.89e-15	[01] 8.98992470e-18+2.94927630e-03j [02] 1.10250415e-16+2.22528967e-02j [03] 2.38223101e-17-7.11223224e-02j [04] -3.94346556e-02+3.58860889e-16j [05] -2.97542593e-01+1.64884003e-16j [06] 9.50973732e-01+1.11979823e-33j

Table S3 9-bus system (Ours) Heavy load

Returned 6/6 eigenpairs; Eigenvalue MAE 3.117347e-07; RMSE 5.504864e-07; mean residual 3.767541e-13; mean MAC 1.0000000000.

Table S3. Computed eigenpairs by the proposed method for the 9-bus system under the Heavy load scenario (Supplementary to Table I and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -4.034255592594e-07 Im -1.107474079355e-06	1.31e-12	[01] 5.77350269e-01+5.90097921e-15j [02] 5.77350269e-01-7.18105996e-16j [03] 5.77350269e-01-4.18542217e-33j [04] -2.32917855e-07-6.39400458e-07j [05] -2.32917855e-07-6.39400458e-07j [06] -2.32917855e-07-6.39400458e-07j
2	Re -2.942445697457e-14 Im -1.334671703959e+01	3.31e-14	[01] 3.68818241e-16-3.00182926e-03j [02] 3.07298783e-17-2.20247681e-02j [03] 3.54457723e-16+7.13322165e-02j [04] -4.00645657e-02-5.68466046e-16j [05] -2.93958347e-01-5.21633417e-17j [06] 9.52050909e-01+4.95957883e-36j
3	Re -2.942445697457e-14 Im 1.334671703959e+01	3.31e-14	[01] 3.68818241e-16+3.00182926e-03j [02] 3.07298783e-17+2.20247681e-02j [03] 3.54457723e-16-7.13322165e-02j [04] -4.00645657e-02+5.68466046e-16j [05] -2.93958347e-01+5.21633417e-17j [06] 9.52050909e-01-1.30037472e-52j
4	Re 9.249561368876e-15 Im -8.608272719265e+00	2.02e-14	[01] 1.48075364e-16-3.72566674e-02j [02] 5.58882502e-17+9.47926988e-02j [03] 4.97862719e-18+5.42350988e-02j [04] -3.20715553e-01-1.04256675e-16j [05] 8.16001403e-01+2.12816904e-34j [06] 4.66870521e-01+3.32230329e-17j
5	Re 1.664966489914e-14 Im 8.608272719266e+00	6.15e-14	[01] 1.38311154e-16+3.72566674e-02j [02] 1.61331394e-16-9.47926988e-02j [03] -2.40691757e-16-5.42350988e-02j [04] -3.20715553e-01-1.24134492e-16j [05] 8.16001403e-01+5.99593301e-36j [06] 4.66870521e-01-2.02628269e-16j
6	Re 5.344757708214e-07 Im 7.482742978669e-07	7.98e-13	[01] 5.77350269e-01+5.60008587e-15j [02] 5.77350269e-01-4.97468030e-16j [03] 5.77350269e-01-1.46033435e-33j [04] 3.08579730e-07+4.32016367e-07j [05] 3.08579730e-07+4.32016367e-07j [06] 3.08579730e-07+4.32016367e-07j

Table S4 9-bus system (Ours) Reactive stress

Returned 6/6 eigenpairs; Eigenvalue MAE 1.675801e-07; RMSE 2.902564e-07; mean residual 3.025876e-13; mean MAC 1.0000000000.

Table S4. Computed eigenpairs by the proposed method for the 9-bus system under the Reactive stress scenario (Supplementary to Table I and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re 7.011735741256e-17 Im -1.118075560772e-06	6.84e-13	[01] 5.77350269e-01+9.66488710e-17j [02] 5.77350269e-01+1.98353167e-16j [03] 5.77350269e-01-2.53734063e-34j [04] 4.61030764e-17-6.45521226e-07j [05] 1.04570975e-16-6.45521226e-07j [06] -2.92139993e-17-6.45521226e-07j
2	Re 7.011735741256e-17 Im 1.118075560772e-06	6.84e-13	[01] 5.77350269e-01-9.66488710e-17j [02] 5.77350269e-01-1.98353167e-16j [03] 5.77350269e-01-4.28315941e-51j [04] 4.61030764e-17+6.45521226e-07j [05] 1.04570975e-16+6.45521226e-07j [06] -2.92139993e-17+6.45521226e-07j
3	Re 2.664410400977e-15 Im -1.337660277579e+01	1.90e-13	[01] -9.60849938e-17-2.96343391e-03j [02] -7.81574522e-17-2.21240793e-02j [03] -1.00063859e-16+7.11291213e-02j [04] -3.96406782e-02-1.37761124e-16j [05] -2.95945020e-01+1.31524671e-16j [06] 9.51466002e-01+1.88693613e-51j
4	Re 2.664410400977e-15 Im 1.337660277579e+01	1.90e-13	[01] -9.60849938e-17+2.96343391e-03j [02] -7.81574522e-17+2.21240793e-02j [03] -1.00063859e-16-7.11291213e-02j [04] -3.96406782e-02+1.37761124e-16j [05] -2.95945020e-01-1.31524671e-16j [06] 9.51466002e-01-2.76245428e-34j
5	Re 4.101387886271e-15 Im 8.671194843856e+00	4.75e-14	[01] 1.39672322e-16+3.60726492e-02j [02] -1.63493264e-17-9.43233092e-02j [03] 1.60516760e-16-5.41019852e-02j [04] -3.12792970e-01-2.06921781e-17j [05] 8.17895792e-01-1.70948934e-34j [06] 4.69128855e-01-2.12205639e-17j
6	Re 1.111745907634e-14 Im -8.671194843856e+00	2.11e-14	[01] 7.83941320e-17-3.60726492e-02j [02] -5.45778647e-17+9.43233092e-02j [03] -8.58560620e-17+5.41019852e-02j [04] -3.12792970e-01-1.21914533e-16j [05] 8.17895792e-01-1.39370210e-35j [06] 4.69128855e-01+1.52825916e-16j

Table S5 39-bus system (Ours) Base

Returned 20/20 eigenpairs; Eigenvalue MAE 1.446623e-11; RMSE 5.396819e-11; mean residual 1.075961e-12; mean MAC 1.0000000000.

Table S5. Computed eigenpairs by the proposed method for the 39-bus system under the Base scenario (Supplementary to Table II and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.295455098032e-04 Im -6.545746650406e+00	9.86e-14	[01] -2.58626181e-06-6.88034058e-03j [02] 7.26614087e-08-3.42576885e-03j [03] 5.23258780e-07-5.69731566e-03j [04] -6.93909312e-06+1.37830874e-01j [05] 1.25694925e-06-4.60550776e-02j [06] -5.38143407e-07-3.14834642e-02j [07] 5.35752496e-06-1.56863846e-02j [08] 4.53469935e-06-1.89226056e-02j [09] -7.18852805e-08-1.17917348e-03j [10] 2.42562544e-08-2.64580620e-05j [11] -4.50369655e-02+1.91963999e-05j [12] -2.24242150e-02+6.53323568e-07j [13] -3.72931851e-02-1.54759462e-06j [14] 9.02205983e-01-1.90193319e-33j [15] -3.01464870e-01+6.94957265e-06j [16] -2.06082780e-01+1.38977846e-05j [17] -1.02679101e-01-2.98996234e-05j [18] -1.23862584e-01-2.34471334e-05j [19] -7.71857083e-03+8.59134151e-07j [20] -1.73187779e-04-1.50056160e-07j
2	Re -3.295455095673e-04 Im 6.545746650406e+00	1.73e-14	[01] -2.58626181e-06+6.88034058e-03j [02] 7.26614088e-08+3.42576885e-03j [03] 5.23258781e-07+5.69731566e-03j [04] -6.93909312e-06-1.37830874e-01j [05] 1.25694925e-06+4.60550776e-02j [06] -5.38143410e-07+3.14834642e-02j [07] 5.35752497e-06+1.56863846e-02j [08] 4.53469935e-06+1.89226056e-02j [09] -7.18852791e-08+1.17917348e-03j [10] 2.42562543e-08+2.64580620e-05j [11] -4.50369655e-02-1.91963999e-05j [12] -2.24242150e-02-6.53323568e-07j [13] -3.72931851e-02+1.54759462e-06j [14] 9.02205983e-01-2.65808464e-34j [15] -3.01464870e-01-6.94957265e-06j [16] -2.06082780e-01-1.38977847e-05j [17] -1.02679101e-01+2.98996235e-05j [18] -1.23862584e-01+2.34471334e-05j [19] -7.71857083e-03-8.59134151e-07j [20] -1.73187779e-04+1.50056160e-07j
3	Re -3.237419521066e-04 Im 6.868699396043e+00	1.32e-14	[01] 3.93828545e-07+1.04936876e-03j [02] -8.53330567e-08+2.85873671e-04j [03] -1.00936578e-07+4.20212720e-04j [04] -3.40402314e-06+1.05097137e-02j [05] 7.27334513e-07-1.98784341e-03j [06] 9.01174967e-06+8.17620683e-02j [07] -5.56788543e-06-1.18131527e-01j [08] -8.99137290e-07+3.17126648e-05j [09] -2.75864888e-08+2.40086440e-04j [10] -6.64317883e-09-2.69668215e-07j [11] -7.20779871e-03+2.36536520e-06j [12] -1.96358028e-03-6.78676417e-07j [13] -2.88631483e-03-8.29343500e-07j [14] -7.21880632e-02-2.67836469e-05j [15] 1.36538986e-02+5.63939044e-06j [16] -5.61599072e-01+3.54291879e-05j [17] 8.11409953e-01+2.41938894e-33j [18] -2.17824471e-04-6.18617048e-06j [19] -1.64908158e-03-2.67209352e-07j [20] 1.85227205e-06-4.55426954e-08j
4	Re -3.237419521041e-04 Im -6.868699396043e+00	1.25e-14	[01] 3.93828582e-07-1.04936876e-03j [02] -8.53330572e-08-2.85873671e-04j [03] -1.00936578e-07-4.20212720e-04j [04] -3.40402314e-06-1.05097137e-02j [05] 7.27334514e-07+1.98784341e-03j [06] 9.01174967e-06-8.17620683e-02j [07] -5.56788543e-06+1.18131527e-01j [08] -8.99137342e-07-3.17126649e-05j [09] -2.75864871e-08-2.40086440e-04j [10] -6.64317924e-09+2.69668214e-07j [11] -7.20779871e-03-2.36536545e-06j [12] -1.96358028e-03+6.78676418e-07j [13] -2.88631483e-03+8.29343501e-07j [14] -7.21880632e-02+2.67836469e-05j [15] 1.36538986e-02-5.63939044e-06j [16] -5.61599072e-01-3.54291879e-05j [17] 8.11409953e-01+1.31490927e-33j [18] -2.17824471e-04+6.18617083e-06j [19] -1.64908158e-03+2.67209341e-07j [20] 1.85227205e-06+4.55426971e-08j
5	Re -3.151425852114e-04 Im -6.810560067478e+00	8.31e-14	[01] 1.62522531e-05-8.22907598e-02j [02] -7.49362789e-07+4.74246726e-04j [03] -7.81207312e-07+3.55683280e-04j [04] -3.96671911e-06-4.84289946e-04j [05] 1.01446635e-06-5.96884599e-04j [06] -5.05515801e-07-1.84356374e-03j [07] 2.41553565e-07+1.53385690e-03j [08] -5.53544336e-06+1.19626706e-01j [09] -7.89313400e-07-3.86352325e-03j [10] -3.12960119e-07+5.93535159e-04j [11] -5.60446168e-01-8.47536234e-05j [12] 3.22988605e-03+4.95412494e-06j [13] 2.42240259e-03+5.20836837e-06j [14] -3.29828452e-03+2.71681991e-05j [15] -4.06511873e-03-6.72098027e-06j [16] -1.25557015e-02+4.02383116e-06j [17] 1.04464245e-02-2.12849868e-06j [18] 8.14724871e-01+1.01145900e-33j [19] -2.63127569e-02+6.59322703e-06j [20] 4.04230695e-03+1.94438548e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
6	Re -3.151425852114e-04 Im 6.810560067478e+00	8.31e-14	[01] 1.62522531e-05+8.22907598e-02j [03] -7.81207312e-07-3.55683280e-04j [05] 1.01446635e-06+5.96884599e-04j [07] 2.41553565e-07-1.53385690e-03j [09] -7.89313400e-07+3.86352325e-03j [11] -5.60446168e-01+8.47536234e-05j [13] 2.42240259e-03-5.20836837e-06j [15] -4.06511873e-03+6.72098027e-06j [17] 1.04464245e-02+2.12849868e-06j [19] -2.63127569e-02-6.59322703e-06j	[02] -7.49362789e-07-4.74246726e-04j [04] -3.96671911e-06+4.84289946e-04j [06] -5.05515801e-07+1.84356374e-03j [08] -5.53544336e-06-1.19626706e-01j [10] -3.12960119e-07-5.93535159e-04j [12] 3.22988605e-03-4.95412494e-06j [14] -3.29828452e-03-2.71681991e-05j [16] -1.25557015e-02-4.02383116e-06j [18] 8.14724871e-01+6.75153080e-50j [20] 4.04230695e-03-1.94438548e-06j
7	Re -3.093790131005e-04 Im -5.061482055348e+00	6.67e-12	[01] 9.16706989e-07+9.02229536e-03j [03] 5.79017614e-06-5.88390916e-02j [05] 8.65304390e-07-7.81564688e-02j [07] -1.02095812e-05+1.00147607e-01j [09] 2.24026501e-06-2.46937367e-02j [11] 4.56661858e-02-7.43120454e-06j [13] -2.97813008e-01-1.11032948e-05j [15] -3.95587564e-01+1.98002461e-05j [17] 5.06895319e-01+2.06920472e-05j [19] -1.24986906e-01-3.69933836e-06j	[02] 1.20586623e-05-7.35063930e-02j [04] -3.18469076e-06+1.15004987e-02j [06] -6.60031534e-06+1.07982055e-01j [08] 3.22472624e-07+8.95079096e-03j [10] -4.23161002e-07+9.07914990e-04j [12] -3.72051293e-01-3.82933700e-05j [14] 5.82095689e-02+1.25612425e-05j [16] 5.46549237e-01+1.37556405e-36j [18] 4.53042677e-02-4.40137610e-06j [20] 4.59539556e-03+1.86093251e-06j
8	Re -3.093789966933e-04 Im 5.061482055348e+00	9.27e-16	[01] 9.16706913e-07-9.02229536e-03j [03] 5.79017653e-06+5.88390916e-02j [05] 8.65304623e-07+7.81564688e-02j [07] -1.02095819e-05-1.00147607e-01j [09] 2.24026532e-06+2.46937367e-02j [11] 4.56661858e-02+7.43120428e-06j [13] -2.97813008e-01+1.11032954e-05j [15] -3.95587564e-01-1.98002461e-05j [17] 5.06895319e-01-2.06920494e-05j [19] -1.24986906e-01+3.69933925e-06j	[02] 1.20586631e-05+7.35063930e-02j [04] -3.18469090e-06-1.15004987e-02j [06] -6.60031579e-06-1.07982055e-01j [08] 3.22472545e-07-8.95079096e-03j [10] -4.23161141e-07-9.07914990e-04j [12] -3.72051293e-01+3.82933730e-05j [14] 5.82095689e-02-1.25612431e-05j [16] 5.46549237e-01-2.33632255e-67j [18] 4.53042677e-02+4.40137573e-06j [20] 4.59539556e-03-1.86093269e-06j
9	Re -3.046730835365e-04 Im 4.200799305801e+00	8.89e-12	[01] -4.35255422e-06-2.22301539e-02j [03] -7.31955580e-06+9.97049145e-04j [05] 2.06960165e-05+1.42347439e-01j [07] -2.92559724e-06+2.32478125e-02j [09] -1.21650377e-05-1.67730238e-01j [11] 9.33844162e-02-1.15112782e-05j [13] -4.18840113e-03-3.10517589e-05j [15] -5.97973028e-01+4.35703828e-05j [17] -9.76593939e-02-1.93728290e-05j [19] 7.04601073e-01-1.59171867e-33j	[02] -7.39521113e-06-7.15710686e-04j [04] 3.26608342e-06+4.89449665e-02j [06] -3.63309616e-06+2.40940603e-02j [08] -6.82922228e-06-3.48071648e-02j [10] 1.79853527e-06+2.95013651e-03j [12] 3.00655921e-03-3.08477400e-05j [14] -2.05607982e-01-1.19205146e-06j [16] -1.01214311e-01-2.26027186e-05j [18] 1.46217916e-01-1.80833868e-05j [20] -1.23929319e-02+6.65646014e-06j
10	Re -3.046730678263e-04 Im -4.200799305801e+00	1.21e-12	[01] -4.35255435e-06+2.22301539e-02j [03] -7.31955601e-06-9.97049145e-04j [05] 2.06960169e-05-1.42347439e-01j [07] -2.92559759e-06-2.32478125e-02j [09] -1.21650385e-05+1.67730238e-01j [11] 9.33844162e-02+1.15112782e-05j [13] -4.18840113e-03+3.10517598e-05j [15] -5.97973028e-01-4.35703832e-05j [17] -9.76593939e-02+1.93728306e-05j [19] 7.04601073e-01-1.21704382e-34j	[02] -7.39521129e-06+7.15710686e-04j [04] 3.26608333e-06-4.89449665e-02j [06] -3.63309655e-06-2.40940603e-02j [08] -6.82922259e-06+3.48071648e-02j [10] 1.79853575e-06-2.95013651e-03j [12] 3.00655921e-03+3.08477407e-05j [14] -2.05607982e-01+1.19205231e-06j [16] -1.01214311e-01+2.26027206e-05j [18] 1.46217916e-01+1.80833880e-05j [20] -1.23929319e-02-6.65646082e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
11	Re -3.005670471533e-04 Im -1.517917908053e-21	2.47e-15	[01] 3.16227751e-01+3.53591796e-19j [02] 3.16227752e-01-3.99104666e-20j [03] 3.16227752e-01+2.23595487e-19j [04] 3.16227753e-01+2.91918872e-19j [05] 3.16227754e-01+3.57071765e-69j [06] 3.16227753e-01+3.44316592e-19j [07] 3.16227753e-01+6.78212020e-21j [08] 3.16227752e-01+5.34714582e-19j [09] 3.16227753e-01+4.67558554e-19j [10] 3.16227745e-01+5.83302123e-19j [11] -9.50476413e-05-1.82525330e-22j [12] -9.50476415e-05-1.07384109e-22j [13] -9.50476415e-05-2.62675674e-22j [14] -9.50476420e-05-1.52093119e-22j [15] -9.50476424e-05-2.33453760e-22j [16] -9.50476419e-05-1.93155720e-22j [17] -9.50476420e-05-1.47662360e-22j [18] -9.50476416e-05-1.72926011e-22j [19] -9.50476418e-05-1.64547849e-22j [20] -9.50476397e-05-1.66777902e-22j
12	Re -3.002929663520e-04 Im -4.583815868253e+00	9.02e-13	[01] 4.53693982e-07+1.79393253e-02j [02] -7.61589952e-06+1.16252744e-01j [03] -3.38949150e-06+1.10279628e-01j [04] -3.86409938e-06-1.70160143e-02j [05] 4.96811792e-07-1.11562490e-01j [06] -7.76020744e-06+2.58689101e-02j [07] -8.11610347e-06+2.45356265e-02j [08] 6.07334259e-07+1.22128577e-02j [09] 1.38569914e-05-7.22661698e-02j [10] 1.51975988e-06-7.38028284e-03j [11] 8.22305639e-02-7.46670283e-06j [12] 5.32881177e-01+3.57112399e-35j [13] 5.05501511e-01-1.75793913e-05j [14] -7.79982751e-02+2.28221094e-05j [15] -5.11381913e-01+3.12241369e-05j [16] 1.18578323e-01+2.78031104e-05j [17] 1.12466796e-01+2.98348479e-05j [18] 5.59814908e-02-6.45134366e-06j [19] -3.31254820e-01-4.18168749e-05j [20] -3.38298581e-02-4.75005278e-06j
13	Re -3.002929637752e-04 Im 4.583815868253e+00	1.90e-13	[01] 4.53694029e-07-1.79393253e-02j [02] -7.61589956e-06-1.16252744e-01j [03] -3.38949150e-06-1.10279628e-01j [04] -3.86409942e-06+1.70160143e-02j [05] 4.96811754e-07+1.11562490e-01j [06] -7.76020752e-06-2.58689101e-02j [07] -8.11610355e-06-2.45356265e-02j [08] 6.07334299e-07-1.22128577e-02j [09] 1.38569914e-05+7.22661698e-02j [10] 1.51975999e-06+7.38028284e-03j [11] 8.22305639e-02+7.46670307e-06j [12] 5.32881177e-01+3.57298654e-35j [13] 5.05501511e-01+1.75793916e-05j [14] -7.79982751e-02-2.28221096e-05j [15] -5.11381913e-01-3.12241373e-05j [16] 1.18578323e-01-2.78031107e-05j [17] 1.12466796e-01-2.98348482e-05j [18] 5.59814908e-02+6.45134384e-06j [19] -3.31254820e-01+4.18168748e-05j [20] -3.38298581e-02+4.75005288e-06j
14	Re -2.922696776643e-04 Im -5.711763079524e+00	4.99e-14	[01] 8.89755210e-06+1.04938876e-01j [02] -8.96676667e-06-1.33951926e-02j [03] 7.25175645e-06-2.98328985e-02j [04] -1.39306449e-07+4.31481797e-03j [05] 1.52653066e-07-1.67250294e-03j [06] -2.43024774e-06-2.94993935e-02j [07] -4.26096750e-07-2.56934450e-02j [08] -6.21770638e-06+1.21511291e-01j [09] 1.74729665e-07-3.64417140e-02j [10] 2.63647356e-07-3.75177825e-03j [11] 5.99385995e-01-8.14911610e-05j [12] -7.65101642e-02+5.51310555e-05j [13] -1.70398450e-01-3.27010631e-05j [14] 2.46452180e-02-4.65405028e-07j [15] -9.55294059e-03-3.83096251e-07j [16] -1.68493546e-01+2.25027775e-05j [17] -1.46754871e-01+9.94317858e-06j [18] 6.94043705e-01+1.02314727e-33j [19] -2.08146436e-01+9.65279354e-06j [20] -2.14292686e-02-4.09360213e-07j
15	Re -2.922696776592e-04 Im 5.711763079524e+00	1.36e-14	[01] 8.89755210e-06-1.04938876e-01j [02] -8.96676674e-06+1.33951926e-02j [03] 7.25175651e-06+2.98328985e-02j [04] -1.39306449e-07-4.31481797e-03j [05] 1.52653066e-07+1.67250294e-03j [06] -2.43024774e-06+2.94993935e-02j [07] -4.26096753e-07+2.56934450e-02j [08] -6.21770638e-06-1.21511291e-01j [09] 1.74729665e-07+3.64417140e-02j [10] 2.63647356e-07+3.75177825e-03j [11] 5.99385995e-01+8.14911610e-05j [12] -7.65101642e-02-5.51310558e-05j [13] -1.70398450e-01+3.27010635e-05j [14] 2.46452180e-02+4.65405024e-07j [15] -9.55294059e-03+3.83096251e-07j [16] -1.68493546e-01-2.25027775e-05j [17] -1.46754871e-01-9.94317860e-06j [18] 6.94043705e-01-5.60274835e-34j [19] -2.08146436e-01-9.65279354e-06j [20] -2.14292686e-02+4.09360214e-07j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
16	Re -2.882560430318e-04 Im 5.639387583526e+00	9.56e-14	[01] 8.81132440e-06+6.13597700e-03j [02] -6.68089717e-06-1.30703829e-01j [03] 8.35884296e-06+1.14747872e-01j [04] 4.11147992e-07+2.97638099e-04j [05] -1.53491401e-07-4.88945872e-04j [06] -4.16940388e-06-8.81465993e-03j [07] -4.13409077e-06-7.71205494e-03j [08] 1.08701761e-05+7.19410612e-03j [09] -3.59550088e-06-2.79730645e-03j [10] -4.04347270e-07+7.13166771e-05j [11] -3.46031550e-02+4.79217409e-05j [12] 7.37089552e-01+1.34904860e-34j [13] -6.47107728e-01+1.40619876e-05j [14] -1.67849672e-03+2.23282690e-06j [15] 2.75735532e-03-7.24655898e-07j [16] 4.97092850e-02-2.09720054e-05j [17] 4.34912681e-02-2.10906937e-05j [18] -4.05703559e-02+5.92273917e-05j [19] 1.57750963e-02-1.94700825e-05j [20] -4.02182267e-04-2.30082841e-06j
17	Re -2.882560430295e-04 Im -5.639387583526e+00	1.96e-15	[01] 8.81132480e-06-6.13597700e-03j [02] -6.68089717e-06+1.30703829e-01j [03] 8.35884280e-06-1.14747872e-01j [04] 4.11148008e-07-2.97638099e-04j [05] -1.53491409e-07+4.88945872e-04j [06] -4.16940399e-06+8.81465993e-03j [07] -4.13409086e-06+7.71205494e-03j [08] 1.08701766e-05-7.19410612e-03j [09] -3.59550102e-06+2.79730645e-03j [10] -4.04347280e-07-7.13166771e-05j [11] -3.46031550e-02-4.79217432e-05j [12] 7.37089552e-01-8.08336243e-68j [13] -6.47107728e-01-1.40619867e-05j [14] -1.67849672e-03-2.23282699e-06j [15] 2.75735532e-03+7.24655941e-07j [16] 4.97092850e-02+2.09720061e-05j [17] 4.34912681e-02+2.10906942e-05j [18] -4.05703559e-02-5.92273943e-05j [19] 1.57750963e-02+1.94700833e-05j [20] -4.02182267e-04+2.30082849e-06j
18	Re -1.848636349090e-04 Im -2.753123443455e+00	1.56e-12	[01] 8.71360871e-06+5.45273153e-02j [02] 7.74275353e-06+6.57761475e-02j [03] 7.32303187e-06+7.59601488e-02j [04] -2.17072243e-06+1.26144716e-01j [05] -1.13580391e-05+1.69152164e-01j [06] 2.76688708e-07+1.24639158e-01j [07] -1.07108461e-06+1.23349387e-01j [08] 6.30345558e-06+6.68972555e-02j [09] 2.87295360e-06+1.24318955e-01j [10] 1.75458469e-05-9.09906037e-02j [11] 1.50120429e-01-3.40697581e-05j [12] 1.81089852e-01-3.34763739e-05j [13] 2.09127665e-01-3.42034798e-05j [14] 3.47291975e-01-1.73433038e-05j [15] 4.65696792e-01-7.69535669e-35j [16] 3.43146988e-01-2.38030058e-05j [17] 3.39596089e-01-1.98539878e-05j [18] 1.84176401e-01-2.97210611e-05j [19] 3.42265429e-01-3.08916497e-05j [20] -2.50508367e-01-3.14850302e-05j
19	Re -1.848636349090e-04 Im 2.753123443455e+00	1.56e-12	[01] 8.71360871e-06-5.45273153e-02j [02] 7.74275353e-06-6.57761475e-02j [03] 7.32303187e-06-7.59601488e-02j [04] -2.17072243e-06-1.26144716e-01j [05] -1.13580391e-05-1.69152164e-01j [06] 2.76688708e-07-1.24639158e-01j [07] -1.07108461e-06-1.23349387e-01j [08] 6.30345558e-06-6.68972555e-02j [09] 2.87295360e-06-1.24318955e-01j [10] 1.75458469e-05+9.09906037e-02j [11] 1.50120429e-01+3.40697581e-05j [12] 1.81089852e-01+3.34763739e-05j [13] 2.09127665e-01+3.42034798e-05j [14] 3.47291975e-01+1.73433038e-05j [15] 4.65696792e-01+1.13496270e-51j [16] 3.43146988e-01+2.38030058e-05j [17] 3.39596089e-01+1.98539878e-05j [18] 1.84176401e-01+2.97210611e-05j [19] 3.42265429e-01+3.08916497e-05j [20] -2.50508367e-01+3.14850302e-05j
20	Re -2.238312313567e-10 Im -8.122683918882e-11	5.70e-14	[01] 3.16227766e-01-1.67496702e-15j [02] 3.16227766e-01-1.17202484e-15j [03] 3.16227766e-01-2.00469211e-15j [04] 3.16227766e-01-1.01469063e-15j [05] 3.16227766e-01-9.60135485e-34j [06] 3.16227766e-01-4.97477707e-16j [07] 3.16227766e-01-8.28539525e-16j [08] 3.16227766e-01-1.70378392e-15j [09] 3.16227766e-01-7.55415807e-16j [10] 3.16227766e-01-4.12150811e-15j [11] -7.07815420e-11-2.56863658e-11j [12] -7.07816864e-11-2.56861906e-11j [13] -7.07816737e-11-2.56861396e-11j [14] -7.07820357e-11-2.56861262e-11j [15] -7.07815681e-11-2.56862931e-11j [16] -7.07816821e-11-2.56864594e-11j [17] -7.07813583e-11-2.56862483e-11j [18] -7.07815331e-11-2.56860615e-11j [19] -7.07815189e-11-2.56862669e-11j [20] -7.07819043e-11-2.56856675e-11j

Table S6 39-bus system (Ours) Light load

Returned 20/20 eigenpairs; Eigenvalue MAE 7.861444e-12; RMSE 2.170980e-11; mean residual 1.406302e-12; mean MAC 1.0000000000.

Table S6. Computed eigenpairs by the proposed method for the 39-bus system under the Light load scenario (Supplementary to Table II and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.288322156381e-04 Im 6.550640975015e+00	1.45e-13	[01] -2.56174042e-06+7.67301858e-03j [02] 6.57493963e-08+3.50591629e-03j [03] 5.13049444e-07+5.80103478e-03j [04] -6.92177560e-06-1.37888153e-01j [05] 1.16459566e-06+4.45465709e-02j [06] -4.43585606e-07+3.34776305e-02j [07] 5.51831318e-06+1.44631174e-02j [08] 4.68174739e-06+1.84170537e-02j [09] -7.38353124e-08+1.38204261e-03j [10] 2.00636782e-08+1.75653084e-05j [11] -5.02631891e-02-1.93041774e-05j [12] -2.29659989e-02-7.22157533e-07j [13] -3.80004963e-02+1.45323559e-06j [14] 9.03255788e-01+2.70970606e-33j [15] -2.91808593e-01-7.01949953e-06j [16] -2.19299938e-01-1.39142935e-05j [17] -9.47426914e-02+3.13925495e-05j [18] -1.20643508e-01+2.46123256e-05j [19] -9.05326493e-03-9.38128755e-07j [20] -1.15064035e-04+1.25653906e-07j
2	Re -3.288322155589e-04 Im -6.550640975015e+00	3.36e-14	[01] -2.56174055e-06-7.67301858e-03j [02] 6.57493951e-08-3.50591629e-03j [03] 5.13049444e-07-5.80103478e-03j [04] -6.92177560e-06+1.37888153e-01j [05] 1.16459566e-06-4.45465709e-02j [06] -4.43585602e-07-3.34776305e-02j [07] 5.51831316e-06-1.44631174e-02j [08] 4.68174762e-06-1.84170537e-02j [09] -7.38353224e-08-1.38204261e-03j [10] 2.00636772e-08-1.75653084e-05j [11] -5.02631891e-02+1.93041783e-05j [12] -2.29659989e-02+7.22157538e-07j [13] -3.80004963e-02-1.45323559e-06j [14] 9.03255788e-01-3.55013370e-34j [15] -2.91808593e-01+7.01949953e-06j [16] -2.19299938e-01+1.39142934e-05j [17] -9.47426914e-02-3.13925493e-05j [18] -1.20643508e-01-2.46123272e-05j [19] -9.05326493e-03+9.38128822e-07j [20] -1.15064035e-04-1.25653909e-07j
3	Re -3.246441690241e-04 Im 6.870225271878e+00	1.20e-14	[01] 7.05380430e-07+1.44075147e-03j [02] -8.86682634e-08+3.16453283e-04j [03] -1.00752345e-07+4.69289979e-04j [04] -3.38923143e-06+1.22640385e-02j [05] 7.03501317e-07-2.24568935e-03j [06] 9.03419814e-06+8.04163731e-02j [07] -5.61547089e-06-1.18836418e-01j [08] -1.29060661e-06-2.87724951e-04j [09] -2.15500144e-08+2.71924041e-04j [10] -9.08537318e-09-5.17397546e-07j [11] -9.89828740e-03+4.37839089e-06j [12] -2.17410531e-03-7.11905657e-07j [13] -3.22412784e-03-8.44543559e-07j [14] -8.42567063e-02-2.72662320e-05j [15] 1.54283915e-02+5.56226248e-06j [16] -5.52478601e-01+3.59602698e-05j [17] 8.16432961e-01-2.02595712e-34j [18] 1.97673565e-03-8.77334988e-06j [19] -1.86817941e-03-2.36332007e-07j [20] 3.55464065e-06-6.22505891e-08j
4	Re -3.246441690106e-04 Im -6.870225271878e+00	1.60e-14	[01] 7.05380456e-07-1.44075147e-03j [02] -8.86682636e-08-3.16453283e-04j [03] -1.00752345e-07-4.69289979e-04j [04] -3.38923143e-06-1.22640385e-02j [05] 7.03501317e-07+2.24568935e-03j [06] 9.03419814e-06-8.04163731e-02j [07] -5.61547089e-06+1.18836418e-01j [08] -1.29060664e-06+2.87724951e-04j [09] -2.15500135e-08-2.71924041e-04j [10] -9.08537342e-09+5.17397546e-07j [11] -9.89828740e-03-4.37839106e-06j [12] -2.17410531e-03+7.11905659e-07j [13] -3.22412784e-03+8.44543561e-07j [14] -8.42567063e-02+2.72662320e-05j [15] 1.54283915e-02-5.56226248e-06j [16] -5.52478601e-01-3.59602698e-05j [17] 8.16432961e-01+8.23490161e-34j [18] 1.97673565e-03+8.77335014e-06j [19] -1.86817941e-03+2.36332000e-07j [20] 3.55464065e-06+6.22505904e-08j
5	Re -3.164263431486e-04 Im -6.818223698329e+00	4.00e-12	[01] 1.61673335e-05-8.11385885e-02j [02] -7.79097774e-07+6.10687861e-04j [03] -8.33920643e-07+5.17479686e-04j [04] -4.15360809e-06+7.99762638e-05j [05] 9.80175193e-07-5.93079554e-04j [06] -8.01061369e-07-1.40893228e-03j [07] 5.31679024e-07+1.27488340e-03j [08] -5.57992606e-06+1.20233926e-01j [09] -8.86355531e-07-3.64564816e-03j [10] -3.04264090e-07+5.53386933e-04j [11] -5.53221052e-01-8.45581064e-05j [12] 4.16380669e-03+5.11882514e-06j [13] 3.52829252e-03+5.52211321e-06j [14] 5.45297372e-04+2.82949223e-05j [15] -4.04374938e-03-6.49538768e-06j [16] -9.60641520e-03+5.90763887e-06j [17] 8.69244005e-03-4.02851322e-06j [18] 8.19781806e-01+1.19314035e-33j [19] -2.48568444e-02+7.19694956e-06j [20] 3.77311600e-03+1.89943420e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
6	Re -3.164263431486e-04 Im 6.818223698329e+00	4.00e-12	[01] 1.61673335e-05+8.11385885e-02j [02] -7.79097774e-07-6.10687861e-04j [03] -8.33920643e-07-5.17479686e-04j [04] -4.15360809e-06-7.99762638e-05j [05] 9.80175193e-07+5.93079554e-04j [06] -8.01061369e-07+1.40893228e-03j [07] 5.31679024e-07-1.27488340e-03j [08] -5.57992606e-06-1.20233926e-01j [09] -8.86355531e-07+3.64564816e-03j [10] -3.04264090e-07-5.53386933e-04j [11] -5.53221052e-01+8.45581064e-05j [12] 4.16380669e-03-5.11882514e-06j [13] 3.52829252e-03-5.52211321e-06j [14] 5.45297372e-04-2.82949223e-05j [15] -4.04374938e-03+6.49538768e-06j [16] -9.60641520e-03-5.90763887e-06j [17] 8.69244005e-03+4.02851322e-06j [18] 8.19781806e-01+6.23020154e-50j [19] -2.48568444e-02-7.19694956e-06j [20] 3.77311600e-03-1.89943420e-06j
7	Re -3.084791376913e-04 Im -5.105575755992e+00	1.58e-13	[01] 9.04025121e-07+9.33538526e-03j [02] 1.21017042e-05-7.50045152e-02j [03] 5.19054646e-06-5.78766411e-02j [04] -3.29080884e-06+1.38435046e-02j [05] 7.24252654e-07-7.57239723e-02j [06] -6.48349039e-06+1.07306937e-01j [07] -1.00379999e-05+9.79649976e-02j [08] 2.84160069e-07+9.15410541e-03j [09] 2.60535294e-06-2.75822094e-02j [10] -3.87586905e-07+8.35653625e-04j [11] 4.76625164e-02-7.49534033e-06j [12] -3.82941238e-01-3.86488396e-05j [13] -2.95493577e-01-8.64699188e-06j [14] 7.06790625e-02+1.25310414e-05j [15] -3.86614477e-01+1.96615388e-05j [16] 5.47863699e-01+6.56674806e-35j [17] 5.00167720e-01+2.10296110e-05j [18] 4.67369786e-02-4.27465130e-06j [19] -1.40823060e-01-4.79329069e-06j [20] 4.26649301e-03+1.72107261e-06j
8	Re -3.084791361659e-04 Im 5.105575755992e+00	2.37e-12	[01] 9.04024608e-07-9.33538526e-03j [02] 1.21017067e-05+7.50045152e-02j [03] 5.19054711e-06+5.78766411e-02j [04] -3.29080911e-06-1.38435046e-02j [05] 7.24252267e-07+7.57239723e-02j [06] -6.48349052e-06-1.07306937e-01j [07] -1.00380002e-05-9.79649976e-02j [08] 2.84159653e-07-9.15410541e-03j [09] 2.60535287e-06+2.75822094e-02j [10] -3.87587179e-07-8.35653625e-04j [11] 4.76625164e-02+7.49533685e-06j [12] -3.82941238e-01+3.86488526e-05j [13] -2.95493577e-01+8.64699372e-06j [14] 7.06790625e-02-1.25310430e-05j [15] -3.86614477e-01-1.96615401e-05j [16] 5.47863699e-01-9.21171577e-35j [17] 5.00167720e-01-2.10296132e-05j [18] 4.67369786e-02+4.27464895e-06j [19] -1.40823060e-01+4.79329080e-06j [20] 4.26649301e-03-1.72107301e-06j
9	Re -3.081160357647e-04 Im -4.289686990722e+00	1.37e-14	[01] -4.58354636e-06+2.38404387e-02j [02] -8.50188368e-06+6.73397499e-03j [03] -8.18136123e-06+4.97849685e-03j [04] 3.06753344e-06-4.58753338e-02j [05] 2.26569818e-05-1.47579402e-01j [06] -4.33701556e-06-1.90450692e-02j [07] -3.66628072e-06-1.81991828e-02j [08] -7.05603259e-06+3.52063827e-02j [09] -1.13738688e-05+1.58350528e-01j [10] 1.89551578e-06-3.40647639e-03j [11] 1.02268021e-01+1.23163577e-05j [12] 2.88866475e-02+3.43955741e-05j [13] 2.13561957e-02+3.35615241e-05j [14] -1.96790823e-01+9.76167673e-07j [15] -6.33069449e-01-5.17197798e-05j [16] -8.16973843e-02+2.44725304e-05j [17] -7.80687965e-02+2.13346568e-05j [18] 1.51024364e-01+1.94205201e-05j [19] 6.79274202e-01-3.95412914e-35j [20] -1.46127180e-02-7.08157938e-06j
10	Re -3.081160277218e-04 Im 4.289686990722e+00	3.83e-12	[01] -4.58354641e-06-2.38404387e-02j [02] -8.50188375e-06-6.73397499e-03j [03] -8.18136133e-06-4.97849685e-03j [04] 3.06753336e-06+4.58753338e-02j [05] 2.26569820e-05+1.47579402e-01j [06] -4.33701579e-06+1.90450692e-02j [07] -3.66628092e-06+1.81991828e-02j [08] -7.05603273e-06-3.52063827e-02j [09] -1.13738691e-05-1.58350528e-01j [10] 1.89551605e-06+3.40647639e-03j [11] 1.02268021e-01-1.23163577e-05j [12] 2.88866475e-02-3.43955744e-05j [13] 2.13561957e-02-3.35615245e-05j [14] -1.96790823e-01-9.76168263e-07j [15] -6.33069449e-01+5.17197797e-05j [16] -8.16973843e-02-2.44725316e-05j [17] -7.80687965e-02-2.13346577e-05j [18] 1.51024364e-01-1.94205207e-05j [19] 6.79274202e-01+6.60803759e-34j [20] -1.46127180e-02+7.08157974e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
11	Re -2.981449793202e-04 Im 4.636905704887e+00	5.05e-13	[01] 8.94493049e-07-1.54359855e-02j [03] -2.88942962e-06-1.07545374e-01j [05] -1.68449255e-06+1.05041087e-01j [07] -7.58583012e-06-2.49948043e-02j [09] 1.52534270e-05+8.25368731e-02j [11] 7.15752089e-02+8.74984153e-06j [13] 4.98677757e-01+1.86661007e-05j [15] -4.87065617e-01-3.91283061e-05j [17] 1.15898553e-01-2.77227035e-05j [19] -3.82715702e-01+4.61207480e-05j	[02] -7.37339523e-06-1.14674875e-01j [04] -4.22190697e-06+1.34909429e-02j [06] -7.24644840e-06-2.66152048e-02j [08] 1.28439458e-06-9.05295451e-03j [10] 1.29836929e-06+6.71093718e-03j [12] 5.31736583e-01-4.02192441e-34j [14] -6.25562287e-02-2.35988414e-05j [16] 1.23412197e-01-2.56659082e-05j [18] 4.19776960e-02+8.65470951e-06j [20] -3.11179833e-02+4.01958357e-06j
12	Re -2.981449759482e-04 Im -4.636905704887e+00	1.92e-12	[01] 8.94493106e-07+1.54359855e-02j [03] -2.88942963e-06+1.07545374e-01j [05] -1.68449260e-06-1.05041087e-01j [07] -7.58583022e-06+2.49948043e-02j [09] 1.52534270e-05-8.25368731e-02j [11] 7.15752089e-02-8.74984182e-06j [13] 4.98677757e-01-1.86661010e-05j [15] -4.87065617e-01+3.91283066e-05j [17] 1.15898553e-01+2.77227039e-05j [19] -3.82715702e-01-4.61207477e-05j	[02] -7.37339528e-06+1.14674875e-01j [04] -4.22190703e-06-1.34909429e-02j [06] -7.24644850e-06+2.66152048e-02j [08] 1.28439463e-06+9.05295451e-03j [10] 1.29836943e-06-6.71093718e-03j [12] 5.31736583e-01-7.37836349e-35j [14] -6.25562287e-02+2.35988417e-05j [16] 1.23412197e-01+2.56659086e-05j [18] 4.19776960e-02-8.65470973e-06j [20] -3.11179833e-02-4.01958369e-06j
13	Re -2.913215818676e-04 Im 5.737927004192e+00	2.87e-12	[01] 9.16748184e-06-1.04536117e-01j [03] 4.78383898e-06+2.87979356e-02j [05] 1.62160682e-07+1.65706393e-03j [07] -3.91764952e-07+2.68582410e-02j [09] 4.81742523e-08+3.82515941e-02j [11] 5.99820604e-01+8.30559684e-05j [13] -1.65240454e-01+1.90598606e-05j [15] -9.50811191e-03+4.47727615e-07j [17] -1.54110626e-01-1.00723041e-05j [19] -2.19484855e-01-1.08670946e-05j	[02] -6.38255984e-06+1.46151183e-02j [04] -1.74887787e-07-3.94032656e-03j [06] -2.59599525e-06+3.18021102e-02j [08] -6.06703640e-06-1.19497538e-01j [10] 2.38910621e-07+3.63924037e-03j [12] -8.38604803e-02-4.08803639e-05j [14] 2.26093062e-02+1.44408695e-07j [16] -1.82478186e-01-2.41602720e-05j [18] 6.85668153e-01+2.23651281e-49j [20] -2.08816957e-02+3.10662599e-07j
14	Re -2.913215811288e-04 Im -5.737927004192e+00	1.27e-13	[01] 9.16748197e-06+1.04536117e-01j [03] 4.78384848e-06-2.87979356e-02j [05] 1.62160540e-07-1.65706393e-03j [07] -3.91765326e-07-2.68582410e-02j [09] 4.81740912e-08-3.82515941e-02j [11] 5.99820604e-01-8.30559691e-05j [13] -1.65240454e-01-1.90599132e-05j [15] -9.50811191e-03-4.47726857e-07j [17] -1.54110626e-01+1.00723061e-05j [19] -2.19484855e-01+1.08670954e-05j	[02] -6.38257040e-06-1.46151183e-02j [04] -1.74887681e-07+3.94032656e-03j [06] -2.59599587e-06-3.18021102e-02j [08] -6.06703636e-06+1.19497538e-01j [10] 2.38910667e-07-3.63924037e-03j [12] -8.38604803e-02+4.08804225e-05j [14] 2.26093062e-02-1.44409414e-07j [16] -1.82478186e-01+2.41602758e-05j [18] 6.85668153e-01+4.12029163e-34j [20] -2.08816957e-02-3.10662716e-07j
15	Re -2.877535985350e-04 Im 5.635952357782e+00	6.04e-13	[01] 6.10681680e-06+5.40059028e-03j [03] 9.93213751e-06+1.16152818e-01j [05] -5.17763978e-08-1.42776526e-04j [07] -3.74991651e-06-8.26491019e-03j [09] -2.78209881e-06-2.63405915e-03j [11] -3.04374713e-02+3.28636893e-05j [13] -6.54631750e-01+2.25536630e-05j [15] 8.04681711e-04-2.50724852e-07j [17] 4.65806411e-02-1.87560932e-05j [19] 1.48454327e-02-1.49218164e-05j	[02] -6.61662469e-06-1.29593451e-01j [04] 2.02254975e-07+3.69975051e-05j [06] -3.78349470e-06-9.64119185e-03j [08] 7.51978615e-06+6.16531804e-03j [10] -3.07174490e-07+5.69516323e-05j [12] 7.30382518e-01-4.08185121e-35j [14] -2.08516234e-04+1.12925324e-06j [16] 5.43372990e-02-1.85493083e-05j [18] -3.47474409e-02+4.06070640e-05j [20] -3.20976598e-04-1.74760883e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
16	Re -2.877535967446e-04 Im -5.635952357782e+00	2.47e-12	[01] 6.10682472e-06-5.40059028e-03j [03] 9.93213432e-06-1.16152818e-01j [05] -5.17765199e-08+1.42776526e-04j [07] -3.74991848e-06+8.26491019e-03j [09] -2.78210170e-06+2.63405915e-03j [11] -3.04374713e-02-3.28637355e-05j [13] -6.54631750e-01-2.25536442e-05j [15] 8.04681711e-04+2.50725563e-07j [17] 4.65806411e-02+1.87561047e-05j [19] 1.48454327e-02+1.49218332e-05j	[02] -6.61662467e-06+1.29593451e-01j [04] 2.02255271e-07-3.69975050e-05j [06] -3.78349703e-06+9.64119185e-03j [08] 7.51979497e-06-6.16531804e-03j [10] -3.07174634e-07-5.69516324e-05j [12] 7.30382518e-01+2.48998034e-34j [14] -2.08516234e-04-1.12925494e-06j [16] 5.43372990e-02+1.85493219e-05j [18] -3.47474409e-02-4.06071150e-05j [20] -3.20976598e-04+1.74761033e-06j
17	Re -2.868555294385e-04 Im 8.835518258850e-23	5.66e-15	[01] 3.16227752e-01-1.19383287e-19j [03] 3.16227753e-01-1.03211347e-19j [05] 3.16227755e-01-2.29715900e-69j [07] 3.16227754e-01+7.99049454e-21j [09] 3.16227754e-01-7.58410817e-20j [11] -9.07116793e-05+4.63573788e-23j [13] -9.07116795e-05+4.97185618e-23j [15] -9.07116802e-05+4.23705843e-23j [17] -9.07116799e-05+2.81611443e-23j [19] -9.07116797e-05+3.37677535e-23j	[02] 3.16227753e-01-6.24484091e-20j [04] 3.16227754e-01-8.15546222e-20j [06] 3.16227754e-01-8.11502127e-20j [08] 3.16227753e-01-4.73801981e-20j [10] 3.16227747e-01-5.99141668e-20j [12] -9.07116795e-05+4.42283915e-23j [14] -9.07116799e-05+3.71932130e-23j [16] -9.07116798e-05+1.38127752e-23j [18] -9.07116795e-05+3.12514913e-23j [20] -9.07116779e-05+9.13244267e-24j
18	Re -1.913021096610e-04 Im 2.834139802005e+00	2.48e-13	[01] 7.67039115e-06-5.67349616e-02j [03] 6.46817365e-06-7.64706369e-02j [05] -1.11733584e-05-1.65533248e-01j [07] -1.13342250e-06-1.19760519e-01j [09] 2.67576592e-06-1.20557535e-01j [11] 1.60794811e-01+3.25924787e-05j [13] 2.16728474e-01+3.29607026e-05j [15] 4.69144370e-01+1.65980781e-33j [17] 3.39418054e-01+1.96981622e-05j [19] 3.41676909e-01+3.06464056e-05j	[02] 6.70826110e-06-6.70647190e-02j [04] -2.14666752e-06-1.22391022e-01j [06] 2.00479699e-07-1.21455787e-01j [08] 5.32962376e-06-6.78129890e-02j [10] 1.52763420e-05+8.08780389e-02j [12] 1.90070788e-01+3.18417720e-05j [14] 3.46873266e-01+1.73297048e-05j [16] 3.44222679e-01+2.38029357e-05j [18] 1.92191490e-01+2.80776667e-05j [20] -2.29219672e-01+2.78231492e-05j
19	Re -1.913020983597e-04 Im -2.834139802006e+00	4.72e-12	[01] 7.67039212e-06+5.67349616e-02j [03] 6.46817464e-06+7.64706369e-02j [05] -1.11733578e-05+1.65533248e-01j [07] -1.13342170e-06+1.19760519e-01j [09] 2.67576680e-06+1.20557535e-01j [11] 1.60794811e-01-3.25924810e-05j [13] 2.16728474e-01-3.29607048e-05j [15] 4.69144370e-01-1.39949030e-33j [17] 3.39418054e-01-1.96981633e-05j [19] 3.41676909e-01-3.06464072e-05j	[02] 6.70826210e-06+6.70647190e-02j [04] -2.14666679e-06+1.22391022e-01j [06] 2.00480518e-07+1.21455787e-01j [08] 5.32962471e-06+6.78129890e-02j [10] 1.52763427e-05-8.08780389e-02j [12] 1.90070788e-01-3.18417743e-05j [14] 3.46873266e-01-1.73297058e-05j [16] 3.44222679e-01-2.38029370e-05j [18] 1.92191490e-01-2.80776687e-05j [20] -2.29219672e-01-2.78231484e-05j
20	Re -4.876840426115e-11 Im -8.163951297997e-11	6.46e-14	[01] 3.16227766e-01-2.30859441e-15j [03] 3.16227766e-01-1.06958650e-15j [05] 3.16227766e-01-1.27854201e-35j [07] 3.16227766e-01-6.35490282e-16j [09] 3.16227766e-01-1.01407535e-15j [11] -1.54216981e-11-2.58161854e-11j [13] -1.54219165e-11-2.58167839e-11j [15] -1.54223916e-11-2.58165040e-11j [17] -1.54215071e-11-2.58169159e-11j [19] -1.54219211e-11-2.58165011e-11j	[02] 3.16227766e-01-1.57984441e-15j [04] 3.16227766e-01-1.38200287e-15j [06] 3.16227766e-01-1.30939369e-15j [08] 3.16227766e-01-1.19770683e-15j [10] 3.16227766e-01-3.94106202e-15j [12] -1.54218040e-11-2.58169044e-11j [14] -1.54220997e-11-2.58166640e-11j [16] -1.54221104e-11-2.58169578e-11j [18] -1.54217438e-11-2.58168301e-11j [20] -1.54220432e-11-2.58165617e-11j

Table S7 39-bus system (Ours) Heavy load

Returned 20/20 eigenpairs; Eigenvalue MAE 7.892416e-12; RMSE 2.871642e-11; mean residual 7.425807e-13; mean MAC 1.0000000000.

Table S7. Computed eigenpairs by the proposed method for the 39-bus system under the Heavy load scenario (Supplementary to Table II and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.302836356904e-04 Im -6.539542147604e+00	5.75e-13	[01] -2.60426805e-06-6.01707209e-03j [02] 7.83686987e-08-3.32710917e-03j [03] 5.31668232e-07-5.58115096e-03j [04] -6.95658054e-06+1.37738741e-01j [05] 1.35642651e-06-4.76572492e-02j [06] -6.21877861e-07-2.93702552e-02j [07] 5.17107272e-06-1.68623118e-02j [08] 4.36105605e-06-1.94809932e-02j [09] -7.11171090e-08-9.65005290e-04j [10] 2.87892296e-08-3.69532740e-05j [11] -3.93488957e-02+1.90180611e-05j [12] -2.17577707e-02+5.86394288e-07j [13] -3.64981721e-02-1.63350401e-06j [14] 9.00748305e-01-4.91799216e-33j [15] -3.11656590e-01+6.87000099e-06j [16] -1.92068021e-01+1.37673110e-05j [17] -1.10271800e-01-2.82471024e-05j [18] -1.27396777e-01-2.20850566e-05j [19] -6.31069274e-03+7.83798777e-07j [20] -2.41657502e-04-1.76063317e-07j
2	Re -3.302836355270e-04 Im 6.539542147604e+00	5.22e-13	[01] -2.60426805e-06+6.01707209e-03j [02] 7.83686994e-08+3.32710917e-03j [03] 5.31668233e-07+5.58115096e-03j [04] -6.95658055e-06-1.37738741e-01j [05] 1.35642652e-06+4.76572492e-02j [06] -6.21877864e-07+2.93702552e-02j [07] 5.17107273e-06+1.68623118e-02j [08] 4.36105605e-06+1.94809932e-02j [09] -7.11171095e-08+9.65005290e-04j [10] 2.87892306e-08+3.69532740e-05j [11] -3.93488957e-02-1.90180611e-05j [12] -2.17577707e-02-5.86394290e-07j [13] -3.64981721e-02+1.63350401e-06j [14] 9.00748305e-01-1.36463278e-33j [15] -3.11656590e-01-6.87000100e-06j [16] -1.92068021e-01-1.37673110e-05j [17] -1.10271800e-01+2.82471025e-05j [18] -1.27396777e-01+2.20850566e-05j [19] -6.31069274e-03-7.83798778e-07j [20] -2.41657502e-04+1.76063317e-07j
3	Re -3.228148124379e-04 Im -6.866199362855e+00	1.81e-14	[01] 1.12329954e-07-6.37116913e-04j [02] -8.29081107e-08-2.52797462e-04j [03] -1.02553160e-07-3.67371068e-04j [04] -3.40475901e-06-8.71397788e-03j [05] 7.48530719e-07+1.69684114e-03j [06] 8.97367531e-06-8.31341176e-02j [07] -5.51898674e-06+1.17387622e-01j [08] -5.66501703e-07-3.52535581e-04j [09] -3.32334181e-08-2.09063251e-04j [10] -4.21917588e-09-1.60991393e-07j [11] -4.37457178e-03-5.65609083e-07j [12] -1.73575774e-03+6.50870381e-07j [13] -2.52244296e-03+8.22743264e-07j [14] -5.98319083e-02+2.61907553e-05j [15] 1.16508493e-02-5.68732660e-06j [16] -5.70815428e-01-3.47781191e-05j [17] 8.06006819e-01-2.15057171e-34j [18] -2.42057940e-03+4.00351734e-06j [19] -1.43546995e-03+2.95675990e-07j [20] -1.10539764e-06+2.90216735e-08j
4	Re -3.228148124235e-04 Im 6.866199362855e+00	1.12e-14	[01] 1.12329951e-07+6.37116913e-04j [02] -8.29081107e-08+2.52797462e-04j [03] -1.02553160e-07+3.67371068e-04j [04] -3.40475901e-06+8.71397788e-03j [05] 7.48530719e-07-1.69684114e-03j [06] 8.97367531e-06+8.31341176e-02j [07] -5.51898674e-06-1.17387622e-01j [08] -5.66501699e-07+3.52535581e-04j [09] -3.32334184e-08+2.09063251e-04j [10] -4.21917598e-09+1.60991393e-07j [11] -4.37457178e-03+5.65609063e-07j [12] -1.73575774e-03-6.50870381e-07j [13] -2.52244296e-03-8.22743264e-07j [14] -5.98319083e-02-2.61907553e-05j [15] 1.16508493e-02+5.68732660e-06j [16] -5.70815428e-01+3.47781191e-05j [17] 8.06006819e-01-9.04743930e-34j [18] -2.42057940e-03-4.00351731e-06j [19] -1.43546995e-03-2.95675991e-07j [20] -1.10539764e-06-2.90216734e-08j
5	Re -3.170621367290e-04 Im -1.082442768168e-12	1.92e-14	[01] 3.16227749e-01-6.54115462e-18j [02] 3.16227750e-01+1.80112534e-16j [03] 3.16227750e-01-8.97022212e-17j [04] 3.16227752e-01+1.35615711e-16j [05] 3.16227753e-01-1.26974688e-33j [06] 3.16227751e-01+1.91042028e-16j [07] 3.16227752e-01-4.91126607e-17j [08] 3.16227750e-01+1.73094319e-16j [09] 3.16227751e-01-4.69594508e-17j [10] 3.16227743e-01+9.98097737e-17j [11] -1.00263846e-04-3.42327973e-13j [12] -1.00263846e-04-3.42347402e-13j [13] -1.00263846e-04-3.42234328e-13j [14] -1.00263847e-04-3.42278767e-13j [15] -1.00263847e-04-3.42357003e-13j [16] -1.00263847e-04-3.42300139e-13j [17] -1.00263847e-04-3.42379563e-13j [18] -1.00263846e-04-3.42244927e-13j [19] -1.00263846e-04-3.42256832e-13j [20] -1.00263844e-04-3.42257890e-13j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
6	Re -3.137199659936e-04 Im -6.801734936276e+00	2.80e-16	[01] 1.63421280e-05-8.35727295e-02j [03] -7.24690656e-07+1.83945468e-04j [05] 1.05209532e-06-5.96342963e-04j [07] 2.72656844e-08+1.71915802e-03j [09] -6.87383255e-07-4.09357225e-03j [11] -5.68439559e-01-8.49363889e-05j [13] 1.25114854e-03+4.87144639e-06j [15] -4.05616710e-03-6.96898879e-06j [17] 1.16932572e-02-7.24788155e-07j [19] -2.78433932e-02+5.95963405e-06j	[02] -7.14764377e-07+3.20456866e-04j [04] -3.78758312e-06-1.05901785e-03j [06] -2.50838661e-07-2.25906192e-03j [08] -5.48592587e-06+1.18939875e-01j [10] -3.22434563e-07+6.37788871e-04j [12] 2.17966288e-03+4.76110412e-06j [14] -7.20315750e-03+2.60943715e-05j [16] -1.53655403e-02+2.41485091e-06j [18] 8.08997508e-01-1.44777022e-65j [20] 4.33807095e-03+1.99302733e-06j
7	Re -3.137199659936e-04 Im 6.801734936276e+00	2.80e-16	[01] 1.63421280e-05+8.35727295e-02j [03] -7.24690656e-07-1.83945468e-04j [05] 1.05209532e-06+5.96342963e-04j [07] 2.72656844e-08-1.71915802e-03j [09] -6.87383255e-07+4.09357225e-03j [11] -5.68439559e-01+8.49363889e-05j [13] 1.25114854e-03-4.87144639e-06j [15] -4.05616710e-03+6.96898879e-06j [17] 1.16932572e-02+7.24788155e-07j [19] -2.78433932e-02-5.95963405e-06j	[02] -7.14764377e-07-3.20456866e-04j [04] -3.78758312e-06+1.05901785e-03j [06] -2.50838661e-07+2.25906192e-03j [08] -5.48592587e-06-1.18939875e-01j [10] -3.22434563e-07-6.37788871e-04j [12] 2.17966288e-03-4.76110412e-06j [14] -7.20315750e-03-2.60943715e-05j [16] -1.53655403e-02-2.41485091e-06j [18] 8.08997508e-01-7.12369922e-83j [20] 4.33807095e-03-1.99302733e-06j
8	Re -3.104300766256e-04 Im -5.009898754153e+00	3.40e-12	[01] 9.24367736e-07+8.66151786e-03j [03] 6.34069520e-06-5.93775610e-02j [05] 1.05198046e-06-8.09701003e-02j [07] -1.04023966e-05+1.02536201e-01j [09] 1.93107906e-06-2.20369395e-02j [11] 4.33933273e-02-7.31978511e-06j [13] -2.97475571e-01-1.33336627e-05j [15] -4.05652005e-01+1.98652396e-05j [17] 5.13695987e-01+2.02846329e-05j [19] -1.10402836e-01-2.83358183e-06j	[02] 1.19950346e-05-7.19447081e-02j [04] -3.06393716e-06+8.86219440e-03j [06] -6.73896405e-06+1.08757271e-01j [08] 3.64057854e-07+8.69786939e-03j [10] -4.61336962e-07+9.84262343e-04j [12] -3.60435707e-01-3.77601071e-05j [14] 4.43986976e-02+1.25989237e-05j [16] 5.44862921e-01+2.72900023e-34j [18] 4.35754449e-02-4.52397280e-06j [20] 4.93105483e-03+2.00570780e-06j
9	Re -3.104300748173e-04 Im 5.009898754153e+00	2.20e-14	[01] 9.24367663e-07-8.66151786e-03j [03] 6.34069560e-06+5.93775610e-02j [05] 1.05197946e-06+8.09701003e-02j [07] -1.04023968e-05-1.02536201e-01j [09] 1.93107892e-06+2.20369395e-02j [11] 4.33933273e-02+7.31978404e-06j [13] -2.97475571e-01+1.33336623e-05j [15] -4.05652005e-01-1.98652437e-05j [17] 5.13695987e-01-2.02846344e-05j [19] -1.10402836e-01+2.83358112e-06j	[02] 1.19950374e-05+7.19447081e-02j [04] -3.06393737e-06-8.86219440e-03j [06] -6.73896411e-06-1.08757271e-01j [08] 3.64058041e-07-8.69786939e-03j [10] -4.61337272e-07-9.84262343e-04j [12] -3.60435707e-01+3.77601217e-05j [14] 4.43986976e-02-1.25989244e-05j [16] 5.44862921e-01-2.19315733e-34j [18] 4.35754449e-02+4.52397417e-06j [20] 4.93105483e-03-2.00570839e-06j
10	Re -3.018913360910e-04 Im 4.524290517924e+00	2.30e-13	[01] 3.45392426e-08-2.01113682e-02j [03] -3.88634167e-06-1.13218183e-01j [05] 2.63163970e-06+1.16533729e-01j [07] -8.49546846e-06-2.36136493e-02j [09] 1.22242719e-05+6.25791101e-02j [11] 9.09896726e-02+6.22771333e-06j [13] 5.12231954e-01+1.65966498e-05j [15] -5.27232447e-01-2.32742208e-05j [17] 1.06835012e-01-3.13072112e-05j [19] -2.83126078e-01+3.64140662e-05j	[02] -7.88851527e-06-1.18221130e-01j [04] -3.36223833e-06+2.06039829e-02j [06] -8.11774748e-06-2.46539761e-02j [08] -4.98923213e-08-1.50824704e-02j [10] 1.75777018e-06+8.09941710e-03j [12] 5.34866739e-01+1.87759023e-33j [14] -9.32184036e-02-2.14319069e-05j [16] 1.11541753e-01-2.92842261e-05j [18] 6.82374780e-02+4.32753970e-06j [20] -3.66441165e-02+5.50751920e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
11	Re -3.018913336901e-04 Im -4.524290517924e+00	9.80e-13	[01] 3.45393102e-08+2.01113682e-02j [02] -7.88851530e-06+1.18221130e-01j [03] -3.88634167e-06+1.13218183e-01j [04] -3.36223838e-06-2.06039829e-02j [05] 2.63163974e-06-1.16533729e-01j [06] -8.11774765e-06+2.46539761e-02j [07] -8.49546864e-06+2.36136493e-02j [08] -4.98922504e-08+1.50824704e-02j [09] 1.22242719e-05-6.25791101e-02j [10] 1.75777031e-06-8.09941710e-03j [11] 9.09896726e-02-6.22771372e-06j [12] 5.34866739e-01+5.09534681e-34j [13] 5.12231954e-01-1.65966501e-05j [14] -9.32184036e-02+2.14319072e-05j [15] -5.27232447e-01+2.32742207e-05j [16] 1.11541753e-01+2.92842269e-05j [17] 1.06835012e-01+3.13072121e-05j [18] 6.82374780e-02-4.32754012e-06j [19] -2.83126078e-01-3.64140663e-05j [20] -3.66441165e-02-5.50751932e-06j
12	Re -3.017063801294e-04 Im -4.095220235418e+00	4.09e-13	[01] -4.10001530e-06+2.03865549e-02j [02] -6.26953565e-06-5.13424677e-03j [03] -6.38606106e-06-6.92527266e-03j [04] 3.45843216e-06-5.23626272e-02j [05] 1.88061559e-05-1.37934169e-01j [06] -2.89050453e-06-2.94751438e-02j [07] -2.13982614e-06-2.87186649e-02j [08] -6.57233587e-06+3.41729471e-02j [09] -1.30022134e-05+1.76485919e-01j [10] 1.68613176e-06-2.40160388e-03j [11] 8.34874335e-02+1.06397120e-05j [12] -2.10258694e-02+2.72241642e-05j [13] -2.83605148e-02+2.82417254e-05j [14] -2.14436492e-01+1.63509728e-06j [15] -5.64870805e-01-3.53997316e-05j [16] -1.20707204e-01+2.07300915e-05j [17] -1.17609257e-01+1.74276637e-05j [18] 1.39945746e-01+1.66049667e-05j [19] 7.22748712e-01+3.76599472e-34j [20] -9.83509734e-03-6.18050173e-06j
13	Re -3.017063739199e-04 Im 4.095220235418e+00	2.74e-12	[01] -4.10001536e-06-2.03865549e-02j [02] -6.26953572e-06+5.13424677e-03j [03] -6.38606115e-06+6.92527266e-03j [04] 3.45843215e-06+5.23626272e-02j [05] 1.88061561e-05+1.37934169e-01j [06] -2.89050466e-06+2.94751438e-02j [07] -2.13982626e-06+2.87186649e-02j [08] -6.57233601e-06-3.41729471e-02j [09] -1.30022138e-05-1.76485919e-01j [10] 1.68613193e-06+2.40160388e-03j [11] 8.34874335e-02-1.06397121e-05j [12] -2.10258694e-02-2.72241645e-05j [13] -2.83605148e-02-2.82417258e-05j [14] -2.14436492e-01-1.63509752e-06j [15] -5.64870805e-01+3.53997319e-05j [16] -1.20707204e-01-2.07300922e-05j [17] -1.17609257e-01-1.74276642e-05j [18] 1.39945746e-01-1.66049673e-05j [19] 7.22748712e-01+3.08993981e-33j [20] -9.83509734e-03+6.18050200e-06j
14	Re -2.931845829598e-04 Im 5.682189721349e+00	3.86e-14	[01] 8.58502135e-06-1.05202514e-01j [02] -1.57169138e-05+9.98998444e-03j [03] 1.34357108e-05+3.25633715e-02j [04] -8.97378217e-08-4.69797181e-03j [05] 1.08219256e-07+1.69174132e-03j [06] -2.45151393e-06+2.70014975e-02j [07] -6.36547659e-07+2.43065592e-02j [08] -6.37454870e-06-1.23544679e-01j [09] 2.75346193e-07+3.45050957e-02j [10] 3.02712702e-07+3.87954825e-03j [11] 5.97780642e-01+7.96254753e-05j [12] -5.67649823e-02-9.22353955e-05j [13] -1.85031259e-01+6.67971794e-05j [14] 2.66947672e-02+8.67465579e-07j [15] -9.61279515e-03+1.18929872e-07j [16] -1.53427631e-01-2.18463901e-05j [17] -1.38114481e-01-1.07432930e-05j [18] 7.02004305e-01-5.96118735e-35j [19] -1.96064500e-01-8.55179277e-06j [20] -2.20443293e-02+5.82647265e-07j
15	Re -2.931845829589e-04 Im -5.682189721349e+00	1.86e-14	[01] 8.58502135e-06+1.05202514e-01j [02] -1.57169137e-05-9.98998444e-03j [03] 1.34357107e-05-3.25633715e-02j [04] -8.97378220e-08+4.69797181e-03j [05] 1.08219257e-07-1.69174132e-03j [06] -2.45151393e-06-2.70014975e-02j [07] -6.36547651e-07-2.43065592e-02j [08] -6.37454870e-06+1.23544679e-01j [09] 2.75346194e-07-3.45050957e-02j [10] 3.02712701e-07-3.87954825e-03j [11] 5.97780642e-01-7.96254753e-05j [12] -5.67649823e-02+9.22353947e-05j [13] -1.85031259e-01-6.67971787e-05j [14] 2.66947672e-02-8.67465577e-07j [15] -9.61279515e-03-1.18929876e-07j [16] -1.53427631e-01+2.18463900e-05j [17] -1.38114481e-01+1.07432930e-05j [18] 7.02004305e-01+1.44358976e-34j [19] -1.96064500e-01+8.55179277e-06j [20] -2.20443293e-02-5.82647262e-07j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
16	Re -2.888538933847e-04 Im -5.640131056641e+00	2.25e-12	[01] 1.56432975e-05-7.94893842e-03j [02] -6.74979240e-06+1.31795748e-01j [03] 5.21606726e-06-1.13205611e-01j [04] 8.18887662e-07-5.74599647e-04j [05] -3.19229122e-07+8.10066785e-04j [06] -5.49599223e-06+8.36774545e-03j [07] -5.42169625e-06+7.49396666e-03j [08] 1.93329583e-05-9.52883721e-03j [09] -5.68163804e-06+3.27555944e-03j [10] -6.68044674e-07-4.28382501e-05j [11] -4.48330590e-02-8.59341665e-05j [12] 7.43345295e-01+2.75578356e-36j [13] -6.38494482e-01+3.28057712e-06j [14] -3.24081755e-03-4.45265837e-06j [15] 4.56888293e-03+1.56650312e-06j [16] 4.71951826e-02+2.85810607e-05j [17] 4.22669556e-02+2.84144161e-05j [18] -5.37438963e-02-1.06287977e-04j [19] 1.84745862e-02+3.10990251e-05j [20] -2.41613152e-04+3.78023350e-06j
17	Re -2.888538933847e-04 Im 5.640131056641e+00	2.25e-12	[01] 1.56432975e-05+7.94893842e-03j [02] -6.74979240e-06-1.31795748e-01j [03] 5.21606726e-06+1.13205611e-01j [04] 8.18887662e-07+5.74599647e-04j [05] -3.19229122e-07-8.10066785e-04j [06] -5.49599223e-06-8.36774545e-03j [07] -5.42169625e-06-7.49396666e-03j [08] 1.93329583e-05+9.52883721e-03j [09] -5.68163804e-06-3.27555944e-03j [10] -6.68044674e-07+4.28382501e-05j [11] -4.48330590e-02+8.59341665e-05j [12] 7.43345295e-01-8.39682503e-53j [13] -6.38494482e-01-3.28057712e-06j [14] -3.24081755e-03+4.45265837e-06j [15] 4.56888293e-03-1.56650312e-06j [16] 4.71951826e-02-2.85810607e-05j [17] 4.22669556e-02-2.84144161e-05j [18] -5.37438963e-02+1.06287977e-04j [19] 1.84745862e-02-3.10990251e-05j [20] -2.41613152e-04-3.78023350e-06j
18	Re -1.770322092396e-04 Im -2.663607833865e+00	1.06e-12	[01] 1.01429294e-05+5.12306823e-02j [02] 9.15525465e-06+6.37108729e-02j [03] 8.51696365e-06+7.48050205e-02j [04] -2.09649612e-06+1.30218247e-01j [05] -1.15194133e-05+1.73319870e-01j [06] 4.74311000e-07+1.28082289e-01j [07] -8.94325266e-07+1.27253677e-01j [08] 7.63891506e-06+6.51883914e-02j [09] 3.19496697e-06+1.28545749e-01j [10] 2.05027336e-05-1.04344949e-01j [11] 1.36458445e-01-3.60862671e-05j [12] 1.69700779e-01-3.56648845e-05j [13] 1.99251237e-01-3.59287491e-05j [14] 3.46850342e-01-1.74685803e-05j [15] 4.61656167e-01+1.94935391e-34j [16] 3.41160988e-01-2.39380690e-05j [17] 3.38953891e-01-2.01458677e-05j [18] 1.73636309e-01-3.18875189e-05j [19] 3.42395464e-01-3.12668769e-05j [20] -2.77934026e-01-3.61388260e-05j
19	Re -1.770322061395e-04 Im 2.663607833865e+00	2.70e-13	[01] 1.01429297e-05-5.12306823e-02j [02] 9.15525495e-06-6.37108729e-02j [03] 8.51696394e-06-7.48050205e-02j [04] -2.09649590e-06-1.30218247e-01j [05] -1.15194131e-05-1.73319870e-01j [06] 4.74311233e-07-1.28082289e-01j [07] -8.94325036e-07-1.27253677e-01j [08] 7.63891534e-06-6.51883914e-02j [09] 3.19496723e-06-1.28545749e-01j [10] 2.05027338e-05+1.04344949e-01j [11] 1.36458445e-01+3.60862677e-05j [12] 1.69700779e-01+3.56648852e-05j [13] 1.99251237e-01+3.59287497e-05j [14] 3.46850342e-01+1.74685806e-05j [15] 4.61656167e-01-4.91488190e-36j [16] 3.41160988e-01+2.39380693e-05j [17] 3.38953891e-01+2.01458680e-05j [18] 1.73636309e-01+3.18875195e-05j [19] 3.42395464e-01+3.12668773e-05j [20] -2.77934026e-01+3.61388258e-05j
20	Re -1.121027352431e-10 Im 5.000831504217e-11	4.02e-14	[01] 3.16227766e-01+1.83655833e-15j [02] 3.16227766e-01+1.39714887e-15j [03] 3.16227766e-01+1.53007441e-15j [04] 3.16227766e-01+6.45403159e-16j [05] 3.16227766e-01-8.23367479e-35j [06] 3.16227766e-01+6.70985615e-16j [07] 3.16227766e-01+9.14933461e-16j [08] 3.16227766e-01+1.28462047e-15j [09] 3.16227766e-01+1.22944283e-15j [10] 3.16227766e-01+3.57379903e-15j [11] -3.54502204e-11+1.58139604e-11j [12] -3.54499042e-11+1.58140318e-11j [13] -3.54499487e-11+1.58137321e-11j [14] -3.54500142e-11+1.58141190e-11j [15] -3.54501015e-11+1.58140581e-11j [16] -3.54499128e-11+1.58138694e-11j [17] -3.54496864e-11+1.58140483e-11j [18] -3.54500553e-11+1.58142604e-11j [19] -3.54502806e-11+1.58140730e-11j [20] -3.54498511e-11+1.58140249e-11j

Table S8 39-bus system (Ours) Reactive stress

Returned 20/20 eigenpairs; Eigenvalue MAE 5.696524e-12; RMSE 1.620968e-11; mean residual 2.292462e-12; mean MAC 1.0000000000.

Table S8. Computed eigenpairs by the proposed method for the 39-bus system under the Reactive stress scenario (Supplementary to Table II and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.298832966135e-04 Im 6.564343772335e+00	7.78e-14	[01] -2.53529146e-06+6.73039721e-03j [02] 7.10187758e-08+3.42466613e-03j [03] 5.27109040e-07+5.67937363e-03j [04] -6.89854764e-06-1.37274117e-01j [05] 1.29602310e-06+4.71400640e-02j [06] -6.22785108e-07+3.04442712e-02j [07] 5.30696716e-06+1.61983546e-02j [08] 4.54201607e-06+1.82814524e-02j [09] -7.37999409e-08+1.20964858e-03j [10] 2.29021421e-08+2.51653041e-05j [11] -4.41806402e-02-1.88627703e-05j [12] -2.24806858e-02-6.63548494e-07j [13] -3.72813611e-02+1.58659445e-06j [14] 9.01114499e-01-3.35809546e-34j [15] -3.09443586e-01-7.04317851e-06j [16] -1.99846662e-01-1.41312321e-05j [17] -1.06331570e-01+2.94931903e-05j [18] -1.20005740e-01+2.37846091e-05j [19] -7.94054907e-03-8.83491041e-07j [20] -1.65193715e-04+1.42035920e-07j
2	Re -3.298832944678e-04 Im -6.564343772336e+00	1.09e-12	[01] -2.53529150e-06-6.73039721e-03j [02] 7.10187812e-08-3.42466613e-03j [03] 5.27109047e-07-5.67937363e-03j [04] -6.89854769e-06+1.37274117e-01j [05] 1.29602312e-06-4.71400640e-02j [06] -6.22785116e-07-3.04442712e-02j [07] 5.30696721e-06-1.61983546e-02j [08] 4.54201613e-06-1.82814524e-02j [09] -7.37999374e-08-1.20964858e-03j [10] 2.29021453e-08-2.51653041e-05j [11] -4.41806402e-02+1.88627707e-05j [12] -2.24806858e-02+6.63548508e-07j [13] -3.72813611e-02-1.58659445e-06j [14] 9.01114499e-01-6.09653253e-34j [15] -3.09443586e-01+7.04317851e-06j [16] -1.99846662e-01+1.41312322e-05j [17] -1.06331570e-01-2.94931905e-05j [18] -1.20005740e-01-2.37846096e-05j [19] -7.94054907e-03+8.83491053e-07j [20] -1.65193715e-04-1.42035923e-07j
3	Re -3.233130477491e-04 Im -6.884868287455e+00	1.29e-14	[01] 2.32744891e-07-7.50366459e-04j [02] -8.33213819e-08-2.64437184e-04j [03] -1.00623384e-07-3.86234807e-04j [04] -3.44217331e-06-9.59435945e-03j [05] 7.53746219e-07+1.86752925e-03j [06] 8.99316769e-06-8.21944211e-02j [07] -5.51818030e-06+1.17508232e-01j [08] -7.19330136e-07-2.78731540e-04j [09] -3.05956967e-08-2.18573937e-04j [10] -4.91406822e-09-4.80689559e-07j [11] -5.16617431e-03-1.35981465e-06j [12] -1.82061515e-03+6.59152731e-07j [13] -2.65917574e-03+8.17653499e-07j [14] -6.60559000e-02+2.68008915e-05j [15] 1.28576927e-02-5.79324001e-06j [16] -5.65897766e-01-3.53422462e-05j [17] 8.09028700e-01+3.63659494e-35j [18] -1.91902971e-03+5.04261078e-06j [19] -1.50485276e-03+2.81315147e-07j [20] -3.30948271e-06+3.39881259e-08j
4	Re -3.233130477447e-04 Im 6.884868287455e+00	1.29e-14	[01] 2.32744924e-07+7.50366459e-04j [02] -8.33213821e-08+2.64437184e-04j [03] -1.00623384e-07+3.86234807e-04j [04] -3.44217331e-06+9.59435945e-03j [05] 7.53746219e-07-1.86752925e-03j [06] 8.99316769e-06+8.21944211e-02j [07] -5.51818030e-06-1.17508232e-01j [08] -7.19330183e-07+2.78731540e-04j [09] -3.05956954e-08+2.18573937e-04j [10] -4.91406849e-09+4.80689559e-07j [11] -5.16617431e-03+1.35981487e-06j [12] -1.82061515e-03-6.59152733e-07j [13] -2.65917574e-03-8.17653501e-07j [14] -6.60559000e-02-2.68008915e-05j [15] 1.28576927e-02+5.79324001e-06j [16] -5.65897766e-01+3.53422462e-05j [17] 8.09028700e-01+1.19881132e-33j [18] -1.91902971e-03-5.04261110e-06j [19] -1.50485276e-03-2.81315137e-07j [20] -3.30948271e-06-3.39881274e-08j
5	Re -3.158410793776e-04 Im -6.823011459430e+00	9.29e-15	[01] 1.59968936e-05-8.16369625e-02j [02] -7.43710456e-07+4.50352452e-04j [03] -7.62499753e-07+3.19994154e-04j [04] -3.92668455e-06-7.03814522e-04j [05] 1.03652864e-06-5.24532384e-04j [06] -4.32844354e-07-1.81559931e-03j [07] 1.83336891e-07+1.42906325e-03j [08] -5.54367468e-06+1.19758190e-01j [09] -7.67522528e-07-3.94814797e-03j [10] -3.05258710e-07+5.76728337e-04j [11] -5.57009936e-01-8.33626818e-05j [12] 3.07276018e-03+4.93210516e-06j [13] 2.18332402e-03+5.10147725e-06j [14] -4.80213331e-03+2.70141072e-05j [15] -3.57889079e-03-6.90657793e-06j [16] -1.23878548e-02+3.52674283e-06j [17] 9.75051486e-03-1.70226659e-06j [18] 8.17111502e-01-2.11932021e-36j [19] -2.69382586e-02+6.48380232e-06j [20] 3.93502415e-03+1.90062918e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
6	Re -3.158410793776e-04 Im 6.823011459430e+00	9.29e-15	[01] 1.59968936e-05+8.16369625e-02j [03] -7.62499753e-07-3.19994154e-04j [05] 1.03652864e-06+5.24532384e-04j [07] 1.83336891e-07-1.42906325e-03j [09] -7.67522528e-07+3.94814797e-03j [11] -5.57009936e-01+8.33626818e-05j [13] 2.18332402e-03-5.10147725e-06j [15] -3.57889079e-03+6.90657793e-06j [17] 9.75051486e-03+1.70226659e-06j [19] -2.69382586e-02-6.48380232e-06j	[02] -7.43710456e-07-4.50352452e-04j [04] -3.92668455e-06+7.03814522e-04j [06] -4.32844354e-07+1.81559931e-03j [08] -5.54367468e-06-1.19758190e-01j [10] -3.05258710e-07-5.76728337e-04j [12] 3.07276018e-03-4.93210516e-06j [14] -4.80213331e-03-2.70141072e-05j [16] -1.23878548e-02-3.52674283e-06j [18] 8.17111502e-01-2.65555895e-53j [20] 3.93502415e-03-1.90062918e-06j
7	Re -3.099940896929e-04 Im -5.068979941776e+00	3.48e-11	[01] 9.90215354e-07+8.48342986e-03j [03] 6.07241770e-06-5.86864499e-02j [05] 8.25745135e-07-8.01162986e-02j [07] -1.02278320e-05+1.00518380e-01j [09] 2.10921615e-06-2.38526131e-02j [11] 4.30023355e-02-7.64920327e-06j [13] -2.97480439e-01-1.25885320e-05j [15] -4.06107911e-01+2.06499081e-05j [17] 5.09525657e-01+2.06845697e-05j [19] -1.20908418e-01-3.29739912e-06j	[02] 1.19103620e-05-7.15379075e-02j [04] -3.12679126e-06+9.58484989e-03j [06] -6.57867147e-06+1.07573524e-01j [08] 4.31083697e-07+8.40923606e-03j [10] -4.41875073e-07+9.57988545e-04j [12] -3.62624222e-01-3.81970592e-05j [14] 4.85854128e-02+1.28783935e-05j [16] 5.45288037e-01+6.90763103e-34j [18] 4.26262488e-02-4.79197046e-06j [20] 4.85602485e-03+1.94289666e-06j
8	Re -3.099940832345e-04 Im 5.068979941779e+00	1.95e-12	[01] 9.90212364e-07-8.48342986e-03j [03] 6.07243114e-06+5.86864499e-02j [05] 8.25733612e-07+8.01162986e-02j [07] -1.02278335e-05-1.00518380e-01j [09] 2.10921210e-06+2.38526131e-02j [11] 4.30023355e-02+7.64917975e-06j [13] -2.97480439e-01+1.25885796e-05j [15] -4.06107911e-01-2.06499514e-05j [17] 5.09525657e-01-2.06845799e-05j [19] -1.20908418e-01+3.29738490e-06j	[02] 1.19103929e-05+7.15379075e-02j [04] -3.12679605e-06-9.58484989e-03j [06] -6.57867187e-06-1.07573524e-01j [08] 4.31081369e-07-8.40923605e-03j [10] -4.41878611e-07-9.57988545e-04j [12] -3.62624222e-01+3.81972148e-05j [14] 4.85854128e-02-1.28784198e-05j [16] 5.45288037e-01+1.10921566e-35j [18] 4.26262488e-02+4.79195616e-06j [20] 4.85602485e-03-1.94290305e-06j
9	Re -3.034620989976e-04 Im -4.208214869649e+00	3.84e-12	[01] -4.27450318e-06+2.21024270e-02j [03] -7.03602298e-06-2.75832517e-03j [05] 2.00479372e-05-1.38812664e-01j [07] -2.58101815e-06-2.51489553e-02j [09] -1.22128927e-05+1.69360457e-01j [11] 9.30117632e-02+1.12807794e-05j [13] -1.16076229e-02+3.04461436e-05j [15] -5.84153523e-01-4.22416466e-05j [17] -1.05832207e-01+1.84932334e-05j [19] 7.12705197e-01-1.10352653e-33j	[02] -7.03454209e-06-1.01843678e-03j [04] 3.43203117e-06-4.97548337e-02j [06] -3.30871562e-06-2.59674437e-02j [08] -6.73898685e-06+3.47990935e-02j [10] 1.70730233e-06-2.76077148e-03j [12] -4.28579867e-03+2.99119216e-05j [14] -2.09379032e-01+6.55981000e-07j [16] -1.09276582e-01+2.18039208e-05j [18] 1.46442065e-01+1.77988991e-05j [20] -1.16179201e-02-6.34690619e-06j
10	Re -3.034620909584e-04 Im 4.208214869649e+00	8.25e-13	[01] -4.27450343e-06-2.21024270e-02j [03] -7.03602350e-06+2.75832517e-03j [05] 2.00479383e-05+1.38812664e-01j [07] -2.58101840e-06+2.51489553e-02j [09] -1.22128932e-05-1.69360457e-01j [11] 9.30117632e-02-1.12807802e-05j [13] -1.16076229e-02-3.04461460e-05j [15] -5.84153523e-01+4.22416511e-05j [17] -1.05832207e-01-1.84932347e-05j [19] 7.12705197e-01+5.75357757e-35j	[02] -7.03454258e-06+1.01843678e-03j [04] 3.43203134e-06+4.97548337e-02j [06] -3.30871590e-06+2.59674437e-02j [08] -6.73898724e-06-3.47990935e-02j [10] 1.70730254e-06+2.76077148e-03j [12] -4.28579867e-03-2.99119238e-05j [14] -2.09379032e-01-6.55980490e-07j [16] -1.09276582e-01-2.18039224e-05j [18] 1.46442065e-01-1.77989009e-05j [20] -1.16179201e-02+6.34690665e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
11	Re -3.008362141204e-04 Im -4.597391141537e+00	8.38e-13	[01] 3.79419140e-07+1.82590449e-02j [03] -3.47546014e-06+1.10771328e-01j [05] 1.24384811e-06-1.12672761e-01j [07] -8.26474921e-06+2.30370774e-02j [09] 1.34746975e-05-6.89716613e-02j [11] 8.39439712e-02-7.23732008e-06j [13] 5.09259125e-01-1.73459770e-05j [15] -5.18000752e-01+2.81775901e-05j [17] 1.05910458e-01+3.10658977e-05j [19] -3.17089709e-01-4.11992818e-05j	[02] -7.58262515e-06+1.15877984e-01j [04] -3.66417932e-06-1.89959643e-02j [06] -7.92579939e-06+2.41857633e-02j [08] 4.79799842e-07+1.26984695e-02j [10] 1.51651626e-06-7.34061398e-03j [12] 5.32736420e-01-1.89982303e-34j [14] -8.73318770e-02+2.25603395e-05j [16] 1.11191416e-01+2.91620465e-05j [18] 5.83798311e-02-6.02598701e-06j [20] -3.37476741e-02-4.76369620e-06j
12	Re -3.008362133861e-04 Im 4.597391141537e+00	5.30e-13	[01] 3.79419153e-07-1.82590449e-02j [03] -3.47546014e-06-1.10771328e-01j [05] 1.24384810e-06+1.12672761e-01j [07] -8.26474924e-06-2.30370774e-02j [09] 1.34746975e-05+6.89716613e-02j [11] 8.39439712e-02+7.23732015e-06j [13] 5.09259125e-01+1.73459771e-05j [15] -5.18000752e-01-2.81775902e-05j [17] 1.05910458e-01-3.10658978e-05j [19] -3.17089709e-01+4.11992818e-05j	[02] -7.58262516e-06-1.15877984e-01j [04] -3.66417933e-06+1.89959643e-02j [06] -7.92579941e-06-2.41857633e-02j [08] 4.79799853e-07-1.26984695e-02j [10] 1.51651629e-06+7.34061398e-03j [12] 5.32736420e-01-2.15341705e-38j [14] -8.73318770e-02-2.25603395e-05j [16] 1.11191416e-01-2.91620466e-05j [18] 5.83798311e-02+6.02598706e-06j [20] -3.37476741e-02+4.76369623e-06j
13	Re -2.999330170165e-04 Im 2.900166290965e-11	2.51e-14	[01] 3.16227751e-01-4.75629387e-17j [03] 3.16227752e-01-4.60795415e-17j [05] 3.16227754e-01+8.90548734e-34j [07] 3.16227753e-01+1.02456765e-16j [09] 3.16227753e-01+5.04448830e-17j [11] -9.48471434e-05+9.17155704e-12j [13] -9.48471436e-05+9.17095214e-12j [15] -9.48471445e-05+9.17097464e-12j [17] -9.48471441e-05+9.17096258e-12j [19] -9.48471439e-05+9.17111592e-12j	[02] 3.16227752e-01-8.99380396e-17j [04] 3.16227753e-01-2.20057316e-17j [06] 3.16227753e-01-1.39304679e-16j [08] 3.16227752e-01-5.70994015e-17j [10] 3.16227746e-01+1.11925818e-16j [12] -9.48471436e-05+9.17160540e-12j [14] -9.48471441e-05+9.17108214e-12j [16] -9.48471440e-05+9.17089836e-12j [18] -9.48471437e-05+9.17105792e-12j [20] -9.48471418e-05+9.17110324e-12j
14	Re -2.913079366769e-04 Im 5.712180903185e+00	1.20e-15	[01] 8.76422065e-06-1.05270958e-01j [03] 1.39987020e-05+3.37239205e-02j [05] 1.27363667e-07+1.70896081e-03j [07] -8.05464299e-07+2.52239002e-02j [09] 1.31192054e-07+3.67933536e-02j [11] 6.01326752e-01+8.07290794e-05j [13] -1.92637139e-01+7.01390727e-05j [15] -9.76189333e-03+2.29690460e-07j [17] -1.44083481e-01-1.19488801e-05j [19] -2.10170292e-01-9.96880316e-06j	[02] -1.64832820e-05+9.70337734e-03j [04] -1.35536720e-07-4.12063425e-03j [06] -2.76026617e-06+2.84875635e-02j [08] -6.15939041e-06-1.20777871e-01j [10] 2.83242547e-07+3.75034084e-03j [12] -5.54274419e-02-9.69821595e-05j [14] 2.35378083e-02+4.26163199e-07j [16] -1.62726115e-01-2.40657930e-05j [18] 6.89905048e-01+1.15674921e-65j [20] -2.14226254e-02+5.25428616e-07j
15	Re -2.913079366743e-04 Im -5.712180903185e+00	2.27e-14	[01] 8.76422065e-06+1.05270958e-01j [03] 1.39987020e-05-3.37239205e-02j [05] 1.27363667e-07-1.70896081e-03j [07] -8.05464297e-07-2.52239002e-02j [09] 1.31192054e-07-3.67933536e-02j [11] 6.01326752e-01-8.07290794e-05j [13] -1.92637139e-01-7.01390724e-05j [15] -9.76189333e-03-2.29690462e-07j [17] -1.44083481e-01+1.19488801e-05j [19] -2.10170292e-01+9.96880316e-06j	[02] -1.64832820e-05-9.70337734e-03j [04] -1.35536720e-07+4.12063425e-03j [06] -2.76026616e-06-2.84875635e-02j [08] -6.15939041e-06+1.20777871e-01j [10] 2.83242547e-07-3.75034084e-03j [12] -5.54274419e-02+9.69821592e-05j [14] 2.35378083e-02-4.26163198e-07j [16] -1.62726115e-01+2.40657930e-05j [18] 6.89905048e-01+4.19579873e-34j [20] -2.14226254e-02-5.25428615e-07j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
16	Re -2.888449669853e-04 Im -5.672456898736e+00	1.42e-14	[01] 1.60180471e-05-8.77900062e-03j [02] -6.66856534e-06+1.30960044e-01j [03] 5.02979074e-06-1.12499355e-01j [04] 7.28608461e-07-5.17319779e-04j [05] -3.15788975e-07+7.16957239e-04j [06] -5.89306048e-06+8.83194600e-03j [07] -5.72311782e-06+7.80689308e-03j [08] 1.94574660e-05-1.02155253e-02j [09] -6.21082889e-06+3.71150027e-03j [10] -6.66559780e-07-6.26887826e-06j [11] -4.97985072e-02-8.83259115e-05j [12] 7.42865204e-01-8.78867448e-34j [13] -6.38147746e-01+3.96360143e-06j [14] -2.93447436e-03-3.98357488e-06j [15] 4.06690913e-03+1.58420986e-06j [16] 5.00988347e-02+3.08770684e-05j [17] 4.42842661e-02+3.02091574e-05j [18] -5.79471327e-02-1.07420934e-04j [19] 2.10533271e-02+3.41586110e-05j [20] -3.55597492e-05+3.78284236e-06j
17	Re -2.888449669853e-04 Im 5.672456898736e+00	1.42e-14	[01] 1.60180471e-05+8.77900062e-03j [02] -6.66856534e-06-1.30960044e-01j [03] 5.02979074e-06+1.12499355e-01j [04] 7.28608461e-07+5.17319779e-04j [05] -3.15788975e-07-7.16957239e-04j [06] -5.89306048e-06-8.83194600e-03j [07] -5.72311782e-06-7.80689308e-03j [08] 1.94574660e-05+1.02155253e-02j [09] -6.21082889e-06-3.71150027e-03j [10] -6.66559780e-07+6.26887826e-06j [11] -4.97985072e-02+8.83259115e-05j [12] 7.42865204e-01-1.99272709e-50j [13] -6.38147746e-01-3.96360143e-06j [14] -2.93447436e-03+3.98357488e-06j [15] 4.06690913e-03-1.58420986e-06j [16] 5.00988347e-02-3.08770684e-05j [17] 4.42842661e-02-3.02091574e-05j [18] -5.79471327e-02+1.07420934e-04j [19] 2.10533271e-02-3.41586110e-05j [20] -3.55597492e-05-3.78284236e-06j
18	Re -1.849987582156e-04 Im -2.748388587592e+00	7.03e-13	[01] 8.58158502e-06+5.48461360e-02j [02] 7.59542710e-06+6.57714005e-02j [03] 7.14423635e-06+7.61156918e-02j [04] -2.43883999e-06+1.26790354e-01j [05] -1.13249171e-05+1.68245847e-01j [06] -4.14977440e-08+1.25421170e-01j [07] -1.40276983e-06+1.24299203e-01j [08] 6.15813858e-06+6.71235086e-02j [09] 2.63183869e-06+1.24392991e-01j [10] 1.76904439e-05-9.07706843e-02j [11] 1.50738493e-01-3.37319973e-05j [12] 1.80765365e-01-3.30428127e-05j [13] 2.09195497e-01-3.37164462e-05j [14] 3.48469162e-01-1.67531780e-05j [15] 4.62404968e-01+1.31969234e-33j [16] 3.44706112e-01-2.30887087e-05j [17] 3.41622510e-01-1.91398416e-05j [18] 1.84481484e-01-2.93427236e-05j [19] 3.41880277e-01-3.02458643e-05j [20] -2.49473116e-01-3.18277502e-05j
19	Re -1.849987557103e-04 Im 2.748388587592e+00	1.03e-12	[01] 8.58158526e-06-5.48461360e-02j [02] 7.59542731e-06-6.57714005e-02j [03] 7.14423657e-06-7.61156918e-02j [04] -2.43883981e-06-1.26790354e-01j [05] -1.13249169e-05-1.68245847e-01j [06] -4.14975594e-08-1.25421170e-01j [07] -1.40276964e-06-1.24299203e-01j [08] 6.15813882e-06-6.71235086e-02j [09] 2.63183891e-06-1.24392991e-01j [10] 1.76904441e-05+9.07706843e-02j [11] 1.50738493e-01+3.37319980e-05j [12] 1.80765365e-01+3.30428130e-05j [13] 2.09195497e-01+3.37164466e-05j [14] 3.48469162e-01+1.67531783e-05j [15] 4.62404968e-01-1.32253256e-50j [16] 3.44706112e-01+2.30887090e-05j [17] 3.41622510e-01+1.91398418e-05j [18] 1.84481484e-01+2.93427240e-05j [19] 3.41880277e-01+3.02458648e-05j [20] -2.49473116e-01+3.18277500e-05j
20	Re 2.169513167211e-11 Im 1.128854237797e-11	2.14e-14	[01] 3.16227766e-01-4.09818613e-16j [02] 3.16227766e-01-4.63187852e-16j [03] 3.16227766e-01-4.56762115e-16j [04] 3.16227766e-01-5.93398080e-16j [05] 3.16227766e-01-5.86972343e-16j [06] 3.16227766e-01-7.34317871e-16j [07] 3.16227766e-01-6.84116858e-16j [08] 3.16227766e-01-8.21868422e-16j [09] 3.16227766e-01-3.11558499e-16j [10] 3.16227766e-01-1.64500286e-34j [11] 6.86040307e-12+3.56970200e-12j [12] 6.86061457e-12+3.56958075e-12j [13] 6.86058293e-12+3.56999475e-12j [14] 6.86054775e-12+3.56981828e-12j [15] 6.86036884e-12+3.56977913e-12j [16] 6.86059529e-12+3.56955951e-12j [17] 6.86079738e-12+3.56979648e-12j [18] 6.86077279e-12+3.56978319e-12j [19] 6.86062846e-12+3.56975297e-12j [20] 6.86071915e-12+3.56973832e-12j

Table S9 9-bus system Base RVVQE

Returned 6/6 eigenpairs; Eigenvalue MAE 8.753269e-04; RMSE 1.213156e-03; mean residual 1.335630e-04; mean MAC 1.0000000000.

Table S9. Computed eigenpairs by the RVVQE baseline method for the 9-bus system under the Base scenario (Supplementary to Table I and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -1.665925829920e-05 Im -8.690417759678e+00	1.18e-04	[01] 3.60211659e-02+1.52892659e-07j [02] -9.41480066e-02-8.50433873e-07j [03] -5.39352750e-02-5.47404211e-08j [04] 2.45529524e-06-3.12947709e-01j [05] -9.71613624e-06+8.18114455e-01j [06] 3.93813130e-06+4.68702554e-01j
2	Re -1.480990796038e-05 Im -1.335821287497e+01	2.81e-04	[01] 3.86318881e-04+2.94393282e-03j [02] 2.88981451e-03+2.19468561e-02j [03] -9.29250087e-03-7.06027926e-02j [04] 3.93932638e-02-5.18119963e-03j [05] 2.93324882e-01-3.86171118e-02j [06] -9.43347808e-01+1.24159314e-01j
3	Re -1.119115580276e-05 Im 1.335819226670e+01	2.84e-04	[01] 3.85514392e-04-2.94378432e-03j [02] 2.89017132e-03-2.19468562e-02j [03] -9.28964278e-03+7.06031089e-02j [04] 3.93933221e-02+5.17727610e-03j [05] 2.93326499e-01+3.86151329e-02j [06] -9.43352008e-01-1.24124372e-01j
4	Re -1.122990155101e-06 Im 7.257274584666e-12	2.01e-07	[01] 5.77350270e-01+0.00000000e+00j [02] 5.77350268e-01+0.00000000e+00j [03] 5.77350270e-01+0.00000000e+00j [04] -5.84155528e-07+0.00000000e+00j [05] -7.26498584e-07+0.00000000e+00j [06] -6.34441033e-07+0.00000000e+00j
5	Re -1.057407769005e-06 Im 8.690415894670e+00	1.18e-04	[01] 3.60210779e-02-6.18582634e-07j [02] -9.41480293e-02+9.71038228e-07j [03] -5.39353692e-02+1.53514296e-08j [04] 3.32675753e-06+3.12947729e-01j [05] -1.18716647e-05-8.18113956e-01j [06] 2.35930900e-06-4.68703404e-01j
6	Re 6.771993135033e-07 Im 2.204085035257e-12	5.17e-07	[01] 5.77350271e-01+0.00000000e+00j [02] 5.77350266e-01+0.00000000e+00j [03] 5.77350271e-01+0.00000000e+00j [04] 4.99556336e-07+0.00000000e+00j [05] 2.48280654e-07+0.00000000e+00j [06] 4.25143186e-07+0.00000000e+00j

Table S10 39-bus system Base RVVQE

Returned 18/20 eigenpairs; Eigenvalue MAE 1.301950e-01; RMSE 2.926423e-01; mean residual 2.865506e-04; mean MAC 0.6665040623.

Notably, under the minimum-total-error assignment, two reference modes were not successfully recovered.

Table S10. Computed eigenpairs by the RVVQE baseline method for the 39-bus system under the Base scenario (Supplementary to Table II and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.782865208858e-04 Im 6.811192377570e+00	1.43e-03	[01] 2.73348633e-03-8.19068605e-02j [02] -1.75596081e-05+4.66115688e-04j [03] 6.92349907e-05+3.55319225e-04j [04] 7.77622149e-04-1.20744272e-03j [05] -1.11638805e-04-4.17187687e-04j [06] 6.13465409e-03-5.22169662e-03j [07] -8.90840417e-03+6.57662816e-03j [08] -3.88376059e-03+1.19020887e-01j [09] 1.61752565e-04-3.84545867e-03j [10] -2.08765201e-05+5.90504174e-04j [11] 5.57768412e-01+1.86664198e-02j [12] -3.16516566e-03-6.88731245e-05j [13] -2.39005865e-03+3.75584730e-04j [14] 8.14391007e-03+5.43552437e-03j [15] 2.88071221e-03-7.97694666e-04j [16] 3.60283698e-02+4.25977310e-02j [17] -4.53009551e-02-6.15610883e-02j [18] -8.10533141e-01-2.64608843e-02j [19] 2.62058121e-02+1.04951511e-03j [20] -4.02276655e-03-1.29520268e-04j
2	Re -3.344892411915e-04 Im -6.545718664777e+00	2.42e-05	[01] -7.32316494e-04+6.84998422e-03j [02] -3.75907254e-04+3.40345721e-03j [03] -6.12619786e-04+5.66745760e-03j [04] 1.48620878e-02-1.37027205e-01j [05] -4.96479647e-03+4.57873115e-02j [06] -3.39176259e-03+3.13085753e-02j [07] -1.69950614e-03+1.55800063e-02j [08] -2.04460314e-03+1.88005477e-02j [09] -1.26954248e-04+1.17242801e-03j [10] -3.07167195e-06+2.59034067e-05j [11] 4.48425843e-02+4.79446967e-03j [12] 2.22815623e-02+2.44648844e-03j [13] 3.70921075e-02+4.00950826e-03j [14] -8.96953042e-01-9.72393530e-02j [15] 2.99713948e-01+3.24839125e-02j [16] 2.04948804e-01+2.21885629e-02j [17] 1.01977612e-01+1.11196360e-02j [18] 1.23057376e-01+1.33860273e-02j [19] 7.67550484e-03+8.32574849e-04j [20] 1.71424009e-04+1.93963150e-05j
3	Re -3.232126284089e-04 Im -6.810747086020e+00	1.18e-04	[01] 1.32412350e-03+8.22858257e-02j [02] -4.99427896e-06-4.74576021e-04j [03] -2.99559985e-06-3.56717495e-04j [04] 9.48808582e-05+4.85742835e-04j [05] -8.59628082e-06+5.96089920e-04j [06] 5.44090764e-04+1.85214424e-03j [07] -7.81529871e-04-1.54854757e-03j [08] -1.93432589e-03-1.19607122e-01j [09] 6.49208684e-05+3.86236687e-03j [10] -9.32831526e-06-5.94168594e-04j [11] 5.60387182e-01-9.04386319e-03j [12] -3.23132751e-03+3.43978015e-05j [13] -2.42740977e-03+1.90748997e-05j [14] 3.30742376e-03-6.48589745e-04j [15] 4.06088770e-03+5.84164111e-05j [16] 1.26159053e-02-3.77698609e-03j [17] -1.05465125e-02+5.39895000e-03j [18] -8.14578558e-01+1.32098082e-02j [19] 2.63034224e-02-4.43432175e-04j [20] -4.04305963e-03+6.36298918e-05j
4	Re -3.179414383783e-04 Im 6.545709327188e+00	7.27e-05	[01] -7.12211446e-04-6.89100562e-03j [02] -3.59887119e-04-3.40043199e-03j [03] -6.27706361e-04-5.66560768e-03j [04] 1.48663769e-02+1.37033069e-01j [05] -4.96556364e-03-4.57890662e-02j [06] -3.38190600e-03-3.13006331e-02j [07] -1.72500960e-03-1.55987925e-02j [08] -2.04375220e-03-1.87345378e-02j [09] -1.32491861e-04-1.17481196e-03j [10] -4.99004774e-06-2.66780127e-05j [11] 4.51384513e-02-4.67247779e-03j [12] 2.22701133e-02-2.36887073e-03j [13] 3.70863589e-02-4.08437289e-03j [14] -8.96993163e-01+9.72688289e-02j [15] 2.99727567e-01-3.24878631e-02j [16] 2.04886583e-01-2.21039699e-02j [17] 1.02108663e-01-1.12909683e-02j [18] 1.22597199e-01-1.34035335e-02j [19] 7.69325113e-03-8.54312141e-04j [20] 1.70768191e-04-2.32377083e-05j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
5	Re -3.066843988032e-04 Im 5.061479080589e+00	1.70e-05	[01] -1.60130134e-05-9.01657947e-03j [02] 1.38349046e-04+7.35000841e-02j [03] 1.10741678e-04+5.88323410e-02j [04] -2.28458327e-05-1.15007358e-02j [05] 1.31820120e-04+7.81595296e-02j [06] -1.89948732e-04-1.07984156e-01j [07] -1.80334687e-04-1.00151199e-01j [08] -1.59440397e-05-8.94457858e-03j [09] 4.34977826e-05+2.46958228e-02j [10] -2.71372116e-06-9.06488669e-04j [11] 4.56292667e-02-8.12325891e-05j [12] -3.72021958e-01+6.74047396e-04j [13] -2.97780514e-01+5.43284964e-04j [14] 5.82119525e-02-1.10658367e-04j [15] -3.95602666e-01+6.44591888e-04j [16] 5.46557439e-01-9.28212779e-04j [17] 5.06918075e-01-8.82313392e-04j [18] 4.52663638e-02-7.78780601e-05j [19] -1.24990587e-01+2.14347540e-04j [20] 4.59410945e-03-1.06028141e-05j
6	Re -3.063124788117e-04 Im -5.061475822873e+00	2.42e-05	[01] -1.24422623e-05+9.01470948e-03j [02] 1.40226029e-04-7.34962264e-02j [03] 1.12423636e-04-5.88314333e-02j [04] -2.45953484e-05+1.14978686e-02j [05] 1.28483885e-04-7.81601490e-02j [06] -1.90196381e-04+1.07987700e-01j [07] -1.80408706e-04+1.00150681e-01j [08] -1.41979696e-05+8.94351301e-03j [09] 4.30948766e-05-2.46977199e-02j [10] -2.63064588e-06+9.06082728e-04j [11] 4.56172662e-02+5.77809134e-05j [12] -3.72001142e-01-6.80771459e-04j [13] -2.97777821e-01-5.49205197e-04j [14] 5.81890912e-02+1.21678359e-04j [15] -3.95601866e-01-6.32945947e-04j [16] 5.46584765e-01+9.29917138e-04j [17] 5.06908695e-01+8.82437785e-04j [18] 4.52630326e-02+7.18248184e-05j [19] -1.24996045e-01-2.10258913e-04j [20] 4.59378219e-03+1.14744890e-05j
7	Re -3.059875495959e-04 Im -4.200802290842e+00	8.03e-06	[01] -1.51334623e-02+1.62835776e-02j [02] -4.93381702e-04+5.18874743e-04j [03] 6.72998575e-04-7.34751047e-04j [04] 3.33168139e-02-3.58554701e-02j [05] 9.69032852e-02-1.04272415e-01j [06] 1.63978843e-02-1.76524209e-02j [07] 1.58221638e-02-1.70321488e-02j [08] -2.36960219e-02+2.54950591e-02j [09] -1.14175307e-01+1.22871460e-01j [10] 2.00969786e-03-2.16014505e-03j [11] 6.84109281e-02+6.35658759e-02j [12] 2.17733207e-03+2.07503377e-03j [13] -3.08581292e-03-2.82597364e-03j [14] -1.50633060e-01-1.39945843e-01j [15] -4.38058451e-01-4.07038161e-01j [16] -7.41564195e-02-6.88806836e-02j [17] -7.15512757e-02-6.64609786e-02j [18] 1.07103654e-01+9.95337406e-02j [19] 5.16191696e-01+4.79591000e-01j [20] -9.07488857e-03-8.44040454e-03j
8	Re -3.056490379732e-04 Im -4.583800861345e+00	3.26e-05	[01] 9.82381575e-04+1.79102647e-02j [02] 6.38419125e-03+1.16084771e-01j [03] 6.05917754e-03+1.10121174e-01j [04] -9.35197976e-04-1.69883056e-02j [05] -6.12499118e-03-1.11376628e-01j [06] 1.42195299e-03+2.58190761e-02j [07] 1.34759080e-03+2.44876920e-02j [08] 6.66027674e-04+1.21918730e-02j [09] -3.97578415e-03-7.21673144e-02j [10] -4.04319444e-04-7.36790641e-03j [11] 8.20974306e-02-4.50449132e-03j [12] 5.32115557e-01-2.93000772e-02j [13] 5.04783346e-01-2.78057268e-02j [14] -7.78723248e-02+4.29467998e-03j [15] -5.10526786e-01+2.81199087e-02j [16] 1.18338925e-01-6.53132129e-03j [17] 1.12231549e-01-6.18858606e-03j [18] 5.58880997e-02-3.05231599e-03j [19] -3.30790781e-01+1.82329165e-02j [20] -3.37776758e-02+1.85403235e-03j
9	Re -3.047736103015e-04 Im 4.200800120559e+00	9.34e-06	[01] -1.51311371e-02-1.62848237e-02j [02] -4.90127000e-04-5.20620367e-04j [03] 6.75000831e-04+7.34206797e-04j [04] 3.33135960e-02+3.58590530e-02j [05] 9.68906568e-02+1.04281608e-01j [06] 1.63965728e-02+1.76565333e-02j [07] 1.58212414e-02+1.70362824e-02j [08] -2.36931471e-02-2.54975166e-02j [09] -1.14166275e-01-1.22881234e-01j [10] 2.01004353e-03+2.15908856e-03j [11] 6.84148898e-02-6.35556359e-02j [12] 2.18794622e-03-2.05882026e-03j [13] -3.08288707e-03+2.83374192e-03j [14] -1.50644714e-01+1.39934887e-01j [15] -4.38094146e-01+4.06982170e-01j [16] -7.41761194e-02+6.88737044e-02j [17] -7.15724162e-02+6.64581047e-02j [18] 1.07116615e-01-9.95219092e-02j [19] 5.16234300e-01-4.79555144e-01j [20] -9.07413460e-03+8.43949904e-03j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
10	Re -2.973796201517e-04 Im 4.583799963609e+00	4.53e-05	[01] 9.84113559e-04-1.79091328e-02j [02] 6.38350718e-03-1.16090429e-01j [03] 6.05767354e-03-1.10127138e-01j [04] -9.38134834e-04+1.69873777e-02j [05] -6.12797465e-03+1.11367485e-01j [06] 1.41339920e-03-2.58122147e-02j [07] 1.34025291e-03-2.44788525e-02j [08] 6.70214116e-04-1.21903507e-02j [09] -3.96271613e-03+7.21702742e-02j [10] -4.02734692e-04+7.36811675e-03j [11] 8.20924849e-02+4.51035869e-03j [12] 5.32145601e-01+2.92987669e-02j [13] 5.04816929e-01+2.77999450e-02j [14] -7.78666836e-02-4.30380255e-03j [15] -5.10482412e-01-2.81243540e-02j [16] 1.18304184e-01+6.48574041e-03j [17] 1.12174206e-01+6.15259233e-03j [18] 5.58814276e-02+3.07689317e-03j [19] -3.30799404e-01-1.81814232e-02j [20] -3.37781543e-02-1.85256485e-03j
11	Re -2.850930864133e-04 Im 6.868689598956e+00	6.65e-04	[01] -3.06268534e-03-5.64822034e-04j [02] 2.02751129e-05+2.98289390e-04j [03] 1.57378445e-05+4.29737384e-04j [04] 6.03895936e-05+1.04874000e-02j [05] -4.57553300e-05-1.99397550e-03j [06] -6.11109391e-05+8.16958555e-02j [07] 3.73310324e-05-1.18044143e-01j [08] 4.32030044e-03+2.31875205e-03j [09] -1.34156666e-04+1.69145925e-04j [10] 2.83678249e-05+1.48066775e-05j [11] 3.65122619e-03-2.06001573e-02j [12] -2.04374427e-03+1.32691780e-04j [13] -2.94903767e-03+1.01900264e-04j [14] -7.20453805e-02+3.72419371e-04j [15] 1.37012892e-02-3.00485017e-04j [16] -5.61142569e-01-4.39271537e-04j [17] 8.10817725e-01+2.91472199e-04j [18] -1.57243410e-02+2.92936303e-02j [19] -1.17240192e-03-9.04739163e-04j [20] -8.20841233e-05+1.58971108e-04j
12	Re -2.818637707083e-04 Im -5.711800575348e+00	6.65e-05	[01] 3.74259227e-05+1.04936687e-01j [02] -3.46495035e-04-1.33506207e-02j [03] 3.05784827e-04-2.98719829e-02j [04] 1.36018934e-06+4.31622484e-03j [05] -1.64460635e-07-1.67352497e-03j [06] -3.15692548e-05-2.94960995e-02j [07] -2.81560150e-05-2.56897225e-02j [08] 2.91994288e-05+1.21510537e-01j [09] -1.17436780e-05-3.64411013e-02j [10] -9.27458428e-08-3.75264006e-03j [11] 5.99368282e-01-2.40225290e-04j [12] -7.62613238e-02+1.94257520e-03j [13] -1.70615609e-01-1.68960196e-03j [14] 2.46547287e-02-1.02696890e-05j [15] -9.55905029e-03+2.82388937e-06j [16] -1.68476938e-01+1.81005651e-04j [17] -1.46733869e-01+1.66562290e-04j [18] 6.94037275e-01-2.02928003e-04j [19] -2.08142032e-01+7.71422043e-05j [20] -2.14301179e-02+1.85102012e-06j
13	Re -2.776530009035e-04 Im 5.711793747140e+00	4.00e-05	[01] 3.95210315e-04-1.04940912e-01j [02] -1.09906437e-04+1.33851064e-02j [03] -5.43821100e-05+2.98505136e-02j [04] 2.55148922e-05-4.31128037e-03j [05] -6.23986486e-06+1.67966145e-03j [06] -1.20779354e-04+2.94909359e-02j [07] -9.46767472e-05+2.56829383e-02j [08] 4.32659027e-04-1.21510699e-01j [09] -1.32867053e-04+3.64407750e-02j [10] -1.35255820e-05+3.75264717e-03j [11] 5.99397054e-01+2.29476146e-03j [12] -7.64471483e-02-6.26248021e-04j [13] -1.70490463e-01-3.28167051e-04j [14] 2.46196728e-02+1.61277651e-04j [15] -9.57855054e-03-4.79241967e-05j [16] -1.68454944e-01-7.09860196e-04j [17] -1.46696866e-01-5.36726566e-04j [18] 6.94031569e-01+2.49513442e-03j [19] -2.08143722e-01-7.70791152e-04j [20] -2.14299447e-02-7.78154310e-05j
14	Re -2.775844032573e-04 Im -6.868418280720e+00	1.56e-03	[01] -2.76041976e-03+6.18620778e-03j [02] 1.28456479e-05-3.35885244e-04j [03] 5.19482777e-06-5.18996079e-04j [04] -3.44377025e-04-9.90491669e-03j [05] 4.38952522e-05+1.84908131e-03j [06] -2.45583182e-03-8.14988138e-02j [07] 3.54699184e-03+1.17587619e-01j [08] 3.85888371e-03-1.03892716e-02j [09] -1.23575832e-04+7.56313938e-05j [10] 2.53318902e-05-6.51241196e-05j [11] 4.14821962e-02+1.85680319e-02j [12] -2.30232298e-03-7.81404875e-05j [13] -3.44972503e-03-2.76595118e-05j [14] -6.83927962e-02+2.34850854e-03j [15] 1.28225698e-02-3.07274522e-04j [16] -5.59754031e-01+1.69081486e-02j [17] 8.07752814e-01-2.43892176e-02j [18] -7.03983561e-02-2.61616268e-02j [19] 4.72921605e-04+8.42403570e-04j [20] -3.73778657e-04-1.42374852e-04j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
15	Re -2.463786533469e-04 Im 6.810352957321e+00	3.50e-04	[01] 2.09787787e-02-7.95203274e-02j [02] -1.21333378e-04+4.49043477e-04j [03] -9.88381322e-05+3.38661138e-04j [04] 2.64566586e-04-7.67946981e-04j [05] 1.05891238e-04-5.09591623e-04j [06] 3.97806949e-04-3.38863589e-03j [07] -3.46387752e-04+3.86450284e-03j [08] -3.05584350e-02+1.15595079e-01j [09] 9.87066500e-04-3.73848575e-03j [10] -1.50088132e-04+5.72785281e-04j [11] 5.41605236e-01+1.42925463e-01j [12] -3.06821917e-03-8.29083130e-04j [13] -2.30297224e-03-6.63264187e-04j [14] 5.21863501e-03+1.72929007e-03j [15] 3.47674606e-03+7.42066881e-04j [16] 2.33057801e-02+2.72063369e-03j [17] -2.65577695e-02-2.34707123e-03j [18] -7.87259206e-01-2.08149554e-01j [19] 2.54615368e-02+6.72302326e-03j [20] -3.90467205e-03-1.02858914e-03j
16	Re -1.915605953233e-04 Im -6.810433674660e+00	6.96e-04	[01] 1.92537898e-02+7.98873314e-02j [02] -1.27933848e-04-4.50218527e-04j [03] -9.09497395e-05-3.37274719e-04j [04] 3.00493802e-04+1.07940684e-03j [05] 6.19034702e-05+4.40537930e-04j [06] -5.49577924e-04+4.88323240e-03j [07] 9.37376282e-04-6.09528242e-03j [08] -2.81126645e-02-1.16104006e-01j [09] 9.03544446e-04+3.75804235e-03j [10] -1.37704869e-04-5.75708323e-04j [11] 5.44092419e-01-1.31189488e-01j [12] -3.06794425e-03+8.55191168e-04j [13] -2.28670230e-03+6.33661874e-04j [14] 7.31210208e-03-1.87590836e-03j [15] 3.01377948e-03-4.69502327e-04j [16] 3.37055216e-02+3.82500798e-03j [17] -4.19655255e-02-6.53756541e-03j [18] -7.90722045e-01+1.91476302e-01j [19] 2.55938432e-02-6.15526433e-03j [20] -3.92261077e-03+9.44657559e-04j
17	Re -1.018560585833e-06 Im 0.000000000000e+00	3.46e-07	[01] 3.16227767e-01+0.00000000e+00j [02] 3.16227767e-01+0.00000000e+00j [03] 3.16227767e-01+0.00000000e+00j [04] 3.16227767e-01+0.00000000e+00j [05] 3.16227766e-01+0.00000000e+00j [06] 3.16227768e-01+0.00000000e+00j [07] 3.16227767e-01+0.00000000e+00j [08] 3.16227766e-01+0.00000000e+00j [09] 3.16227759e-01+0.00000000e+00j [10] 3.16227766e-01+0.00000000e+00j [11] -5.84900625e-07+0.00000000e+00j [12] -3.29588853e-07+0.00000000e+00j [13] -3.66816593e-07+0.00000000e+00j [14] -3.73577587e-07+0.00000000e+00j [15] -3.06443936e-07+0.00000000e+00j [16] -3.06852394e-07+0.00000000e+00j [17] -3.01455940e-07+0.00000000e+00j [18] -3.76395114e-07+0.00000000e+00j [19] -3.50248486e-07+0.00000000e+00j [20] -1.85457352e-07+0.00000000e+00j
18	Re 9.989551485527e-07 Im 0.000000000000e+00	4.55e-07	[01] 3.16227767e-01+0.00000000e+00j [02] 3.16227767e-01+0.00000000e+00j [03] 3.16227766e-01+0.00000000e+00j [04] 3.16227766e-01+0.00000000e+00j [05] 3.16227766e-01+0.00000000e+00j [06] 3.16227767e-01+0.00000000e+00j [07] 3.16227766e-01+0.00000000e+00j [08] 3.16227763e-01+0.00000000e+00j [09] 3.16227767e-01+0.00000000e+00j [10] 3.16227765e-01+0.00000000e+00j [11] 2.82023155e-07+0.00000000e+00j [12] 4.81854163e-07+0.00000000e+00j [13] 2.50812651e-07+0.00000000e+00j [14] 3.87216591e-07+0.00000000e+00j [15] 4.20155027e-07+0.00000000e+00j [16] 2.59580456e-07+0.00000000e+00j [17] 3.28428287e-07+0.00000000e+00j [18] 3.21497005e-07+0.00000000e+00j [19] -5.16548431e-08+0.00000000e+00j [20] 2.82618880e-07+0.00000000e+00j

4 Component ablation experiments

The following data correspond to the ablation studies presented in Tables III and IV of the main paper. These experiments were conducted on the Base scenarios of 9-bus system and 39-bus system. Four specific algorithmic configurations are evaluated: the full proposed pipeline, the pipeline without Step-1 optimization, initialization using only $\widehat{H}$ seeds, and initialization using only $\widehat{K}$ seeds.

Table S11 9-bus system Base Ablation Full pipeline

Returned 6/6 eigenpairs; Eigenvalue MAE 3.268396e-07; RMSE 5.710774e-07; mean residual 3.623158e-13; mean MAC 1.0000000000. Modal align rate 100.00% (6/6).

Table D01. Ablation study results for the 9-bus system using the Full pipeline (Supplementary to Table III and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -9.049113375636e-15 Im -8.689799888785e+00	1.26e-14	[01] -1.48156394e-16-3.60131621e-02j [02] -5.38284587e-17+9.41468493e-02j [03] -2.51307019e-17+5.39366471e-02j [04] -3.12947172e-01-1.89960548e-16j [05] 8.18117281e-01+2.78735747e-35j [06] 4.68698670e-01-1.07683353e-16j
2	Re -4.664608679178e-17 Im -9.658385477926e-07	9.58e-13	[01] 5.77350269e-01+2.05616948e-18j [02] 5.77350269e-01-1.77008577e-16j [03] 5.77350269e-01+2.08270950e-33j [04] -7.42202599e-17-5.57627146e-07j [05] -6.85767024e-17-5.57627145e-07j [06] 6.20019100e-17-5.57627146e-07j
3	Re 6.282010963885e-16 Im 8.689799888785e+00	2.67e-13	[01] 7.90554384e-16+3.60131621e-02j [02] -9.28083374e-15-9.41468493e-02j [03] -2.06048098e-14-5.39366471e-02j [04] -3.12947172e-01+4.06225573e-14j [05] 8.18117281e-01+6.71040727e-35j [06] 4.68698670e-01-3.04101982e-13j
4	Re 5.178718390205e-14 Im -1.336021110593e+01	4.98e-14	[01] 4.52684804e-16-2.97384465e-03j [02] 3.42024000e-16-2.21449322e-02j [03] 4.95002509e-17+7.12175300e-02j [04] -3.97311924e-02+3.87081745e-17j [05] -2.95860969e-01-1.10671382e-16j [06] 9.51481235e-01+3.24907636e-34j
5	Re 5.178718390205e-14 Im 1.336021110593e+01	4.98e-14	[01] 4.52684804e-16+2.97384465e-03j [02] 3.42024000e-16+2.21449322e-02j [03] 4.95002509e-17-7.12175300e-02j [04] -3.97311924e-02-3.87081745e-17j [05] -2.95860969e-01+1.10671382e-16j [06] 9.51481235e-01-8.08277946e-51j
6	Re 3.971587095095e-07 Im 8.366362606539e-07	8.37e-13	[01] 5.77350269e-01+4.96350612e-35j [02] 5.77350269e-01-5.21378896e-15j [03] 5.77350269e-01-4.63878596e-15j [04] 2.29299688e-07+4.83032170e-07j [05] 2.29299688e-07+4.83032171e-07j [06] 2.29299688e-07+4.83032170e-07j

Table S12 9-bus system Base Ablation Without Step-1

Returned 6/6 eigenpairs; Eigenvalue MAE 2.076457e-07; RMSE 3.655854e-07; mean residual 4.403019e-13; mean MAC 1.0000000000. Modal align rate 83.33% (5/6).

One reference mode remains unmatched under the minimum-total-error assignment.

Table D02. Ablation study results for the 9-bus system Without Step-1 optimization (Supplementary to Table III and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -1.041580190041e-06 Im 1.281039451924e-07	6.59e-13	[01] 5.77350269e-01+2.51951192e-34j [02] 5.77350269e-01+2.09134411e-15j [03] 5.77350269e-01+1.79071168e-15j [04] -6.01356603e-07+7.39608474e-08j [05] -6.01356603e-07+7.39608473e-08j [06] -6.01356603e-07+7.39608470e-08j
2	Re -1.347988161248e-12 Im 1.336021110593e+01	2.01e-13	[01] -3.21921350e-15+2.97384465e-03j [02] -5.92796692e-15+2.21449322e-02j [03] 3.98188067e-15-7.12175300e-02j [04] -3.97311924e-02+1.14282979e-15j [05] -2.95860969e-01-3.84803093e-15j [06] 9.51481235e-01-1.48114514e-33j
3	Re -8.872654609458e-14 Im 8.689799888785e+00	3.75e-14	[01] -1.15728001e-15+3.60131621e-02j [02] 5.43076537e-16-9.41468493e-02j [03] 1.02330653e-15-5.39366471e-02j [04] -3.12947172e-01-2.30630742e-15j [05] 8.18117281e-01-1.15716064e-33j [06] 4.68698670e-01+1.58247156e-14j
4	Re 5.414965394998e-15 Im -8.689799888785e+00	1.46e-14	[01] 1.01837989e-16-3.60131621e-02j [02] -4.26871614e-17+9.41468493e-02j [03] 8.44169386e-17+5.39366471e-02j [04] -3.12947172e-01-7.23708386e-17j [05] 8.18117281e-01+5.69506241e-35j [06] 4.68698670e-01-7.58245596e-18j
5	Re 5.865946710293e-14 Im -1.336021110592e+01	9.51e-13	[01] 3.45664411e-16-2.97384465e-03j [02] 1.77995878e-16-2.21449322e-02j [03] -1.38091425e-16+7.12175300e-02j [04] -3.97311924e-02+1.41450783e-16j [05] -2.95860969e-01-1.78398762e-16j [06] 9.51481235e-01+5.11285936e-34j
6	Re 5.774564940618e-07 Im -7.360023736098e-07	7.79e-13	[01] 5.77350269e-01-5.90799083e-34j [02] 5.77350269e-01+5.77355901e-15j [03] 5.77350269e-01+4.99626361e-15j [04] 3.33394662e-07-4.24931169e-07j [05] 3.33394662e-07-4.24931168e-07j [06] 3.33394662e-07-4.24931169e-07j

Table S13 9-bus system Base Ablation $\widehat{H}$ only

Returned 6/6 eigenpairs; Eigenvalue MAE 7.350003e+00; RMSE 9.201591e+00; mean residual 1.174677e-12; mean MAC 0.3349650167. Modal align rate 33.33% (2/6).

Four reference modes remain unmatched under the minimum-total-error assignment.

Table D03. Ablation study results for the 9-bus system using $\widehat{H}$ seeds only (Supplementary to Table III and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -1.147904394141e-06 Im 2.067684494766e-07	8.72e-13	[01] 5.77350269e-01+4.10892759e-33j [02] 5.77350269e-01+3.30191067e-15j [03] 5.77350269e-01+3.16917963e-15j [04] -6.62742911e-07+1.19377820e-07j [05] -6.62742911e-07+1.19377820e-07j [06] -6.62742911e-07+1.19377820e-07j
2	Re -6.068784249393e-07 Im -8.116860619198e-07	8.99e-13	[01] 5.77350269e-01-1.75371916e-34j [02] 5.77350269e-01-7.54015110e-15j [03] 5.77350269e-01-6.58334988e-15j [04] -3.50381422e-07-4.68627167e-07j [05] -3.50381422e-07-4.68627166e-07j [06] -3.50381422e-07-4.68627166e-07j
3	Re -4.431600810626e-07 Im -1.359139546911e-06	1.80e-12	[01] 5.77350269e-01+8.09060497e-15j [02] 5.77350269e-01-9.96506079e-16j [03] 5.77350269e-01-3.86253311e-49j [04] -2.55858592e-07-7.84699583e-07j [05] -2.55858592e-07-7.84699584e-07j [06] -2.55858592e-07-7.84699583e-07j
4	Re -4.431600810626e-07 Im 1.359139546911e-06	1.80e-12	[01] 5.77350269e-01-8.09060497e-15j [02] 5.77350269e-01+9.96506079e-16j [03] 5.77350269e-01-3.78067260e-33j [04] -2.55858592e-07+7.84699583e-07j [05] -2.55858592e-07+7.84699584e-07j [06] -2.55858592e-07+7.84699583e-07j
5	Re 2.430697184453e-07 Im -9.391126379277e-07	9.43e-13	[01] 5.77350269e-01-2.95065020e-15j [02] 5.77350269e-01+3.56666466e-16j [03] 5.77350269e-01+5.84608357e-34j [04] 1.40336367e-07-5.42196934e-07j [05] 1.40336367e-07-5.42196934e-07j [06] 1.40336367e-07-5.42196934e-07j
6	Re 8.025399324442e-07 Im -5.861684080593e-07	7.43e-13	[01] 5.77350269e-01+2.15668806e-33j [02] 5.77350269e-01+7.06417881e-15j [03] 5.77350269e-01+6.45211212e-15j [04] 4.63346646e-07-3.38424488e-07j [05] 4.63346646e-07-3.38424488e-07j [06] 4.63346646e-07-3.38424488e-07j

Table S14 9-bus system Base Ablation $\widehat{K}$ only

Returned 6/6 eigenpairs; Eigenvalue MAE 2.226702e+00; RMSE 4.027517e+00; mean residual 8.120736e-13; mean MAC 0.6753001051. Modal align rate 66.67% (4/6).

Two reference modes remain unmatched under the minimum-total-error assignment.

Table D04. Ablation study results for the 9-bus system using $\widehat{K}$ seeds only (Supplementary to Table III and Section IV-B in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -9.049113375636e-15 Im -8.689799888785e+00	1.26e-14	[01] -1.48156394e-16-3.60131621e-02j [02] -5.38284587e-17+9.41468493e-02j [03] -2.51307019e-17+5.39366471e-02j [04] -3.12947172e-01-1.89960548e-16j [05] 8.18117281e-01+2.78735747e-35j [06] 4.68698670e-01-1.07683353e-16j
2	Re -4.664608679178e-17 Im -9.658385477926e-07	9.58e-13	[01] 5.77350269e-01+2.05616948e-18j [02] 5.77350269e-01-1.77008577e-16j [03] 5.77350269e-01+2.08270950e-33j [04] -7.42202599e-17-5.57627146e-07j [05] -6.85767024e-17-5.57627145e-07j [06] 6.20019100e-17-5.57627146e-07j
3	Re 2.447931642076e-16 Im -1.112185670482e-06	1.20e-12	[01] 5.77350269e-01-5.77629157e-17j [02] 5.77350269e-01-4.73216645e-17j [03] 5.77350269e-01+9.25578602e-50j [04] 3.39409101e-16-6.42120696e-07j [05] -1.91272709e-17-6.42120696e-07j [06] 1.03705404e-16-6.42120696e-07j
4	Re 2.447931642076e-16 Im 1.112185670482e-06	1.20e-12	[01] 5.77350269e-01+5.77629157e-17j [02] 5.77350269e-01+4.73216645e-17j [03] 5.77350269e-01+1.60018468e-33j [04] 3.39409101e-16+6.42120696e-07j [05] -1.91272709e-17+6.42120696e-07j [06] 1.03705404e-16+6.42120696e-07j
5	Re 6.282010963885e-16 Im 8.689799888785e+00	2.67e-13	[01] 7.90554384e-16+3.60131621e-02j [02] -9.28083374e-15-9.41468493e-02j [03] -2.06048098e-14-5.39366471e-02j [04] -3.12947172e-01+4.06225573e-14j [05] 8.18117281e-01+6.71040727e-35j [06] 4.68698670e-01-3.04101982e-13j
6	Re 2.622243097995e-14 Im -1.336021110593e+01	1.24e-12	[01] 1.59502494e-16-2.97384465e-03j [02] 3.84679695e-17-2.21449322e-02j [03] -7.80867759e-17+7.12175300e-02j [04] -3.97311924e-02-9.62695193e-17j [05] -2.95860969e-01+1.87957946e-16j [06] 9.51481235e-01-3.32681401e-50j

Table S15 39-bus system Base Ablation Full pipeline

Returned 20/20 eigenpairs; Eigenvalue MAE 1.446623e-11; RMSE 5.396819e-11; mean residual 1.075961e-12; mean MAC 1.0000000000. Modal align rate 100.00% (20/20).

Table S15. Ablation study results for the 39-bus system using the Full pipeline (Supplementary to Table IV and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.295455098032e-04 Im -6.545746650406e+00	9.86e-14	[01] -2.58626181e-06-6.88034058e-03j [02] 7.26614087e-08-3.42576885e-03j [03] 5.23258780e-07-5.69731566e-03j [04] -6.93909312e-06+1.37830874e-01j [05] 1.25694925e-06-4.60550776e-02j [06] -5.38143407e-07-3.14834642e-02j [07] 5.35752496e-06-1.56863846e-02j [08] 4.53469935e-06-1.89226056e-02j [09] -7.18852805e-08-1.17917348e-03j [10] 2.42562544e-08-2.64580620e-05j [11] -4.50369655e-02+1.91963999e-05j [12] -2.24242150e-02+6.53323568e-07j [13] -3.72931851e-02-1.54759462e-06j [14] 9.02205983e-01-1.90193319e-33j [15] -3.01464870e-01+6.94957265e-06j [16] -2.06082780e-01+1.38977846e-05j [17] -1.02679101e-01-2.98996234e-05j [18] -1.23862584e-01-2.34471334e-05j [19] -7.71857083e-03+8.59134151e-07j [20] -1.73187779e-04-1.50056160e-07j
2	Re -3.295455095673e-04 Im 6.545746650406e+00	1.73e-14	[01] -2.58626181e-06+6.88034058e-03j [02] 7.26614088e-08+3.42576885e-03j [03] 5.23258781e-07+5.69731566e-03j [04] -6.93909312e-06-1.37830874e-01j [05] 1.25694925e-06+4.60550776e-02j [06] -5.38143410e-07+3.14834642e-02j [07] 5.35752497e-06+1.56863846e-02j [08] 4.53469935e-06+1.89226056e-02j [09] -7.18852791e-08+1.17917348e-03j [10] 2.42562543e-08+2.64580620e-05j [11] -4.50369655e-02-1.91963999e-05j [12] -2.24242150e-02-6.53323568e-07j [13] -3.72931851e-02+1.54759462e-06j [14] 9.02205983e-01-2.65808464e-34j [15] -3.01464870e-01-6.94957265e-06j [16] -2.06082780e-01-1.38977847e-05j [17] -1.02679101e-01+2.98996235e-05j [18] -1.23862584e-01+2.34471334e-05j [19] -7.71857083e-03-8.59134151e-07j [20] -1.73187779e-04+1.50056160e-07j
3	Re -3.237419521066e-04 Im 6.868699396043e+00	1.32e-14	[01] 3.93828545e-07+1.04936876e-03j [02] -8.53330567e-08+2.85873671e-04j [03] -1.00936578e-07+4.20212720e-04j [04] -3.40402314e-06+1.05097137e-02j [05] 7.27334513e-07-1.98784341e-03j [06] 9.01174967e-06+8.17620683e-02j [07] -5.56788543e-06-1.18131527e-01j [08] -8.99137290e-07+3.17126648e-05j [09] -2.75864888e-08+2.40086440e-04j [10] -6.64317883e-09-2.69668215e-07j [11] -7.20779871e-03+2.36536520e-06j [12] -1.96358028e-03-6.78676417e-07j [13] -2.88631483e-03-8.29343500e-07j [14] -7.21880632e-02-2.67836469e-05j [15] 1.36538986e-02+5.63939044e-06j [16] -5.61599072e-01+3.54291879e-05j [17] 8.11409953e-01+2.41938894e-33j [18] -2.17824471e-04-6.18617048e-06j [19] -1.64908158e-03-2.67209352e-07j [20] 1.85227205e-06-4.55426954e-08j
4	Re -3.237419521041e-04 Im -6.868699396043e+00	1.25e-14	[01] 3.93828582e-07-1.04936876e-03j [02] -8.53330572e-08-2.85873671e-04j [03] -1.00936578e-07-4.20212720e-04j [04] -3.40402314e-06-1.05097137e-02j [05] 7.27334514e-07+1.98784341e-03j [06] 9.01174967e-06-8.17620683e-02j [07] -5.56788543e-06+1.18131527e-01j [08] -8.99137342e-07-3.17126649e-05j [09] -2.75864871e-08-2.40086440e-04j [10] -6.64317924e-09+2.69668214e-07j [11] -7.20779871e-03-2.36536545e-06j [12] -1.96358028e-03+6.78676418e-07j [13] -2.88631483e-03+8.29343501e-07j [14] -7.21880632e-02+2.67836469e-05j [15] 1.36538986e-02-5.63939044e-06j [16] -5.61599072e-01-3.54291879e-05j [17] 8.11409953e-01+1.31490927e-33j [18] -2.17824471e-04+6.18617083e-06j [19] -1.64908158e-03+2.67209341e-07j [20] 1.85227205e-06+4.55426971e-08j
5	Re -3.151425852114e-04 Im -6.810560067478e+00	8.31e-14	[01] 1.62522531e-05-8.22907598e-02j [02] -7.49362789e-07+4.74246726e-04j [03] -7.81207312e-07+3.55683280e-04j [04] -3.96671911e-06-4.84289946e-04j [05] 1.01446635e-06-5.96884599e-04j [06] -5.05515801e-07-1.84356374e-03j [07] 2.41553565e-07+1.53385690e-03j [08] -5.53544336e-06+1.19626706e-01j [09] -7.89313400e-07-3.86352325e-03j [10] -3.12960119e-07+5.93535159e-04j [11] -5.60446168e-01-8.47536234e-05j [12] 3.22988605e-03+4.95412494e-06j [13] 2.42240259e-03+5.20836837e-06j [14] -3.29828452e-03+2.71681991e-05j [15] -4.06511873e-03-6.72098027e-06j [16] -1.25557015e-02+4.02383116e-06j [17] 1.04464245e-02-2.12849868e-06j [18] 8.14724871e-01+1.01145900e-33j [19] -2.63127569e-02+6.59322703e-06j [20] 4.04230695e-03+1.94438548e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
6	Re -3.151425852114e-04 Im 6.810560067478e+00	8.31e-14	[01] 1.62522531e-05+8.22907598e-02j [02] -7.49362789e-07-4.74246726e-04j [03] -7.81207312e-07-3.55683280e-04j [04] -3.96671911e-06+4.84289946e-04j [05] 1.01446635e-06+5.96884599e-04j [06] -5.05515801e-07+1.84356374e-03j [07] 2.41553565e-07-1.53385690e-03j [08] -5.53544336e-06-1.19626706e-01j [09] -7.89313400e-07+3.86352325e-03j [10] -3.12960119e-07-5.93535159e-04j [11] -5.60446168e-01+8.47536234e-05j [12] 3.22988605e-03-4.95412494e-06j [13] 2.42240259e-03-5.20836837e-06j [14] -3.29828452e-03-2.71681991e-05j [15] -4.06511873e-03+6.72098027e-06j [16] -1.25557015e-02-4.02383116e-06j [17] 1.04464245e-02+2.12849868e-06j [18] 8.14724871e-01+6.75153080e-50j [19] -2.63127569e-02-6.59322703e-06j [20] 4.04230695e-03-1.94438548e-06j
7	Re -3.093790131005e-04 Im -5.061482055348e+00	6.67e-12	[01] 9.16706989e-07+9.02229536e-03j [02] 1.20586623e-05-7.35063930e-02j [03] 5.79017614e-06-5.88390916e-02j [04] -3.18469076e-06+1.15004987e-02j [05] 8.65304390e-07-7.81564688e-02j [06] -6.60031534e-06+1.07982055e-01j [07] -1.02095812e-05+1.00147607e-01j [08] 3.22472624e-07+8.95079096e-03j [09] 2.24026501e-06-2.46937367e-02j [10] -4.23161002e-07+9.07914990e-04j [11] 4.56661858e-02-7.43120454e-06j [12] -3.72051293e-01-3.82933700e-05j [13] -2.97813008e-01-1.11032948e-05j [14] 5.82095689e-02+1.25612425e-05j [15] -3.95587564e-01+1.98002461e-05j [16] 5.46549237e-01+1.37556405e-36j [17] 5.06895319e-01+2.06920472e-05j [18] 4.53042677e-02-4.40137610e-06j [19] -1.24986906e-01-3.69933836e-06j [20] 4.59539556e-03+1.86093251e-06j
8	Re -3.093789966933e-04 Im 5.061482055348e+00	9.27e-16	[01] 9.16706913e-07-9.02229536e-03j [02] 1.20586631e-05+7.35063930e-02j [03] 5.79017653e-06+5.88390916e-02j [04] -3.18469090e-06-1.15004987e-02j [05] 8.65304623e-07+7.81564688e-02j [06] -6.60031579e-06-1.07982055e-01j [07] -1.02095819e-05-1.00147607e-01j [08] 3.22472545e-07-8.95079096e-03j [09] 2.24026532e-06+2.46937367e-02j [10] -4.23161141e-07-9.07914990e-04j [11] 4.56661858e-02+7.43120428e-06j [12] -3.72051293e-01+3.82933730e-05j [13] -2.97813008e-01+1.11032954e-05j [14] 5.82095689e-02-1.25612431e-05j [15] -3.95587564e-01-1.98002461e-05j [16] 5.46549237e-01-2.33632255e-67j [17] 5.06895319e-01-2.06920494e-05j [18] 4.53042677e-02+4.40137573e-06j [19] -1.24986906e-01+3.69933925e-06j [20] 4.59539556e-03-1.86093269e-06j
9	Re -3.046730835365e-04 Im 4.200799305801e+00	8.89e-12	[01] -4.35255422e-06-2.22301539e-02j [02] -7.39521113e-06-7.15710686e-04j [03] -7.31955580e-06+9.97049145e-04j [04] 3.26608342e-06+4.89449665e-02j [05] 2.06960165e-05+1.42347439e-01j [06] -3.63309616e-06+2.40940603e-02j [07] -2.92559724e-06+2.32478125e-02j [08] -6.82922228e-06-3.48071648e-02j [09] -1.21650377e-05-1.67730238e-01j [10] 1.79853527e-06+2.95013651e-03j [11] 9.33844162e-02-1.15112782e-05j [12] 3.00655921e-03-3.08477400e-05j [13] -4.18840113e-03-3.10517589e-05j [14] -2.05607982e-01-1.19205146e-06j [15] -5.97973028e-01+4.35703828e-05j [16] -1.01214311e-01-2.26027186e-05j [17] -9.76593939e-02-1.93728290e-05j [18] 1.46217916e-01-1.80833868e-05j [19] 7.04601073e-01-1.59171867e-33j [20] -1.23929319e-02+6.65646014e-06j
10	Re -3.046730678263e-04 Im -4.200799305801e+00	1.21e-12	[01] -4.35255435e-06+2.22301539e-02j [02] -7.39521129e-06+7.15710686e-04j [03] -7.31955601e-06-9.97049145e-04j [04] 3.26608333e-06-4.89449665e-02j [05] 2.06960169e-05-1.42347439e-01j [06] -3.63309655e-06-2.40940603e-02j [07] -2.92559759e-06-2.32478125e-02j [08] -6.82922259e-06+3.48071648e-02j [09] -1.21650385e-05+1.67730238e-01j [10] 1.79853575e-06-2.95013651e-03j [11] 9.33844162e-02+1.15112782e-05j [12] 3.00655921e-03+3.08477407e-05j [13] -4.18840113e-03+3.10517598e-05j [14] -2.05607982e-01+1.19205231e-06j [15] -5.97973028e-01-4.35703832e-05j [16] -1.01214311e-01+2.26027206e-05j [17] -9.76593939e-02+1.93728306e-05j [18] 1.46217916e-01+1.80833880e-05j [19] 7.04601073e-01-1.21704382e-34j [20] -1.23929319e-02-6.65646082e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
11	Re -3.005670471533e-04 Im -1.517917908053e-21	2.47e-15	[01] 3.16227751e-01+3.53591796e-19j [02] 3.16227752e-01-3.99104666e-20j [03] 3.16227752e-01+2.23595487e-19j [04] 3.16227753e-01+2.91918872e-19j [05] 3.16227754e-01+3.57071765e-69j [06] 3.16227753e-01+3.44316592e-19j [07] 3.16227753e-01+6.78212020e-21j [08] 3.16227752e-01+5.34714582e-19j [09] 3.16227753e-01+4.67558554e-19j [10] 3.16227745e-01+5.83302123e-19j [11] -9.50476413e-05-1.82525330e-22j [12] -9.50476415e-05-1.07384109e-22j [13] -9.50476415e-05-2.62675674e-22j [14] -9.50476420e-05-1.52093119e-22j [15] -9.50476424e-05-2.33453760e-22j [16] -9.50476419e-05-1.93155720e-22j [17] -9.50476420e-05-1.47662360e-22j [18] -9.50476416e-05-1.72926011e-22j [19] -9.50476418e-05-1.64547849e-22j [20] -9.50476397e-05-1.66777902e-22j
12	Re -3.002929663520e-04 Im -4.583815868253e+00	9.02e-13	[01] 4.53693982e-07+1.79393253e-02j [02] -7.61589952e-06+1.16252744e-01j [03] -3.38949150e-06+1.10279628e-01j [04] -3.86409938e-06-1.70160143e-02j [05] 4.96811792e-07-1.11562490e-01j [06] -7.76020744e-06+2.58689101e-02j [07] -8.11610347e-06+2.45356265e-02j [08] 6.07334259e-07+1.22128577e-02j [09] 1.38569914e-05-7.22661698e-02j [10] 1.51975988e-06-7.38028284e-03j [11] 8.22305639e-02-7.46670283e-06j [12] 5.32881177e-01+3.57112399e-35j [13] 5.05501511e-01-1.75793913e-05j [14] -7.79982751e-02+2.28221094e-05j [15] -5.11381913e-01+3.12241369e-05j [16] 1.18578323e-01+2.78031104e-05j [17] 1.12466796e-01+2.98348479e-05j [18] 5.59814908e-02-6.45134366e-06j [19] -3.31254820e-01-4.18168749e-05j [20] -3.38298581e-02-4.75005278e-06j
13	Re -3.002929637752e-04 Im 4.583815868253e+00	1.90e-13	[01] 4.53694029e-07-1.79393253e-02j [02] -7.61589956e-06-1.16252744e-01j [03] -3.38949150e-06-1.10279628e-01j [04] -3.86409942e-06+1.70160143e-02j [05] 4.96811754e-07+1.11562490e-01j [06] -7.76020752e-06-2.58689101e-02j [07] -8.11610355e-06-2.45356265e-02j [08] 6.07334299e-07-1.22128577e-02j [09] 1.38569914e-05+7.22661698e-02j [10] 1.51975999e-06+7.38028284e-03j [11] 8.22305639e-02+7.46670307e-06j [12] 5.32881177e-01+3.57298654e-35j [13] 5.05501511e-01+1.75793916e-05j [14] -7.79982751e-02-2.28221096e-05j [15] -5.11381913e-01-3.12241373e-05j [16] 1.18578323e-01-2.78031107e-05j [17] 1.12466796e-01-2.98348482e-05j [18] 5.59814908e-02+6.45134384e-06j [19] -3.31254820e-01+4.18168748e-05j [20] -3.38298581e-02+4.75005288e-06j
14	Re -2.922696776643e-04 Im -5.711763079524e+00	4.99e-14	[01] 8.89755210e-06+1.04938876e-01j [02] -8.96676667e-06-1.33951926e-02j [03] 7.25175645e-06-2.98328985e-02j [04] -1.39306449e-07+4.31481797e-03j [05] 1.52653066e-07-1.67250294e-03j [06] -2.43024774e-06-2.94993935e-02j [07] -4.26096750e-07-2.56934450e-02j [08] -6.21770638e-06+1.21511291e-01j [09] 1.74729665e-07-3.64417140e-02j [10] 2.63647356e-07-3.75177825e-03j [11] 5.99385995e-01-8.14911610e-05j [12] -7.65101642e-02+5.51310555e-05j [13] -1.70398450e-01-3.27010631e-05j [14] 2.46452180e-02-4.65405028e-07j [15] -9.55294059e-03-3.83096251e-07j [16] -1.68493546e-01+2.25027775e-05j [17] -1.46754871e-01+9.94317858e-06j [18] 6.94043705e-01+1.02314727e-33j [19] -2.08146436e-01+9.65279354e-06j [20] -2.14292686e-02-4.09360213e-07j
15	Re -2.922696776592e-04 Im 5.711763079524e+00	1.36e-14	[01] 8.89755210e-06-1.04938876e-01j [02] -8.96676674e-06+1.33951926e-02j [03] 7.25175651e-06+2.98328985e-02j [04] -1.39306449e-07-4.31481797e-03j [05] 1.52653066e-07+1.67250294e-03j [06] -2.43024774e-06+2.94993935e-02j [07] -4.26096753e-07+2.56934450e-02j [08] -6.21770638e-06-1.21511291e-01j [09] 1.74729665e-07+3.64417140e-02j [10] 2.63647356e-07+3.75177825e-03j [11] 5.99385995e-01+8.14911610e-05j [12] -7.65101642e-02-5.51310558e-05j [13] -1.70398450e-01+3.27010635e-05j [14] 2.46452180e-02+4.65405024e-07j [15] -9.55294059e-03+3.83096251e-07j [16] -1.68493546e-01-2.25027775e-05j [17] -1.46754871e-01-9.94317860e-06j [18] 6.94043705e-01-5.60274835e-34j [19] -2.08146436e-01-9.65279354e-06j [20] -2.14292686e-02+4.09360214e-07j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
16	Re -2.882560430318e-04 Im 5.639387583526e+00	9.56e-14	[01] 8.81132440e-06+6.13597700e-03j [02] -6.68089717e-06-1.30703829e-01j [03] 8.35884296e-06+1.14747872e-01j [04] 4.11147992e-07+2.97638099e-04j [05] -1.53491401e-07-4.88945872e-04j [06] -4.16940388e-06-8.81465993e-03j [07] -4.13409077e-06-7.71205494e-03j [08] 1.08701761e-05+7.19410612e-03j [09] -3.59550088e-06-2.79730645e-03j [10] -4.04347270e-07+7.13166771e-05j [11] -3.46031550e-02+4.79217409e-05j [12] 7.37089552e-01+1.34904860e-34j [13] -6.47107728e-01+1.40619876e-05j [14] -1.67849672e-03+2.23282690e-06j [15] 2.75735532e-03-7.24655898e-07j [16] 4.97092850e-02-2.09720054e-05j [17] 4.34912681e-02-2.10906937e-05j [18] -4.05703559e-02+5.92273917e-05j [19] 1.57750963e-02-1.94700825e-05j [20] -4.02182267e-04-2.30082841e-06j
17	Re -2.882560430295e-04 Im -5.639387583526e+00	1.96e-15	[01] 8.81132480e-06-6.13597700e-03j [02] -6.68089717e-06+1.30703829e-01j [03] 8.35884280e-06-1.14747872e-01j [04] 4.11148008e-07-2.97638099e-04j [05] -1.53491409e-07+4.88945872e-04j [06] -4.16940399e-06+8.81465993e-03j [07] -4.13409086e-06+7.71205494e-03j [08] 1.08701766e-05-7.19410612e-03j [09] -3.59550102e-06+2.79730645e-03j [10] -4.04347280e-07-7.13166771e-05j [11] -3.46031550e-02-4.79217432e-05j [12] 7.37089552e-01-8.08336243e-68j [13] -6.47107728e-01-1.40619867e-05j [14] -1.67849672e-03-2.23282699e-06j [15] 2.75735532e-03+7.24655941e-07j [16] 4.97092850e-02+2.09720061e-05j [17] 4.34912681e-02+2.10906942e-05j [18] -4.05703559e-02-5.92273943e-05j [19] 1.57750963e-02+1.94700833e-05j [20] -4.02182267e-04+2.30082849e-06j
18	Re -1.848636349090e-04 Im -2.753123443455e+00	1.56e-12	[01] 8.71360871e-06+5.45273153e-02j [02] 7.74275353e-06+6.57761475e-02j [03] 7.32303187e-06+7.59601488e-02j [04] -2.17072243e-06+1.26144716e-01j [05] -1.13580391e-05+1.69152164e-01j [06] 2.76688708e-07+1.24639158e-01j [07] -1.07108461e-06+1.23349387e-01j [08] 6.30345558e-06+6.68972555e-02j [09] 2.87295360e-06+1.24318955e-01j [10] 1.75458469e-05-9.09906037e-02j [11] 1.50120429e-01-3.40697581e-05j [12] 1.81089852e-01-3.34763739e-05j [13] 2.09127665e-01-3.42034798e-05j [14] 3.47291975e-01-1.73433038e-05j [15] 4.65696792e-01-7.69535669e-35j [16] 3.43146988e-01-2.38030058e-05j [17] 3.39596089e-01-1.98539878e-05j [18] 1.84176401e-01-2.97210611e-05j [19] 3.42265429e-01-3.08916497e-05j [20] -2.50508367e-01-3.14850302e-05j
19	Re -1.848636349090e-04 Im 2.753123443455e+00	1.56e-12	[01] 8.71360871e-06-5.45273153e-02j [02] 7.74275353e-06-6.57761475e-02j [03] 7.32303187e-06-7.59601488e-02j [04] -2.17072243e-06-1.26144716e-01j [05] -1.13580391e-05-1.69152164e-01j [06] 2.76688708e-07-1.24639158e-01j [07] -1.07108461e-06-1.23349387e-01j [08] 6.30345558e-06-6.68972555e-02j [09] 2.87295360e-06-1.24318955e-01j [10] 1.75458469e-05+9.09906037e-02j [11] 1.50120429e-01+3.40697581e-05j [12] 1.81089852e-01+3.34763739e-05j [13] 2.09127665e-01+3.42034798e-05j [14] 3.47291975e-01+1.73433038e-05j [15] 4.65696792e-01+1.13496270e-51j [16] 3.43146988e-01+2.38030058e-05j [17] 3.39596089e-01+1.98539878e-05j [18] 1.84176401e-01+2.97210611e-05j [19] 3.42265429e-01+3.08916497e-05j [20] -2.50508367e-01+3.14850302e-05j
20	Re -2.238312313567e-10 Im -8.122683918882e-11	5.70e-14	[01] 3.16227766e-01-1.67496702e-15j [02] 3.16227766e-01-1.17202484e-15j [03] 3.16227766e-01-2.00469211e-15j [04] 3.16227766e-01-1.01469063e-15j [05] 3.16227766e-01-9.60135485e-34j [06] 3.16227766e-01-4.97477707e-16j [07] 3.16227766e-01-8.28539525e-16j [08] 3.16227766e-01-1.70378392e-15j [09] 3.16227766e-01-7.55415807e-16j [10] 3.16227766e-01-4.12150811e-15j [11] -7.07815420e-11-2.56863658e-11j [12] -7.07816864e-11-2.56861906e-11j [13] -7.07816737e-11-2.56861396e-11j [14] -7.07820357e-11-2.56861262e-11j [15] -7.07815681e-11-2.56862931e-11j [16] -7.07816821e-11-2.56864594e-11j [17] -7.07813583e-11-2.56862483e-11j [18] -7.07815331e-11-2.56860615e-11j [19] -7.07815189e-11-2.56862669e-11j [20] -7.07819043e-11-2.56856675e-11j

Table S16 39-bus system Base Ablation Without Step-1

Returned 16/20 eigenpairs; Eigenvalue MAE 7.917010e-12; RMSE 2.812498e-11; mean residual 5.199405e-14; mean MAC 1.0000000000. Modal align rate 80.00% (16/20).

Four reference modes remain unmatched under the minimum-total-error assignment and are not enumerated below.

Table S16. Ablation study results for the 39-bus system Without Step-1 optimization (Supplementary to Table IV and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.295455095543e-04 Im 6.545746650406e+00	3.00e-14	[01] -2.58626182e-06+6.88034058e-03j [02] 7.26614087e-08+3.42576885e-03j [03] 5.23258781e-07+5.69731566e-03j [04] -6.93909312e-06-1.37830874e-01j [05] 1.25694925e-06+4.60550776e-02j [06] -5.38143408e-07+3.14834642e-02j [07] 5.35752496e-06+1.56863846e-02j [08] 4.53469936e-06+1.89226056e-02j [09] -7.18852797e-08+1.17917348e-03j [10] 2.42562541e-08+2.64580620e-05j [11] -4.50369655e-02-1.91964000e-05j [12] -2.24242150e-02-6.53323568e-07j [13] -3.72931851e-02+1.54759462e-06j [14] 9.02205983e-01+1.32707307e-33j [15] -3.01464870e-01-6.94957265e-06j [16] -2.06082780e-01-1.38977847e-05j [17] -1.02679101e-01+2.98996235e-05j [18] -1.23862584e-01+2.34471335e-05j [19] -7.71857083e-03-8.59134154e-07j [20] -1.73187779e-04+1.50056160e-07j
2	Re -3.295455095486e-04 Im -6.545746650406e+00	4.46e-14	[01] -2.58626181e-06-6.88034058e-03j [02] 7.26614089e-08-3.42576885e-03j [03] 5.23258781e-07-5.69731566e-03j [04] -6.93909312e-06+1.37830874e-01j [05] 1.25694925e-06-4.60550776e-02j [06] -5.38143410e-07-3.14834642e-02j [07] 5.35752497e-06-1.56863846e-02j [08] 4.53469935e-06-1.89226056e-02j [09] -7.18852789e-08-1.17917348e-03j [10] 2.42562544e-08-2.64580620e-05j [11] -4.50369655e-02+1.91963999e-05j [12] -2.24242150e-02+6.53323568e-07j [13] -3.72931851e-02-1.54759462e-06j [14] 9.02205983e-01+1.12556598e-33j [15] -3.01464870e-01+6.94957265e-06j [16] -2.06082780e-01+1.38977847e-05j [17] -1.02679101e-01-2.98996235e-05j [18] -1.23862584e-01-2.34471334e-05j [19] -7.71857083e-03+8.59134151e-07j [20] -1.73187779e-04-1.50056160e-07j
3	Re -3.093789966957e-04 Im -5.061482055348e+00	2.03e-14	[01] 9.16706913e-07+9.02229536e-03j [02] 1.20586631e-05-7.35063930e-02j [03] 5.79017653e-06-5.88390916e-02j [04] -3.18469090e-06+1.15004987e-02j [05] 8.65304623e-07-7.81564688e-02j [06] -6.60031579e-06+1.07982055e-01j [07] -1.02095819e-05+1.00147607e-01j [08] 3.22472546e-07+8.95079096e-03j [09] 2.24026532e-06-2.46937367e-02j [10] -4.23161141e-07+9.07914990e-04j [11] 4.56661858e-02-7.43120428e-06j [12] -3.72051293e-01-3.82933730e-05j [13] -2.97813008e-01-1.11032954e-05j [14] 5.82095689e-02+1.25612431e-05j [15] -3.95587564e-01+1.98002461e-05j [16] 5.46549237e-01+2.58524256e-33j [17] 5.06895319e-01+2.06920494e-05j [18] 4.53042677e-02-4.40137573e-06j [19] -1.24986906e-01-3.69933925e-06j [20] 4.59539556e-03+1.86093269e-06j
4	Re -3.093789966730e-04 Im 5.061482055348e+00	3.28e-13	[01] 9.16706919e-07-9.02229536e-03j [02] 1.20586631e-05+7.35063930e-02j [03] 5.79017644e-06+5.88390916e-02j [04] -3.18469090e-06-1.15004987e-02j [05] 8.65304642e-07+7.81564688e-02j [06] -6.60031579e-06-1.07982055e-01j [07] -1.02095819e-05-1.00147607e-01j [08] 3.22472569e-07-8.95079096e-03j [09] 2.24026536e-06+2.46937367e-02j [10] -4.23161134e-07-9.07914990e-04j [11] 4.56661858e-02+7.43120430e-06j [12] -3.72051293e-01+3.82933730e-05j [13] -2.97813008e-01+1.11032950e-05j [14] 5.82095689e-02-1.25612431e-05j [15] -3.95587564e-01-1.98002460e-05j [16] 5.46549237e-01-1.32712337e-33j [17] 5.06895319e-01-2.06920494e-05j [18] 4.53042677e-02+4.40137589e-06j [19] -1.24986906e-01+3.69933939e-06j [20] 4.59539556e-03-1.86093268e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
5	Re -3.046730653566e-04 Im -4.200799305801e+00	7.25e-14	[01] -4.35255437e-06+2.22301539e-02j [02] -7.39521131e-06+7.15710686e-04j [03] -7.31955604e-06-9.97049145e-04j [04] 3.26608332e-06-4.89449665e-02j [05] 2.06960170e-05-1.42347439e-01j [06] -3.63309661e-06-2.40940603e-02j [07] -2.92559764e-06-2.32478125e-02j [08] -6.82922264e-06+3.48071648e-02j [09] -1.21650386e-05+1.67730238e-01j [10] 1.79853583e-06-2.95013651e-03j [11] 9.33844162e-02+1.15112782e-05j [12] 3.00655921e-03+3.08477408e-05j [13] -4.18840113e-03+3.10517600e-05j [14] -2.05607982e-01+1.19205245e-06j [15] -5.97973028e-01-4.35703833e-05j [16] -1.01214311e-01+2.26027209e-05j [17] -9.76593939e-02+1.93728308e-05j [18] 1.46217916e-01+1.80833881e-05j [19] 7.04601073e-01+7.73405053e-34j [20] -1.23929319e-02-6.65646093e-06j
6	Re -3.046730653544e-04 Im 4.200799305801e+00	8.96e-15	[01] -4.35255437e-06-2.22301539e-02j [02] -7.39521131e-06-7.15710686e-04j [03] -7.31955605e-06+9.97049145e-04j [04] 3.26608332e-06+4.89449665e-02j [05] 2.06960170e-05+1.42347439e-01j [06] -3.63309661e-06+2.40940603e-02j [07] -2.92559764e-06+2.32478125e-02j [08] -6.82922264e-06-3.48071648e-02j [09] -1.21650386e-05-1.67730238e-01j [10] 1.79853583e-06+2.95013651e-03j [11] 9.33844162e-02-1.15112782e-05j [12] 3.00655921e-03-3.08477408e-05j [13] -4.18840113e-03-3.10517600e-05j [14] -2.05607982e-01-1.19205244e-06j [15] -5.97973028e-01+4.35703833e-05j [16] -1.01214311e-01-2.26027209e-05j [17] -9.76593939e-02-1.93728309e-05j [18] 1.46217916e-01-1.80833881e-05j [19] 7.04601073e-01+5.15644078e-34j [20] -1.23929319e-02+6.65646093e-06j
7	Re -3.005670382991e-04 Im -1.282937057402e-11	1.96e-14	[01] 3.16227751e-01-1.85062342e-16j [02] 3.16227752e-01+9.30130722e-17j [03] 3.16227752e-01+9.69461848e-17j [04] 3.16227753e-01+3.70527371e-19j [05] 3.16227754e-01-3.64569543e-33j [06] 3.16227753e-01+1.01684925e-16j [07] 3.16227753e-01+2.06493631e-16j [08] 3.16227752e-01+1.36860468e-16j [09] 3.16227753e-01-3.81000371e-17j [10] 3.16227745e-01-9.51715803e-17j [11] -9.50476385e-05-4.05721218e-12j [12] -9.50476387e-05-4.05710029e-12j [13] -9.50476387e-05-4.05695044e-12j [14] -9.50476392e-05-4.05686127e-12j [15] -9.50476396e-05-4.05689200e-12j [16] -9.50476391e-05-4.05708297e-12j [17] -9.50476392e-05-4.05686404e-12j [18] -9.50476388e-05-4.05689422e-12j [19] -9.50476390e-05-4.05699058e-12j [20] -9.50476369e-05-4.05718490e-12j
8	Re -3.002929642299e-04 Im 4.583815868253e+00	2.60e-14	[01] 4.53694020e-07-1.79393253e-02j [02] -7.61589955e-06-1.16252744e-01j [03] -3.38949150e-06-1.10279628e-01j [04] -3.86409941e-06+1.70160143e-02j [05] 4.96811761e-07+1.11562490e-01j [06] -7.76020750e-06-2.58689101e-02j [07] -8.11610353e-06-2.45356265e-02j [08] 6.07334291e-07-1.22128577e-02j [09] 1.38569914e-05+7.22661698e-02j [10] 1.51975997e-06+7.38028284e-03j [11] 8.22305639e-02+7.46670302e-06j [12] 5.32881177e-01-9.26939433e-35j [13] 5.05501511e-01+1.75793915e-05j [14] -7.79982751e-02-2.28221096e-05j [15] -5.11381913e-01-3.12241372e-05j [16] 1.18578323e-01-2.78031106e-05j [17] 1.12466796e-01-2.98348482e-05j [18] 5.59814908e-02+6.45134380e-06j [19] -3.31254820e-01+4.18168748e-05j [20] -3.38298581e-02+4.75005286e-06j
9	Re -3.002929642219e-04 Im -4.583815868253e+00	2.16e-14	[01] 4.53694020e-07+1.79393253e-02j [02] -7.61589955e-06+1.16252744e-01j [03] -3.38949150e-06+1.10279628e-01j [04] -3.86409941e-06-1.70160143e-02j [05] 4.96811762e-07-1.11562490e-01j [06] -7.76020751e-06+2.58689101e-02j [07] -8.11610354e-06+2.45356265e-02j [08] 6.07334291e-07+1.22128577e-02j [09] 1.38569914e-05-7.22661698e-02j [10] 1.51975997e-06-7.38028284e-03j [11] 8.22305639e-02-7.46670302e-06j [12] 5.32881177e-01+1.46506573e-34j [13] 5.05501511e-01-1.75793915e-05j [14] -7.79982751e-02+2.28221096e-05j [15] -5.11381913e-01+3.12241372e-05j [16] 1.18578323e-01+2.78031106e-05j [17] 1.12466796e-01+2.98348482e-05j [18] 5.59814908e-02-6.45134380e-06j [19] -3.31254820e-01-4.18168748e-05j [20] -3.38298581e-02-4.75005286e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
10	Re -2.922696776661e-04 Im 5.711763079524e+00	3.63e-14	[01] 8.89755210e-06-1.04938876e-01j [02] -8.96676670e-06+1.33951926e-02j [03] 7.25175648e-06+2.98328985e-02j [04] -1.39306449e-07-4.31481797e-03j [05] 1.52653066e-07+1.67250294e-03j [06] -2.43024774e-06+2.94993935e-02j [07] -4.26096752e-07+2.56934450e-02j [08] -6.21770638e-06-1.21511291e-01j [09] 1.74729666e-07+3.64417140e-02j [10] 2.63647356e-07+3.75177825e-03j [11] 5.99385995e-01+8.14911610e-05j [12] -7.65101642e-02-5.51310556e-05j [13] -1.70398450e-01+3.27010633e-05j [14] 2.46452180e-02+4.65405025e-07j [15] -9.55294059e-03+3.83096251e-07j [16] -1.68493546e-01-2.25027775e-05j [17] -1.46754871e-01-9.94317859e-06j [18] 6.94043705e-01-9.19158895e-35j [19] -2.08146436e-01-9.65279354e-06j [20] -2.14292686e-02+4.09360213e-07j
11	Re -2.922696776563e-04 Im -5.711763079524e+00	1.82e-14	[01] 8.89755210e-06+1.04938876e-01j [02] -8.96676674e-06-1.33951926e-02j [03] 7.25175652e-06-2.98328985e-02j [04] -1.39306449e-07+4.31481797e-03j [05] 1.52653066e-07-1.67250294e-03j [06] -2.43024774e-06-2.94993935e-02j [07] -4.26096753e-07-2.56934450e-02j [08] -6.21770638e-06+1.21511291e-01j [09] 1.74729665e-07-3.64417140e-02j [10] 2.63647356e-07-3.75177825e-03j [11] 5.99385995e-01-8.14911610e-05j [12] -7.65101642e-02+5.51310559e-05j [13] -1.70398450e-01-3.27010635e-05j [14] 2.46452180e-02-4.65405024e-07j [15] -9.55294059e-03-3.83096251e-07j [16] -1.68493546e-01+2.25027775e-05j [17] -1.46754871e-01+9.94317860e-06j [18] 6.94043705e-01+4.02692510e-34j [19] -2.08146436e-01+9.65279354e-06j [20] -2.14292686e-02-4.09360214e-07j
12	Re -2.882560430336e-04 Im -5.639387583526e+00	4.96e-14	[01] 8.81132479e-06-6.13597700e-03j [02] -6.68089717e-06+1.30703829e-01j [03] 8.35884280e-06-1.14747872e-01j [04] 4.11148007e-07-2.97638099e-04j [05] -1.53491408e-07+4.88945872e-04j [06] -4.16940398e-06+8.81465993e-03j [07] -4.13409086e-06+7.71205494e-03j [08] 1.08701766e-05-7.19410612e-03j [09] -3.59550102e-06+2.79730645e-03j [10] -4.04347280e-07-7.13166771e-05j [11] -3.46031550e-02-4.79217432e-05j [12] 7.37089552e-01+1.91574747e-34j [13] -6.47107728e-01-1.40619867e-05j [14] -1.67849672e-03-2.23282699e-06j [15] 2.75735532e-03+7.24655940e-07j [16] 4.97092850e-02+2.09720061e-05j [17] 4.34912681e-02+2.10906942e-05j [18] -4.05703559e-02-5.92273942e-05j [19] 1.57750963e-02+1.94700833e-05j [20] -4.02182267e-04+2.30082849e-06j
13	Re -2.882560430296e-04 Im 5.639387583526e+00	1.71e-15	[01] 8.81132480e-06+6.13597700e-03j [02] -6.68089717e-06-1.30703829e-01j [03] 8.35884280e-06+1.14747872e-01j [04] 4.11148008e-07+2.97638099e-04j [05] -1.53491409e-07-4.88945872e-04j [06] -4.16940399e-06-8.81465993e-03j [07] -4.13409086e-06-7.71205494e-03j [08] 1.08701766e-05+7.19410612e-03j [09] -3.59550102e-06-2.79730645e-03j [10] -4.04347280e-07+7.13166771e-05j [11] -3.46031550e-02+4.79217432e-05j [12] 7.37089552e-01+1.25159183e-65j [13] -6.47107728e-01+1.40619867e-05j [14] -1.67849672e-03+2.23282699e-06j [15] 2.75735532e-03-7.24655941e-07j [16] 4.97092850e-02-2.09720061e-05j [17] 4.34912681e-02-2.10906942e-05j [18] -4.05703559e-02+5.92273943e-05j [19] 1.57750963e-02-1.94700833e-05j [20] -4.02182267e-04-2.30082849e-06j
14	Re -1.848636309393e-04 Im -2.753123443455e+00	3.16e-14	[01] 8.71360906e-06+5.45273153e-02j [02] 7.74275390e-06+6.57761475e-02j [03] 7.32303223e-06+7.59601488e-02j [04] -2.17072216e-06+1.26144716e-01j [05] -1.13580389e-05+1.69152164e-01j [06] 2.76688994e-07+1.24639158e-01j [07] -1.07108433e-06+1.23349387e-01j [08] 6.30345592e-06+6.68972555e-02j [09] 2.87295392e-06+1.24318955e-01j [10] 1.75458472e-05-9.09906037e-02j [11] 1.50120429e-01-3.40697589e-05j [12] 1.81089852e-01-3.34763748e-05j [13] 2.09127665e-01-3.42034806e-05j [14] 3.47291975e-01-1.73433041e-05j [15] 4.65696792e-01-1.30489658e-33j [16] 3.43146988e-01-2.38030063e-05j [17] 3.39596089e-01-1.98539882e-05j [18] 1.84176401e-01-2.97210618e-05j [19] 3.42265429e-01-3.08916502e-05j [20] -2.50508367e-01-3.14850299e-05j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
15	Re -1.848636308992e-04 Im 2.753123443455e+00	8.23e-14	[01] 8.71360907e-06-5.45273153e-02j [02] 7.74275391e-06-6.57761475e-02j [03] 7.32303224e-06-7.59601488e-02j [04] -2.17072215e-06-1.26144716e-01j [05] -1.13580389e-05-1.69152164e-01j [06] 2.76689003e-07-1.24639158e-01j [07] -1.07108432e-06-1.23349387e-01j [08] 6.30345593e-06-6.68972555e-02j [09] 2.87295394e-06-1.24318955e-01j [10] 1.75458472e-05+9.09906037e-02j [11] 1.50120429e-01+3.40697589e-05j [12] 1.81089852e-01+3.34763748e-05j [13] 2.09127665e-01+3.42034807e-05j [14] 3.47291975e-01+1.73433041e-05j [15] 4.65696792e-01-6.98433672e-34j [16] 3.43146988e-01+2.38030063e-05j [17] 3.39596089e-01+1.98539882e-05j [18] 1.84176401e-01+2.97210618e-05j [19] 3.42265429e-01+3.08916503e-05j [20] -2.50508367e-01+3.14850299e-05j
16	Re -6.873899549667e-11 Im -8.616898691581e-11	4.11e-14	[01] 3.16227766e-01-1.45153625e-15j [02] 3.16227766e-01-1.50550122e-15j [03] 3.16227766e-01-1.61136957e-15j [04] 3.16227766e-01-5.31994490e-16j [05] 3.16227766e-01-3.61563258e-33j [06] 3.16227766e-01-8.37305822e-16j [07] 3.16227766e-01-8.33698058e-16j [08] 3.16227766e-01-1.50550122e-15j [09] 3.16227766e-01-3.88062664e-16j [10] 3.16227766e-01-4.48101715e-15j [11] -2.17369826e-11-2.72491299e-11j [12] -2.17372076e-11-2.72491980e-11j [13] -2.17371663e-11-2.72489409e-11j [14] -2.17367299e-11-2.72487095e-11j [15] -2.17376841e-11-2.72493787e-11j [16] -2.17370712e-11-2.72492091e-11j [17] -2.17371634e-11-2.72489334e-11j [18] -2.17371951e-11-2.72488832e-11j [19] -2.17373299e-11-2.72489695e-11j [20] -2.17372598e-11-2.72489102e-11j

Table S17 39-bus system Base Ablation $\widehat{H}$ only

Returned 6/20 eigenpairs; Eigenvalue MAE 3.995904e-12; RMSE 8.026267e-12; mean residual 1.810222e-14; mean MAC 1.0000000000. Modal align rate 30.00% (6/20).

Fourteen reference modes remain unmatched under the minimum-total-error assignment and are not enumerated below.

Table D07. Ablation study results for the 39-bus system using $\widehat{H}$ seeds only (Supplementary to Table IV and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.237419521042e-04 Im 6.868699396043e+00	1.21e-14	[01] 3.93828538e-07+1.04936876e-03j [02] -8.53330568e-08+2.85873671e-04j [03] -1.00936578e-07+4.20212720e-04j [04] -3.40402314e-06+1.05097137e-02j [05] 7.27334513e-07-1.98784341e-03j [06] 9.01174967e-06+8.17620683e-02j [07] -5.56788543e-06-1.18131527e-01j [08] -8.99137280e-07+3.17126648e-05j [09] -2.75864890e-08+2.40086440e-04j [10] -6.64317884e-09-2.69668215e-07j [11] -7.20779871e-03+2.36536515e-06j [12] -1.96358028e-03-6.78676416e-07j [13] -2.88631483e-03-8.29343500e-07j [14] -7.21880632e-02-2.67836469e-05j [15] 1.36538986e-02+5.63939044e-06j [16] -5.61599072e-01+3.54291879e-05j [17] 8.11409953e-01+6.69772136e-33j [18] -2.17824471e-04-6.18617041e-06j [19] -1.64908158e-03-2.67209354e-07j [20] 1.85227205e-06-4.55426948e-08j
2	Re -3.237419521038e-04 Im -6.868699396043e+00	1.84e-14	[01] 3.93828544e-07-1.04936876e-03j [02] -8.53330568e-08-2.85873671e-04j [03] -1.00936578e-07-4.20212720e-04j [04] -3.40402314e-06-1.05097137e-02j [05] 7.27334513e-07+1.98784341e-03j [06] 9.01174967e-06-8.17620683e-02j [07] -5.56788543e-06+1.18131527e-01j [08] -8.99137288e-07-3.17126649e-05j [09] -2.75864889e-08-2.40086440e-04j [10] -6.64317874e-09+2.69668214e-07j [11] -7.20779871e-03-2.36536519e-06j [12] -1.96358028e-03+6.78676416e-07j [13] -2.88631483e-03+8.29343500e-07j [14] -7.21880632e-02+2.67836469e-05j [15] 1.36538986e-02-5.63939044e-06j [16] -5.61599072e-01-3.54291879e-05j [17] 8.11409953e-01+7.66458802e-33j [18] -2.17824471e-04+6.18617047e-06j [19] -1.64908158e-03+2.67209352e-07j [20] 1.85227205e-06+4.55426952e-08j
3	Re -3.151425852267e-04 Im 6.810560067478e+00	2.72e-14	[01] 1.62522531e-05+8.22907598e-02j [02] -7.49362789e-07-4.74246726e-04j [03] -7.81207312e-07-3.55683280e-04j [04] -3.96671913e-06+4.84289946e-04j [05] 1.01446636e-06+5.96884599e-04j [06] -5.05515970e-07+1.84356374e-03j [07] 2.41553812e-07-1.53385690e-03j [08] -5.53544336e-06-1.19626706e-01j [09] -7.89313400e-07+3.86352325e-03j [10] -3.12960119e-07-5.93535159e-04j [11] -5.60446168e-01+8.47536234e-05j [12] 3.22988605e-03-4.95412495e-06j [13] 2.42240259e-03-5.20836838e-06j [14] -3.29828452e-03-2.71681993e-05j [15] -4.06511873e-03+6.72098031e-06j [16] -1.25557015e-02-4.02383234e-06j [17] 1.04464245e-02+2.12850039e-06j [18] 8.14724871e-01-2.82704888e-34j [19] -2.63127569e-02-6.59322703e-06j [20] 4.04230695e-03-1.94438548e-06j
4	Re -3.151425852224e-04 Im -6.810560067478e+00	1.03e-14	[01] 1.62522531e-05-8.22907598e-02j [02] -7.49362789e-07+4.74246726e-04j [03] -7.81207312e-07+3.55683280e-04j [04] -3.96671912e-06-4.84289946e-04j [05] 1.01446636e-06-5.96884599e-04j [06] -5.05515877e-07-1.84356374e-03j [07] 2.41553676e-07+1.53385690e-03j [08] -5.53544336e-06+1.19626706e-01j [09] -7.89313400e-07-3.86352325e-03j [10] -3.12960119e-07+5.93535159e-04j [11] -5.60446168e-01-8.47536234e-05j [12] 3.22988605e-03+4.95412495e-06j [13] 2.42240259e-03+5.20836837e-06j [14] -3.29828452e-03+2.71681992e-05j [15] -4.06511873e-03-6.72098029e-06j [16] -1.25557015e-02+4.02383169e-06j [17] 1.04464245e-02-2.12849945e-06j [18] 8.14724871e-01+1.24544215e-33j [19] -2.63127569e-02+6.59322703e-06j [20] 4.04230695e-03+1.94438548e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
5	Re -3.005670432294e-04 Im 4.033044235121e-12	1.93e-14	[01] 3.16227751e-01-9.25770038e-17j [02] 3.16227752e-01+6.12688349e-17j [03] 3.16227752e-01+4.25690904e-17j [04] 3.16227753e-01-1.86116275e-16j [05] 3.16227754e-01+1.20918328e-33j [06] 3.16227753e-01+8.28216189e-18j [07] 3.16227753e-01-3.79599948e-17j [08] 3.16227752e-01+2.01752255e-16j [09] 3.16227753e-01-7.31999684e-17j [10] 3.16227745e-01-1.30961881e-17j [11] -9.50476400e-05+1.27553879e-12j [12] -9.50476403e-05+1.27539379e-12j [13] -9.50476403e-05+1.27529511e-12j [14] -9.50476407e-05+1.27534176e-12j [15] -9.50476411e-05+1.27535906e-12j [16] -9.50476407e-05+1.27545734e-12j [17] -9.50476408e-05+1.27512034e-12j [18] -9.50476403e-05+1.27539922e-12j [19] -9.50476406e-05+1.27526231e-12j [20] -9.50476384e-05+1.27543671e-12j
6	Re 1.849762197594e-11 Im 9.547571088691e-12	2.12e-14	[01] 3.16227766e-01-3.73600073e-16j [02] 3.16227766e-01-5.80666692e-16j [03] 3.16227766e-01-3.77093642e-16j [04] 3.16227766e-01-5.18168403e-16j [05] 3.16227766e-01-5.55208000e-16j [06] 3.16227766e-01-4.70841075e-16j [07] 3.16227766e-01-5.03190417e-16j [08] 3.16227766e-01-4.32604793e-16j [09] 3.16227766e-01-4.93906396e-16j [10] 3.16227766e-01-2.20081955e-34j [11] 5.84944766e-12+3.01909168e-12j [12] 5.84980462e-12+3.01941697e-12j [13] 5.84946985e-12+3.01925096e-12j [14] 5.84953987e-12+3.01941459e-12j [15] 5.84941682e-12+3.01922562e-12j [16] 5.84934007e-12+3.01906902e-12j [17] 5.84944237e-12+3.01904645e-12j [18] 5.84936645e-12+3.01918169e-12j [19] 5.84936272e-12+3.01910694e-12j [20] 5.84942631e-12+3.01926684e-12j

Table S18 39-bus system Base Ablation $\widehat{K}$ only

Returned 17/20 eigenpairs; Eigenvalue MAE 2.884288e-12; RMSE 6.152357e-12; mean residual 1.253263e-12; mean MAC 1.0000000000. Modal align rate 85.00% (17/20).

Three reference modes remain unmatched under the minimum-total-error assignment and are not enumerated below.

Table S18. Ablation study results for the 39-bus system using $\widehat{K}$ seeds only (Supplementary to Table IV and Section IV-C in the main paper).

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
1	Re -3.295455098032e-04 Im -6.545746650406e+00	9.86e-14	[01] -2.58626181e-06-6.88034058e-03j [02] 7.26614087e-08-3.42576885e-03j [03] 5.23258780e-07-5.69731566e-03j [04] -6.93909312e-06+1.37830874e-01j [05] 1.25694925e-06-4.60550776e-02j [06] -5.38143407e-07-3.14834642e-02j [07] 5.35752496e-06-1.56863846e-02j [08] 4.53469935e-06-1.89226056e-02j [09] -7.18852805e-08-1.17917348e-03j [10] 2.42562544e-08-2.64580620e-05j [11] -4.50369655e-02+1.91963999e-05j [12] -2.24242150e-02+6.53323568e-07j [13] -3.72931851e-02-1.54759462e-06j [14] 9.02205983e-01-1.90193319e-33j [15] -3.01464870e-01+6.94957265e-06j [16] -2.06082780e-01+1.38977846e-05j [17] -1.02679101e-01-2.98996234e-05j [18] -1.23862584e-01-2.34471334e-05j [19] -7.71857083e-03+8.59134151e-07j [20] -1.73187779e-04-1.50056160e-07j
2	Re -3.295455095673e-04 Im 6.545746650406e+00	1.73e-14	[01] -2.58626181e-06+6.88034058e-03j [02] 7.26614088e-08+3.42576885e-03j [03] 5.23258781e-07+5.69731566e-03j [04] -6.93909312e-06-1.37830874e-01j [05] 1.25694925e-06+4.60550776e-02j [06] -5.38143410e-07+3.14834642e-02j [07] 5.35752497e-06+1.56863846e-02j [08] 4.53469935e-06+1.89226056e-02j [09] -7.18852791e-08+1.17917348e-03j [10] 2.42562543e-08+2.64580620e-05j [11] -4.50369655e-02-1.91963999e-05j [12] -2.24242150e-02-6.53323568e-07j [13] -3.72931851e-02+1.54759462e-06j [14] 9.02205983e-01-2.65808464e-34j [15] -3.01464870e-01-6.94957265e-06j [16] -2.06082780e-01-1.38977847e-05j [17] -1.02679101e-01+2.98996235e-05j [18] -1.23862584e-01+2.34471334e-05j [19] -7.71857083e-03-8.59134151e-07j [20] -1.73187779e-04+1.50056160e-07j
3	Re -3.237419521098e-04 Im -6.868699396043e+00	1.75e-14	[01] 3.93828542e-07-1.04936876e-03j [02] -8.53330568e-08-2.85873671e-04j [03] -1.00936578e-07-4.20212720e-04j [04] -3.40402314e-06-1.05097137e-02j [05] 7.27334513e-07+1.98784341e-03j [06] 9.01174967e-06-8.17620683e-02j [07] -5.56788543e-06+1.18131527e-01j [08] -8.99137286e-07-3.17126648e-05j [09] -2.75864890e-08-2.40086440e-04j [10] -6.64317880e-09+2.69668214e-07j [11] -7.20779871e-03-2.36536518e-06j [12] -1.96358028e-03+6.78676416e-07j [13] -2.88631483e-03+8.29343499e-07j [14] -7.21880632e-02+2.67836469e-05j [15] 1.36538986e-02-5.63939044e-06j [16] -5.61599072e-01-3.54291879e-05j [17] 8.11409953e-01+3.04493308e-34j [18] -2.17824471e-04+6.18617046e-06j [19] -1.64908158e-03+2.67209352e-07j [20] 1.85227205e-06+4.55426951e-08j
4	Re -3.237419521095e-04 Im 6.868699396043e+00	1.75e-14	[01] 3.93828535e-07+1.04936876e-03j [02] -8.53330567e-08+2.85873671e-04j [03] -1.00936578e-07+4.20212720e-04j [04] -3.40402314e-06+1.05097137e-02j [05] 7.27334513e-07-1.98784341e-03j [06] 9.01174967e-06+8.17620683e-02j [07] -5.56788543e-06-1.18131527e-01j [08] -8.99137275e-07+3.17126648e-05j [09] -2.75864892e-08+2.40086440e-04j [10] -6.64317857e-09-2.69668215e-07j [11] -7.20779871e-03+2.36536513e-06j [12] -1.96358028e-03-6.78676416e-07j [13] -2.88631483e-03-8.29343499e-07j [14] -7.21880632e-02-2.67836469e-05j [15] 1.36538986e-02+5.63939044e-06j [16] -5.61599072e-01+3.54291879e-05j [17] 8.11409953e-01-1.45144738e-34j [18] -2.17824471e-04-6.18617038e-06j [19] -1.64908158e-03-2.67209354e-07j [20] 1.85227205e-06-4.55426952e-08j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value	
5	Re -3.093790131005e-04 Im -5.061482055348e+00	6.67e-12	[01] 9.16706989e-07+9.02229536e-03j [03] 5.79017614e-06-5.88390916e-02j [05] 8.65304390e-07-7.81564688e-02j [07] -1.02095812e-05+1.00147607e-01j [09] 2.24026501e-06-2.46937367e-02j [11] 4.56661858e-02-7.43120454e-06j [13] -2.97813008e-01-1.11032948e-05j [15] -3.95587564e-01+1.98002461e-05j [17] 5.06895319e-01+2.06920472e-05j [19] -1.24986906e-01-3.69933836e-06j	[02] 1.20586623e-05-7.35063930e-02j [04] -3.18469076e-06+1.15004987e-02j [06] -6.60031534e-06+1.07982055e-01j [08] 3.22472624e-07+8.95079096e-03j [10] -4.23161002e-07+9.07914990e-04j [12] -3.72051293e-01-3.82933700e-05j [14] 5.82095689e-02+1.25612425e-05j [16] 5.46549237e-01+1.37556405e-36j [18] 4.53042677e-02-4.40137610e-06j [20] 4.59539556e-03+1.86093251e-06j
6	Re -3.093789966933e-04 Im 5.061482055348e+00	9.27e-16	[01] 9.16706913e-07-9.02229536e-03j [03] 5.79017653e-06+5.88390916e-02j [05] 8.65304623e-07+7.81564688e-02j [07] -1.02095819e-05-1.00147607e-01j [09] 2.24026532e-06+2.46937367e-02j [11] 4.56661858e-02+7.43120428e-06j [13] -2.97813008e-01+1.11032954e-05j [15] -3.95587564e-01-1.98002461e-05j [17] 5.06895319e-01-2.06920494e-05j [19] -1.24986906e-01+3.69933925e-06j	[02] 1.20586631e-05+7.35063930e-02j [04] -3.18469090e-06-1.15004987e-02j [06] -6.60031579e-06-1.07982055e-01j [08] 3.22472545e-07-8.95079096e-03j [10] -4.23161141e-07-9.07914990e-04j [12] -3.72051293e-01+3.82933730e-05j [14] 5.82095689e-02-1.25612431e-05j [16] 5.46549237e-01-2.33632255e-67j [18] 4.53042677e-02+4.40137573e-06j [20] 4.59539556e-03-1.86093269e-06j
7	Re -3.046730835365e-04 Im 4.200799305801e+00	8.89e-12	[01] -4.35255422e-06-2.22301539e-02j [03] -7.31955580e-06+9.97049145e-04j [05] 2.06960165e-05+1.42347439e-01j [07] -2.92559724e-06+2.32478125e-02j [09] -1.21650377e-05-1.67730238e-01j [11] 9.33844162e-02-1.15112782e-05j [13] -4.18840113e-03-3.10517589e-05j [15] -5.97973028e-01+4.35703828e-05j [17] -9.76593939e-02-1.93728290e-05j [19] 7.04601073e-01-1.59171867e-33j	[02] -7.39521113e-06-7.15710686e-04j [04] 3.26608342e-06+4.89449665e-02j [06] -3.63309616e-06+2.40940603e-02j [08] -6.82922228e-06-3.48071648e-02j [10] 1.79853527e-06+2.95013651e-03j [12] 3.00655921e-03-3.08477400e-05j [14] -2.05607982e-01-1.19205146e-06j [16] -1.01214311e-01-2.26027186e-05j [18] 1.46217916e-01-1.80833868e-05j [20] -1.23929319e-02+6.65646014e-06j
8	Re -3.046730678263e-04 Im -4.200799305801e+00	1.21e-12	[01] -4.35255435e-06+2.22301539e-02j [03] -7.31955601e-06-9.97049145e-04j [05] 2.06960169e-05-1.42347439e-01j [07] -2.92559759e-06-2.32478125e-02j [09] -1.21650385e-05+1.67730238e-01j [11] 9.33844162e-02+1.15112782e-05j [13] -4.18840113e-03+3.10517598e-05j [15] -5.97973028e-01-4.35703832e-05j [17] -9.76593939e-02+1.93728306e-05j [19] 7.04601073e-01-1.21704382e-34j	[02] -7.39521129e-06+7.15710686e-04j [04] 3.26608333e-06-4.89449665e-02j [06] -3.63309655e-06-2.40940603e-02j [08] -6.82922259e-06+3.48071648e-02j [10] 1.79853575e-06-2.95013651e-03j [12] 3.00655921e-03+3.08477407e-05j [14] -2.05607982e-01+1.19205231e-06j [16] -1.01214311e-01+2.26027206e-05j [18] 1.46217916e-01+1.80833880e-05j [20] -1.23929319e-02-6.65646082e-06j
9	Re -3.005670471533e-04 Im -1.517917908053e-21	2.47e-15	[01] 3.16227751e-01+3.53591796e-19j [03] 3.16227752e-01+2.23595487e-19j [05] 3.16227754e-01+3.57071765e-69j [07] 3.16227753e-01+6.78212020e-21j [09] 3.16227753e-01+4.67558554e-19j [11] -9.50476413e-05-1.82525330e-22j [13] -9.50476415e-05-2.62675674e-22j [15] -9.50476424e-05-2.33453760e-22j [17] -9.50476420e-05-1.47662360e-22j [19] -9.50476418e-05-1.64547849e-22j	[02] 3.16227752e-01-3.99104666e-20j [04] 3.16227753e-01+2.91918872e-19j [06] 3.16227753e-01+3.44316592e-19j [08] 3.16227752e-01+5.34714582e-19j [10] 3.16227745e-01+5.83302123e-19j [12] -9.50476415e-05-1.07384109e-22j [14] -9.50476420e-05-1.52093119e-22j [16] -9.50476419e-05-1.93155720e-22j [18] -9.50476416e-05-1.72926011e-22j [20] -9.50476397e-05-1.66777902e-22j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
10	Re -3.002929663520e-04 Im -4.583815868253e+00	9.02e-13	[01] 4.53693982e-07+1.79393253e-02j [02] -7.61589952e-06+1.16252744e-01j [03] -3.38949150e-06+1.10279628e-01j [04] -3.86409938e-06-1.70160143e-02j [05] 4.96811792e-07-1.11562490e-01j [06] -7.76020744e-06+2.58689101e-02j [07] -8.11610347e-06+2.45356265e-02j [08] 6.07334259e-07+1.22128577e-02j [09] 1.38569914e-05-7.22661698e-02j [10] 1.51975988e-06-7.38028284e-03j [11] 8.22305639e-02-7.46670283e-06j [12] 5.32881177e-01+3.57112399e-35j [13] 5.05501511e-01-1.75793913e-05j [14] -7.79982751e-02+2.28221094e-05j [15] -5.11381913e-01+3.12241369e-05j [16] 1.18578323e-01+2.78031104e-05j [17] 1.12466796e-01+2.98348479e-05j [18] 5.59814908e-02-6.45134366e-06j [19] -3.31254820e-01-4.18168749e-05j [20] -3.38298581e-02-4.75005278e-06j
11	Re -3.002929637752e-04 Im 4.583815868253e+00	1.90e-13	[01] 4.53694029e-07-1.79393253e-02j [02] -7.61589956e-06-1.16252744e-01j [03] -3.38949150e-06-1.10279628e-01j [04] -3.86409942e-06+1.70160143e-02j [05] 4.96811754e-07+1.11562490e-01j [06] -7.76020752e-06-2.58689101e-02j [07] -8.11610355e-06-2.45356265e-02j [08] 6.07334299e-07-1.22128577e-02j [09] 1.38569914e-05+7.22661698e-02j [10] 1.51975999e-06+7.38028284e-03j [11] 8.22305639e-02+7.46670307e-06j [12] 5.32881177e-01+3.57298654e-35j [13] 5.05501511e-01+1.75793916e-05j [14] -7.79982751e-02-2.28221096e-05j [15] -5.11381913e-01-3.12241373e-05j [16] 1.18578323e-01-2.78031107e-05j [17] 1.12466796e-01-2.98348482e-05j [18] 5.59814908e-02+6.45134384e-06j [19] -3.31254820e-01+4.18168748e-05j [20] -3.38298581e-02+4.75005288e-06j
12	Re -2.922696776643e-04 Im -5.711763079524e+00	4.99e-14	[01] 8.89755210e-06+1.04938876e-01j [02] -8.96676667e-06-1.33951926e-02j [03] 7.25175645e-06-2.98328985e-02j [04] -1.39306449e-07+4.31481797e-03j [05] 1.52653066e-07-1.67250294e-03j [06] -2.43024774e-06-2.94993935e-02j [07] -4.26096750e-07-2.56934450e-02j [08] -6.21770638e-06+1.21511291e-01j [09] 1.74729665e-07-3.64417140e-02j [10] 2.63647356e-07-3.75177825e-03j [11] 5.99385995e-01-8.14911610e-05j [12] -7.65101642e-02+5.51310555e-05j [13] -1.70398450e-01-3.27010631e-05j [14] 2.46452180e-02-4.65405028e-07j [15] -9.55294059e-03-3.83096251e-07j [16] -1.68493546e-01+2.25027775e-05j [17] -1.46754871e-01+9.94317858e-06j [18] 6.94043705e-01+1.02314727e-33j [19] -2.08146436e-01+9.65279354e-06j [20] -2.14292686e-02-4.09360213e-07j
13	Re -2.922696776592e-04 Im 5.711763079524e+00	1.36e-14	[01] 8.89755210e-06-1.04938876e-01j [02] -8.96676674e-06+1.33951926e-02j [03] 7.25175651e-06+2.98328985e-02j [04] -1.39306449e-07-4.31481797e-03j [05] 1.52653066e-07+1.67250294e-03j [06] -2.43024774e-06+2.94993935e-02j [07] -4.26096753e-07+2.56934450e-02j [08] -6.21770638e-06-1.21511291e-01j [09] 1.74729665e-07+3.64417140e-02j [10] 2.63647356e-07+3.75177825e-03j [11] 5.99385995e-01+8.14911610e-05j [12] -7.65101642e-02-5.51310558e-05j [13] -1.70398450e-01+3.27010635e-05j [14] 2.46452180e-02+4.65405024e-07j [15] -9.55294059e-03+3.83096251e-07j [16] -1.68493546e-01-2.25027775e-05j [17] -1.46754871e-01-9.94317860e-06j [18] 6.94043705e-01-5.60274835e-34j [19] -2.08146436e-01-9.65279354e-06j [20] -2.14292686e-02+4.09360214e-07j
14	Re -2.882560430318e-04 Im 5.639387583526e+00	9.56e-14	[01] 8.81132440e-06+6.13597700e-03j [02] -6.68089717e-06-1.30703829e-01j [03] 8.35884296e-06+1.14747872e-01j [04] 4.11147992e-07+2.97638099e-04j [05] -1.53491401e-07-4.88945872e-04j [06] -4.16940388e-06-8.81465993e-03j [07] -4.13409077e-06-7.71205494e-03j [08] 1.08701761e-05+7.19410612e-03j [09] -3.59550088e-06-2.79730645e-03j [10] -4.04347270e-07+7.13166771e-05j [11] -3.46031550e-02+4.79217409e-05j [12] 7.37089552e-01+1.34904860e-34j [13] -6.47107728e-01+1.40619876e-05j [14] -1.67849672e-03+2.23282690e-06j [15] 2.75735532e-03-7.24655898e-07j [16] 4.97092850e-02-2.09720054e-05j [17] 4.34912681e-02-2.10906937e-05j [18] -4.05703559e-02+5.92273917e-05j [19] 1.57750963e-02-1.94700825e-05j [20] -4.02182267e-04-2.30082841e-06j

ID	Eigenvalue real and imaginary parts	Residual	Complete right eigenvector component index followed by complex value
15	Re -2.882560430295e-04 Im -5.639387583526e+00	1.96e-15	[01] 8.81132480e-06-6.13597700e-03j [02] -6.68089717e-06+1.30703829e-01j [03] 8.35884280e-06-1.14747872e-01j [04] 4.11148008e-07-2.97638099e-04j [05] -1.53491409e-07+4.88945872e-04j [06] -4.16940399e-06+8.81465993e-03j [07] -4.13409086e-06+7.71205494e-03j [08] 1.08701766e-05-7.19410612e-03j [09] -3.59550102e-06+2.79730645e-03j [10] -4.04347280e-07-7.13166771e-05j [11] -3.46031550e-02-4.79217432e-05j [12] 7.37089552e-01-8.08336243e-68j [13] -6.47107728e-01-1.40619867e-05j [14] -1.67849672e-03-2.23282699e-06j [15] 2.75735532e-03+7.24655941e-07j [16] 4.97092850e-02+2.09720061e-05j [17] 4.34912681e-02+2.10906942e-05j [18] -4.05703559e-02-5.92273943e-05j [19] 1.57750963e-02+1.94700833e-05j [20] -4.02182267e-04+2.30082849e-06j
16	Re -1.848636349090e-04 Im -2.753123443455e+00	1.56e-12	[01] 8.71360871e-06+5.45273153e-02j [02] 7.74275353e-06+6.57761475e-02j [03] 7.32303187e-06+7.59601488e-02j [04] -2.17072243e-06+1.26144716e-01j [05] -1.13580391e-05+1.69152164e-01j [06] 2.76688708e-07+1.24639158e-01j [07] -1.07108461e-06+1.23349387e-01j [08] 6.30345558e-06+6.68972555e-02j [09] 2.87295360e-06+1.24318955e-01j [10] 1.75458469e-05-9.09906037e-02j [11] 1.50120429e-01-3.40697581e-05j [12] 1.81089852e-01-3.34763739e-05j [13] 2.09127665e-01-3.42034798e-05j [14] 3.47291975e-01-1.73433038e-05j [15] 4.65696792e-01-7.69535669e-35j [16] 3.43146988e-01-2.38030058e-05j [17] 3.39596089e-01-1.98539878e-05j [18] 1.84176401e-01-2.97210611e-05j [19] 3.42265429e-01-3.08916497e-05j [20] -2.50508367e-01-3.14850302e-05j
17	Re -1.848636349090e-04 Im 2.753123443455e+00	1.56e-12	[01] 8.71360871e-06-5.45273153e-02j [02] 7.74275353e-06-6.57761475e-02j [03] 7.32303187e-06-7.59601488e-02j [04] -2.17072243e-06-1.26144716e-01j [05] -1.13580391e-05-1.69152164e-01j [06] 2.76688708e-07-1.24639158e-01j [07] -1.07108461e-06-1.23349387e-01j [08] 6.30345558e-06-6.68972555e-02j [09] 2.87295360e-06-1.24318955e-01j [10] 1.75458469e-05+9.09906037e-02j [11] 1.50120429e-01+3.40697581e-05j [12] 1.81089852e-01+3.34763739e-05j [13] 2.09127665e-01+3.42034798e-05j [14] 3.47291975e-01+1.73433038e-05j [15] 4.65696792e-01+1.13496270e-51j [16] 3.43146988e-01+2.38030058e-05j [17] 3.39596089e-01+1.98539878e-05j [18] 1.84176401e-01+2.97210611e-05j [19] 3.42265429e-01+3.08916497e-05j [20] -2.50508367e-01+3.14850302e-05j

5 Complete reference eigenpairs

This section details the complete set of classical reference eigenpairs for the evaluated state matrices across all operating conditions. Within each table, the eigenpairs are sorted in ascending order of their real parts, followed by their imaginary parts.

Table S19 9-bus system Base eig reference

Complete reference spectrum with 6 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S19. Classical reference eigenpairs (MATLAB eig) for the 9-bus system under the Base scenario (Serving as the ground-truth benchmark for Table I and Table III in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -5.486540295405e-07 Im 1.692711865184e-36	[01] 5.77350269e-01+0.00000000e+00j [02] 5.77350269e-01-1.19695920e-16j [03] 5.77350269e-01-7.80625564e-17j [04] -3.16765550e-07+1.79452537e-15j [05] -3.16765549e-07-7.20293319e-15j [06] -3.16765550e-07+5.15421033e-15j
2	Re -3.181482854941e-15 Im 8.689799888785e+00	[01] -3.46944695e-17+3.60131621e-02j [02] 6.93889390e-18-9.41468493e-02j [03] 6.93889390e-17-5.39366471e-02j [04] -3.12947172e-01-1.38777878e-16j [05] 8.18117281e-01+0.00000000e+00j [06] 4.68698670e-01+5.55111512e-16j
3	Re -1.179768097869e-15 Im -8.689799888785e+00	[01] -4.59701721e-17-3.60131621e-02j [02] 4.51028104e-17+9.41468493e-02j [03] 8.50014503e-17+5.39366471e-02j [04] -3.12947172e-01+1.94289029e-16j [05] 8.18117281e-01+0.00000000e+00j [06] 4.68698670e-01-7.91033905e-16j
4	Re 0.000000000000e+00 Im 1.336021110593e+01	[01] -1.46328274e-17+2.97384465e-03j [02] 4.35981617e-17+2.21449322e-02j [03] 1.85531623e-17-7.12175300e-02j [04] -3.97311924e-02+2.98772236e-17j [05] -2.95860969e-01+1.93725005e-16j [06] 9.51481235e-01+0.00000000e+00j
5	Re 8.247132617866e-16 Im -1.336021110593e+01	[01] -2.47198095e-17-2.97384465e-03j [02] 1.00613962e-16-2.21449322e-02j [03] 1.38777878e-17+7.12175300e-02j [04] -3.97311924e-02-1.21430643e-16j [05] -2.95860969e-01-5.55111512e-17j [06] 9.51481235e-01+0.00000000e+00j
6	Re 5.486540275200e-07 Im 0.000000000000e+00	[01] 5.77350269e-01+0.00000000e+00j [02] 5.77350269e-01-1.21367974e-16j [03] 5.77350269e-01-6.62364188e-17j [04] 3.16765552e-07+1.87636238e-15j [05] 3.16765553e-07-6.65293540e-15j [06] 3.16765552e-07+5.64491489e-15j

Table S20 9-bus system Light load eig reference

Complete reference spectrum with 6 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S20. Classical reference eigenpairs (MATLAB eig) for the 9-bus system under the Light load scenario (Serving as the ground-truth benchmark for Table I in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -3.272983483189e-16 Im 4.138778984472e-07	[01] 5.77350269e-01-2.47820400e-17j [02] 5.77350269e-01-3.93443350e-16j [03] 5.77350269e-01+0.00000000e+00j [04] 2.34189668e-16+2.38952516e-07j [05] 5.88314914e-16+2.38952515e-07j [06] 2.46848205e-16+2.38952522e-07j
2	Re -3.272983415996e-16 Im -4.138778984472e-07	[01] 5.77350269e-01+8.32667268e-17j [02] 5.77350269e-01-3.33066907e-16j [03] 5.77350269e-01+0.00000000e+00j [04] 3.28937542e-16-2.38952516e-07j [05] 1.15825747e-16-2.38952518e-07j [06] 1.04671494e-15-2.38952510e-07j
3	Re -3.115108048992e-16 Im -1.337096006609e+01	[01] -1.20209361e-17-2.94927630e-03j [02] 5.16101735e-17-2.22528967e-02j [03] 2.23953199e-17+7.11223224e-02j [04] -3.94346556e-02+3.83344468e-17j [05] -2.97542593e-01+3.41161478e-16j [06] 9.50973732e-01+0.00000000e+00j
4	Re 1.193489751472e-15 Im 8.763042094030e+00	[01] 4.19586241e-17+3.49476907e-02j [02] -1.63714528e-17-9.35685399e-02j [03] 1.25333771e-16-5.36522151e-02j [04] -3.06248085e-01-1.60461922e-17j [05] 8.19945054e-01+0.00000000e+00j [06] 4.70156619e-01+6.64832772e-16j
5	Re 1.776356839400e-15 Im 1.337096006609e+01	[01] -2.87545723e-18+2.94927630e-03j [02] 9.30172737e-18+2.22528967e-02j [03] 2.72034696e-17-7.11223224e-02j [04] -3.94346556e-02+6.01814407e-17j [05] -2.97542593e-01-4.86862479e-16j [06] 9.50973732e-01+0.00000000e+00j
6	Re 4.876056949562e-15 Im -8.763042094030e+00	[01] 5.04247184e-17-3.49476907e-02j [02] -6.23755062e-18+9.35685399e-02j [03] 5.58423411e-17+5.36522151e-02j [04] -3.06248085e-01-1.18530403e-16j [05] 8.19945054e-01+0.00000000e+00j [06] 4.70156619e-01-6.53706147e-17j

Table S21 9-bus system Heavy load eig reference

Complete reference spectrum with 6 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S21. Classical reference eigenpairs (MATLAB eig) for the 9-bus system under the Heavy load scenario (Serving as the ground-truth benchmark for Table I in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -5.766802135986e-07 Im 0.000000000000e+00	[01] 5.77350269e-01+3.21743118e-16j [02] 5.77350269e-01+2.46238851e-16j [03] 5.77350269e-01+0.00000000e+00j [04] -3.32946476e-07+5.86441109e-16j [05] -3.32946480e-07-1.59758718e-15j [06] -3.32946477e-07+4.26023053e-16j
2	Re -7.596789450752e-15 Im 1.334671703959e+01	[01] -4.03913230e-17+3.00182926e-03j [02] -4.34987335e-17+2.20247681e-02j [03] 3.97495084e-17-7.13322165e-02j [04] -4.00645657e-02+1.40494605e-16j [05] -2.93958347e-01-4.04255196e-16j [06] 9.52050909e-01+0.00000000e+00j
3	Re 1.146910995373e-15 Im 8.608272719265e+00	[01] 1.38777878e-17+3.72566674e-02j [02] -1.38777878e-17-9.47926988e-02j [03] -1.04083409e-17-5.42350988e-02j [04] -3.20715553e-01+2.77555756e-17j [05] 8.16001403e-01+0.00000000e+00j [06] 4.66870521e-01-3.88578059e-16j
4	Re 1.776356839400e-15 Im -1.334671703959e+01	[01] -4.52368304e-17-3.00182926e-03j [02] -6.56507612e-18-2.20247681e-02j [03] 4.13956022e-17+7.13322165e-02j [04] -4.00645657e-02-7.65627778e-17j [05] -2.93958347e-01+8.40879224e-17j [06] 9.52050909e-01+0.00000000e+00j
5	Re 2.619432448725e-15 Im -8.608272719265e+00	[01] 1.36744999e-17-3.72566674e-02j [02] -3.80012861e-17+9.47926988e-02j [03] 4.88162028e-17+5.42350988e-02j [04] -3.20715553e-01-3.02492406e-17j [05] 8.16001403e-01+0.00000000e+00j [06] 4.66870521e-01-2.27248775e-16j
6	Re 5.766802132930e-07 Im 3.864929481820e-57	[01] 5.77350269e-01+3.46944695e-16j [02] 5.77350269e-01+2.35922393e-16j [03] 5.77350269e-01+0.00000000e+00j [04] 3.32946477e-07+5.17432416e-16j [05] 3.32946474e-07-1.60164675e-15j [06] 3.32946476e-07-1.24976006e-17j

Table S22 9-bus system Reactive stress eig reference

Complete reference spectrum with 6 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S22. Classical reference eigenpairs (MATLAB eig) for the 9-bus system under the Reactive stress scenario (Serving as the ground-truth benchmark for Table I in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -4.281297538711e-15 Im -8.671194843856e+00	[01] -7.28583860e-17-3.60726492e-02j [02] -4.85722573e-17+9.43233092e-02j [03] -1.38777878e-16+5.41019852e-02j [04] -3.12792970e-01-1.11022302e-16j [05] 8.17895792e-01+0.00000000e+00j [06] 4.69128855e-01+7.21644966e-16j
2	Re -3.405676753578e-15 Im 8.671194843856e+00	[01] -8.32667268e-17+3.60726492e-02j [02] -6.93889390e-18-9.43233092e-02j [03] -3.81639165e-17-5.41019852e-02j [04] -3.12792970e-01-5.55111512e-17j [05] 8.17895792e-01+0.00000000e+00j [06] 4.69128855e-01-5.55111512e-17j
3	Re 0.000000000000e+00 Im -1.337660277578e+01	[01] -3.07800341e-17-2.96343391e-03j [02] 6.76184850e-17-2.21240793e-02j [03] 3.94172155e-18+7.11291213e-02j [04] -3.96406782e-02+1.02195999e-16j [05] -2.95945020e-01-6.46648877e-16j [06] 9.51466002e-01+0.00000000e+00j
4	Re 8.037413111274e-16 Im 1.337660277578e+01	[01] -3.79964580e-17+2.96343391e-03j [02] 7.67973744e-17+2.21240793e-02j [03] 2.42192689e-17-7.11291213e-02j [04] -3.96406782e-02-6.32460693e-17j [05] -2.95945020e-01+3.31616478e-16j [06] 9.51466002e-01+0.00000000e+00j
5	Re 1.432779735445e-15 Im 6.153366784303e-07	[01] 5.77350269e-01+0.00000000e+00j [02] 5.77350269e-01+1.26089536e-16j [03] 5.77350269e-01+2.65300248e-17j [04] -1.84017250e-15+3.55264796e-07j [05] 2.07341287e-15+3.55264799e-07j [06] -1.10007339e-15+3.55264801e-07j
6	Re 1.432779759046e-15 Im -6.153366784303e-07	[01] 5.77350269e-01+0.00000000e+00j [02] 5.77350269e-01+1.94289029e-16j [03] 5.77350269e-01+5.55111512e-17j [04] -1.85887741e-15-3.55264799e-07j [05] 2.06669599e-15-3.55264794e-07j [06] -1.82592290e-15-3.55264793e-07j

Table S23 39-bus system Base eig reference

Complete reference spectrum with 20 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S23. Classical reference eigenpairs (MATLAB eig) for the 39-bus system under the Base scenario (Serving as the ground-truth benchmark for Table II and Table IV in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -3.295455095640e-04 Im -6.545746650406e+00	[01] -2.58626181e-06-6.88034058e-03j [02] 7.26614090e-08-3.42576885e-03j [03] 5.23258781e-07-5.69731566e-03j [04] -6.93909312e-06+1.37830874e-01j [05] 1.25694925e-06-4.60550776e-02j [06] -5.38143409e-07-3.14834642e-02j [07] 5.35752497e-06-1.56863846e-02j [08] 4.53469935e-06-1.89226056e-02j [09] -7.18852792e-08-1.17917348e-03j [10] 2.42562543e-08-2.64580620e-05j [11] -4.50369655e-02+1.91963999e-05j [12] -2.24242150e-02+6.53323568e-07j [13] -3.72931851e-02-1.54759462e-06j [14] 9.02205983e-01+0.00000000e+00j [15] -3.01464870e-01+6.94957265e-06j [16] -2.06082780e-01+1.38977847e-05j [17] -1.02679101e-01-2.98996235e-05j [18] -1.23862584e-01-2.34471334e-05j [19] -7.71857083e-03+8.59134150e-07j [20] -1.73187779e-04-1.50056160e-07j
2	Re -3.295455095574e-04 Im 6.545746650406e+00	[01] -2.58626181e-06+6.88034058e-03j [02] 7.26614089e-08+3.42576885e-03j [03] 5.23258781e-07+5.69731566e-03j [04] -6.93909312e-06-1.37830874e-01j [05] 1.25694925e-06+4.60550776e-02j [06] -5.38143410e-07+3.14834642e-02j [07] 5.35752497e-06+1.56863846e-02j [08] 4.53469935e-06+1.89226056e-02j [09] -7.18852792e-08+1.17917348e-03j [10] 2.42562543e-08+2.64580620e-05j [11] -4.50369655e-02-1.91963999e-05j [12] -2.24242150e-02-6.53323568e-07j [13] -3.72931851e-02+1.54759462e-06j [14] 9.02205983e-01+0.00000000e+00j [15] -3.01464870e-01-6.94957265e-06j [16] -2.06082780e-01-1.38977847e-05j [17] -1.02679101e-01+2.98996235e-05j [18] -1.23862584e-01+2.34471334e-05j [19] -7.71857083e-03-8.59134151e-07j [20] -1.73187779e-04+1.50056160e-07j
3	Re -3.237419521118e-04 Im 6.868699396043e+00	[01] 3.93828546e-07+1.04936876e-03j [02] -8.53330569e-08+2.85873671e-04j [03] -1.00936578e-07+4.20212720e-04j [04] -3.40402314e-06+1.05097137e-02j [05] 7.27334513e-07-1.98784341e-03j [06] 9.01174967e-06+8.17620683e-02j [07] -5.56788543e-06-1.18131527e-01j [08] -8.99137290e-07+3.17126648e-05j [09] -2.75864889e-08+2.40086440e-04j [10] -6.64317881e-09-2.69668215e-07j [11] -7.20779871e-03+2.36536520e-06j [12] -1.96358028e-03-6.78676417e-07j [13] -2.88631483e-03-8.29343499e-07j [14] -7.21880632e-02-2.67836469e-05j [15] 1.36538986e-02+5.63939044e-06j [16] -5.61599072e-01+3.54291879e-05j [17] 8.11409953e-01+0.00000000e+00j [18] -2.17824471e-04-6.18617048e-06j [19] -1.64908158e-03-2.67209352e-07j [20] 1.85227205e-06-4.55426953e-08j
4	Re -3.237419521088e-04 Im -6.868699396043e+00	[01] 3.93828546e-07-1.04936876e-03j [02] -8.53330569e-08-2.85873671e-04j [03] -1.00936578e-07-4.20212720e-04j [04] -3.40402314e-06-1.05097137e-02j [05] 7.27334514e-07+1.98784341e-03j [06] 9.01174967e-06-8.17620683e-02j [07] -5.56788543e-06+1.18131527e-01j [08] -8.99137290e-07-3.17126648e-05j [09] -2.75864887e-08-2.40086440e-04j [10] -6.64317883e-09+2.69668215e-07j [11] -7.20779871e-03-2.36536520e-06j [12] -1.96358028e-03+6.78676418e-07j [13] -2.88631483e-03+8.29343499e-07j [14] -7.21880632e-02+2.67836469e-05j [15] 1.36538986e-02-5.63939044e-06j [16] -5.61599072e-01-3.54291879e-05j [17] 8.11409953e-01+0.00000000e+00j [18] -2.17824471e-04+6.18617048e-06j [19] -1.64908158e-03+2.67209351e-07j [20] 1.85227205e-06+4.55426954e-08j
5	Re -3.151425852227e-04 Im -6.810560067478e+00	[01] 1.62522531e-05-8.22907598e-02j [02] -7.49362789e-07+4.74246726e-04j [03] -7.81207312e-07+3.55683280e-04j [04] -3.96671912e-06-4.84289946e-04j [05] 1.01446636e-06-5.96884599e-04j [06] -5.05515880e-07-1.84356374e-03j [07] 2.41553681e-07+1.53385690e-03j [08] -5.53544336e-06+1.19626706e-01j [09] -7.89313400e-07-3.86352325e-03j [10] -3.12960119e-07+5.93535159e-04j [11] -5.60446168e-01-8.47536234e-05j [12] 3.22988605e-03+4.95412495e-06j [13] 2.42240259e-03+5.20836837e-06j [14] -3.29828452e-03+2.71681992e-05j [15] -4.06511873e-03-6.72098029e-06j [16] -1.25557015e-02+4.02383171e-06j [17] 1.04464245e-02-2.12849948e-06j [18] 8.14724871e-01+0.00000000e+00j [19] -2.63127569e-02+6.59322703e-06j [20] 4.04230695e-03+1.94438548e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
6	Re -3.151425852215e-04 Im 6.810560067478e+00	[01] 1.62522531e-05+8.22907598e-02j [03] -7.81207312e-07-3.55683280e-04j [05] 1.01446636e-06+5.96884599e-04j [07] 2.41553681e-07-1.53385690e-03j [09] -7.89313400e-07+3.86352325e-03j [11] -5.60446168e-01+8.47536234e-05j [13] 2.42240259e-03-5.20836837e-06j [15] -4.06511873e-03+6.72098029e-06j [17] 1.04464245e-02+2.12849948e-06j [19] -2.63127569e-02-6.59322703e-06j	[02] -7.49362789e-07-4.74246726e-04j [04] -3.96671912e-06+4.84289946e-04j [06] -5.05515880e-07+1.84356374e-03j [08] -5.53544336e-06-1.19626706e-01j [10] -3.12960119e-07-5.93535159e-04j [12] 3.22988605e-03-4.95412495e-06j [14] -3.29828452e-03-2.71681992e-05j [16] -1.25557015e-02-4.02383171e-06j [18] 8.14724871e-01+0.00000000e+00j [20] 4.04230695e-03-1.94438548e-06j
7	Re -3.093789966944e-04 Im -5.061482055348e+00	[01] 9.16706913e-07+9.02229536e-03j [03] 5.79017653e-06-5.88390916e-02j [05] 8.65304623e-07-7.81564688e-02j [07] -1.02095819e-05+1.00147607e-01j [09] 2.24026532e-06-2.46937367e-02j [11] 4.56661858e-02-7.43120428e-06j [13] -2.97813008e-01-1.11032954e-05j [15] -3.95587564e-01+1.98002461e-05j [17] 5.06895319e-01+2.06920494e-05j [19] -1.24986906e-01-3.69933925e-06j	[02] 1.20586631e-05-7.35063930e-02j [04] -3.18469090e-06+1.15004987e-02j [06] -6.60031579e-06+1.07982055e-01j [08] 3.22472545e-07+8.95079096e-03j [10] -4.23161141e-07+9.07914990e-04j [12] -3.72051293e-01-3.82933730e-05j [14] 5.82095689e-02+1.25612431e-05j [16] 5.46549237e-01+0.00000000e+00j [18] 4.53042677e-02-4.40137573e-06j [20] 4.59539556e-03+1.86093269e-06j
8	Re -3.093789966900e-04 Im 5.061482055348e+00	[01] 9.16706913e-07-9.02229536e-03j [03] 5.79017653e-06+5.88390916e-02j [05] 8.65304623e-07+7.81564688e-02j [07] -1.02095819e-05-1.00147607e-01j [09] 2.24026532e-06+2.46937367e-02j [11] 4.56661858e-02+7.43120428e-06j [13] -2.97813008e-01+1.11032954e-05j [15] -3.95587564e-01-1.98002461e-05j [17] 5.06895319e-01-2.06920494e-05j [19] -1.24986906e-01+3.69933925e-06j	[02] 1.20586631e-05+7.35063930e-02j [04] -3.18469090e-06-1.15004987e-02j [06] -6.60031579e-06-1.07982055e-01j [08] 3.22472546e-07-8.95079096e-03j [10] -4.23161141e-07-9.07914990e-04j [12] -3.72051293e-01+3.82933730e-05j [14] 5.82095689e-02-1.25612431e-05j [16] 5.46549237e-01+0.00000000e+00j [18] 4.53042677e-02+4.40137573e-06j [20] 4.59539556e-03-1.86093269e-06j
9	Re -3.046730653548e-04 Im 4.200799305801e+00	[01] -4.35255437e-06-2.22301539e-02j [03] -7.31955605e-06+9.97049145e-04j [05] 2.06960170e-05+1.42347439e-01j [07] -2.92559764e-06+2.32478125e-02j [09] -1.21650386e-05-1.67730238e-01j [11] 9.33844162e-02-1.15112783e-05j [13] -4.18840113e-03-3.10517600e-05j [15] -5.97973028e-01+4.35703833e-05j [17] -9.76593939e-02-1.93728309e-05j [19] 7.04601073e-01+0.00000000e+00j	[02] -7.39521131e-06-7.15710686e-04j [04] 3.26608332e-06+4.89449665e-02j [06] -3.63309661e-06+2.40940603e-02j [08] -6.82922264e-06-3.48071648e-02j [10] 1.79853583e-06+2.95013651e-03j [12] 3.00655921e-03-3.08477408e-05j [14] -2.05607982e-01-1.19205244e-06j [16] -1.01214311e-01-2.26027209e-05j [18] 1.46217916e-01-1.80833881e-05j [20] -1.23929319e-02+6.65646093e-06j
10	Re -3.046730653523e-04 Im -4.200799305801e+00	[01] -4.35255437e-06+2.22301539e-02j [03] -7.31955605e-06-9.97049145e-04j [05] 2.06960170e-05-1.42347439e-01j [07] -2.92559764e-06-2.32478125e-02j [09] -1.21650386e-05+1.67730238e-01j [11] 9.33844162e-02+1.15112783e-05j [13] -4.18840113e-03+3.10517600e-05j [15] -5.97973028e-01-4.35703833e-05j [17] -9.76593939e-02+1.93728309e-05j [19] 7.04601073e-01+0.00000000e+00j	[02] -7.39521131e-06+7.15710686e-04j [04] 3.26608332e-06-4.89449665e-02j [06] -3.63309661e-06-2.40940603e-02j [08] -6.82922264e-06+3.48071648e-02j [10] 1.79853583e-06-2.95013651e-03j [12] 3.00655921e-03+3.08477408e-05j [14] -2.05607982e-01+1.19205244e-06j [16] -1.01214311e-01+2.26027209e-05j [18] 1.46217916e-01+1.80833881e-05j [20] -1.23929319e-02-6.65646093e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
11	Re -3.005670460009e-04 Im 0.000000000000e+00	[01] 3.16227751e-01+1.48930192e-16j [02] 3.16227752e-01-8.34663701e-17j [03] 3.16227752e-01-1.27654449e-16j [04] 3.16227753e-01+5.23710565e-17j [05] 3.16227754e-01+0.00000000e+00j [06] 3.16227753e-01-4.74612693e-17j [07] 3.16227753e-01-4.58246739e-17j [08] 3.16227752e-01+9.65591354e-17j [09] 3.16227753e-01+3.07679955e-16j [10] 3.16227745e-01-1.53839975e-16j [11] -9.50476409e-05-4.18946805e-16j [12] -9.50476412e-05+1.04744277e-15j [13] -9.50476411e-05-4.18946805e-16j [14] -9.50476416e-05+2.09505869e-16j [15] -9.50476420e-05-2.61833637e-16j [16] -9.50476415e-05+2.16438937e-20j [17] -9.50476416e-05+2.09505869e-16j [18] -9.50476412e-05+7.33216430e-16j [19] -9.50476415e-05-9.42657367e-16j [20] -9.50476393e-05-3.66575749e-16j
12	Re -3.002929642228e-04 Im 4.583815868253e+00	[01] 4.53694021e-07-1.79393253e-02j [02] -7.61589955e-06-1.16252744e-01j [03] -3.38949150e-06-1.10279628e-01j [04] -3.86409941e-06+1.70160143e-02j [05] 4.96811760e-07+1.11562490e-01j [06] -7.76020751e-06-2.58689101e-02j [07] -8.11610354e-06-2.45356265e-02j [08] 6.07334292e-07-1.22128577e-02j [09] 1.38569914e-05+7.22661698e-02j [10] 1.51975997e-06+7.38028284e-03j [11] 8.22305639e-02+7.46670302e-06j [12] 5.32881177e-01+0.00000000e+00j [13] 5.05501511e-01+1.75793915e-05j [14] -7.79982751e-02-2.28221096e-05j [15] -5.11381913e-01-3.12241372e-05j [16] 1.18578323e-01-2.78031106e-05j [17] 1.12466796e-01-2.98348482e-05j [18] 5.59814908e-02+6.45134381e-06j [19] -3.31254820e-01+4.18168748e-05j [20] -3.38298581e-02+4.75005286e-06j
13	Re -3.002929642226e-04 Im -4.583815868253e+00	[01] 4.53694021e-07+1.79393253e-02j [02] -7.61589955e-06+1.16252744e-01j [03] -3.38949150e-06+1.10279628e-01j [04] -3.86409941e-06-1.70160143e-02j [05] 4.96811760e-07-1.11562490e-01j [06] -7.76020751e-06+2.58689101e-02j [07] -8.11610353e-06+2.45356265e-02j [08] 6.07334292e-07+1.22128577e-02j [09] 1.38569914e-05-7.22661698e-02j [10] 1.51975997e-06-7.38028284e-03j [11] 8.22305639e-02-7.46670302e-06j [12] 5.32881177e-01+0.00000000e+00j [13] 5.05501511e-01-1.75793915e-05j [14] -7.79982751e-02+2.28221096e-05j [15] -5.11381913e-01+3.12241372e-05j [16] 1.18578323e-01+2.78031106e-05j [17] 1.12466796e-01+2.98348482e-05j [18] 5.59814908e-02-6.45134381e-06j [19] -3.31254820e-01-4.18168748e-05j [20] -3.38298581e-02-4.75005286e-06j
14	Re -2.922696776578e-04 Im -5.711763079524e+00	[01] 8.89755210e-06+1.04938876e-01j [02] -8.96676674e-06-1.33951926e-02j [03] 7.25175652e-06-2.98328985e-02j [04] -1.39306449e-07+4.31481797e-03j [05] 1.52653066e-07-1.67250294e-03j [06] -2.43024774e-06-2.94993935e-02j [07] -4.26096753e-07-2.56934450e-02j [08] -6.21770638e-06+1.21511291e-01j [09] 1.74729665e-07-3.64417140e-02j [10] 2.63647356e-07-3.75177825e-03j [11] 5.99385995e-01-8.14911610e-05j [12] -7.65101642e-02+5.51310559e-05j [13] -1.70398450e-01-3.27010635e-05j [14] 2.46452180e-02-4.65405027e-07j [15] -9.55294059e-03-3.83096251e-07j [16] -1.68493546e-01+2.25027775e-05j [17] -1.46754871e-01+9.94317860e-06j [18] 6.94043705e-01+0.00000000e+00j [19] -2.08146436e-01+9.65279354e-06j [20] -2.14292686e-02-4.09360214e-07j
15	Re -2.922696776537e-04 Im 5.711763079524e+00	[01] 8.89755210e-06-1.04938876e-01j [02] -8.96676674e-06+1.33951926e-02j [03] 7.25175652e-06+2.98328985e-02j [04] -1.39306449e-07-4.31481797e-03j [05] 1.52653066e-07+1.67250294e-03j [06] -2.43024774e-06+2.94993935e-02j [07] -4.26096754e-07+2.56934450e-02j [08] -6.21770638e-06-1.21511291e-01j [09] 1.74729665e-07+3.64417140e-02j [10] 2.63647356e-07+3.75177825e-03j [11] 5.99385995e-01+8.14911610e-05j [12] -7.65101642e-02-5.51310559e-05j [13] -1.70398450e-01+3.27010635e-05j [14] 2.46452180e-02+4.65405026e-07j [15] -9.55294059e-03+3.83096250e-07j [16] -1.68493546e-01-2.25027775e-05j [17] -1.46754871e-01-9.94317860e-06j [18] 6.94043705e-01+0.00000000e+00j [19] -2.08146436e-01-9.65279354e-06j [20] -2.14292686e-02+4.09360214e-07j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
16	Re -2.882560430304e-04 Im -5.639387583526e+00	[01] 8.81132480e-06-6.13597700e-03j [03] 8.35884280e-06-1.14747872e-01j [05] -1.53491408e-07+4.88945872e-04j [07] -4.13409086e-06+7.71205494e-03j [09] -3.59550102e-06+2.79730645e-03j [11] -3.46031550e-02-4.79217432e-05j [13] -6.47107728e-01-1.40619867e-05j [15] 2.75735532e-03+7.24655941e-07j [17] 4.34912681e-02+2.10906942e-05j [19] 1.57750963e-02+1.94700833e-05j	[02] -6.68089717e-06+1.30703829e-01j [04] 4.11148008e-07-2.97638099e-04j [06] -4.16940399e-06+8.81465993e-03j [08] 1.08701766e-05-7.19410612e-03j [10] -4.04347280e-07-7.13166771e-05j [12] 7.37089552e-01+0.00000000e+00j [14] -1.67849672e-03-2.23282699e-06j [16] 4.97092850e-02+2.09720061e-05j [18] -4.05703559e-02-5.92273943e-05j [20] -4.02182267e-04+2.30082849e-06j
17	Re -2.882560430285e-04 Im 5.639387583526e+00	[01] 8.81132480e-06+6.13597700e-03j [03] 8.35884280e-06+1.14747872e-01j [05] -1.53491409e-07-4.88945872e-04j [07] -4.13409086e-06-7.71205494e-03j [09] -3.59550102e-06-2.79730645e-03j [11] -3.46031550e-02+4.79217432e-05j [13] -6.47107728e-01+1.40619867e-05j [15] 2.75735532e-03-7.24655942e-07j [17] 4.34912681e-02-2.10906942e-05j [19] 1.57750963e-02-1.94700833e-05j	[02] -6.68089717e-06-1.30703829e-01j [04] 4.11148008e-07+2.97638099e-04j [06] -4.16940399e-06-8.81465993e-03j [08] 1.08701766e-05+7.19410612e-03j [10] -4.04347280e-07+7.13166771e-05j [12] 7.37089552e-01+0.00000000e+00j [14] -1.67849672e-03+2.23282699e-06j [16] 4.97092850e-02-2.09720061e-05j [18] -4.05703559e-02+5.92273943e-05j [20] -4.02182267e-04-2.30082849e-06j
18	Re -1.848636309416e-04 Im -2.753123443455e+00	[01] 8.71360906e-06+5.45273153e-02j [03] 7.32303223e-06+7.59601488e-02j [05] -1.13580389e-05+1.69152164e-01j [07] -1.07108433e-06+1.23349387e-01j [09] 2.87295392e-06+1.24318955e-01j [11] 1.50120429e-01-3.40697589e-05j [13] 2.09127665e-01-3.42034806e-05j [15] 4.65696792e-01+0.00000000e+00j [17] 3.39596089e-01-1.98539882e-05j [19] 3.42265429e-01-3.08916502e-05j	[02] 7.74275390e-06+6.57761475e-02j [04] -2.17072216e-06+1.26144716e-01j [06] 2.76688995e-07+1.24639158e-01j [08] 6.30345592e-06+6.68972555e-02j [10] 1.75458472e-05-9.09906037e-02j [12] 1.81089852e-01-3.34763747e-05j [14] 3.47291975e-01-1.73433041e-05j [16] 3.43146988e-01-2.38030063e-05j [18] 1.84176401e-01-2.97210618e-05j [20] -2.50508367e-01-3.14850299e-05j
19	Re -1.848636309393e-04 Im 2.753123443455e+00	[01] 8.71360906e-06-5.45273153e-02j [03] 7.32303223e-06-7.59601488e-02j [05] -1.13580389e-05-1.69152164e-01j [07] -1.07108433e-06-1.23349387e-01j [09] 2.87295392e-06-1.24318955e-01j [11] 1.50120429e-01+3.40697589e-05j [13] 2.09127665e-01+3.42034806e-05j [15] 4.65696792e-01+0.00000000e+00j [17] 3.39596089e-01+1.98539882e-05j [19] 3.42265429e-01+3.08916502e-05j	[02] 7.74275390e-06-6.57761475e-02j [04] -2.17072216e-06-1.26144716e-01j [06] 2.76688995e-07-1.24639158e-01j [08] 6.30345592e-06-6.68972555e-02j [10] 1.75458472e-05+9.09906037e-02j [12] 1.81089852e-01+3.34763747e-05j [14] 3.47291975e-01+1.73433041e-05j [16] 3.43146988e-01+2.38030063e-05j [18] 1.84176401e-01+2.97210618e-05j [20] -2.50508367e-01+3.14850299e-05j
20	Re 2.022708344001e-12 Im 0.000000000000e+00	[01] 3.16227766e-01-1.37728786e-16j [03] 3.16227766e-01-4.12262005e-16j [05] 3.16227766e-01-2.95793367e-16j [07] 3.16227766e-01-3.30918829e-16j [09] 3.16227766e-01+0.00000000e+00j [11] 6.38928462e-13-2.36634694e-16j [13] 6.40422219e-13-8.28221427e-16j [15] 6.40762381e-13-5.17638392e-16j [17] 6.39989621e-13-1.18317347e-16j [19] 6.38662248e-13-1.47896683e-15j	[02] 3.16227766e-01-3.64195583e-16j [04] 3.16227766e-01-2.44029528e-16j [06] 3.16227766e-01-3.31843183e-16j [08] 3.16227766e-01-1.91341334e-16j [10] 3.16227766e-01-4.45538759e-16j [12] 6.39875001e-13+5.47217729e-16j [14] 6.39091148e-13-7.32297113e-28j [16] 6.38588300e-13-7.31720928e-28j [18] 6.40111636e-13+9.46538774e-16j [20] 6.39828783e-13-2.95793367e-16j

Table S24 39-bus system Light load eig reference

Complete reference spectrum with 20 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S24. Classical reference eigenpairs (MATLAB eig) for the 39-bus system under the Light load scenario (Serving as the ground-truth benchmark for Table II in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -3.288322155560e-04 Im -6.550640975015e+00	[01] -2.56174055e-06-7.67301858e-03j [02] 6.57493953e-08-3.50591629e-03j [03] 5.13049445e-07-5.80103478e-03j [04] -6.92177560e-06+1.37888153e-01j [05] 1.16459566e-06-4.45465709e-02j [06] -4.43585600e-07-3.34776305e-02j [07] 5.51831316e-06-1.44631174e-02j [08] 4.68174762e-06-1.84170537e-02j [09] -7.38353224e-08-1.38204261e-03j [10] 2.00636775e-08-1.75653084e-05j [11] -5.02631891e-02+1.93041783e-05j [12] -2.29659989e-02+7.22157537e-07j [13] -3.80004963e-02-1.45323559e-06j [14] 9.03255788e-01+0.00000000e+00j [15] -2.91808593e-01+7.01949953e-06j [16] -2.19299938e-01+1.39142934e-05j [17] -9.47426914e-02-3.13925493e-05j [18] -1.20643508e-01-2.46123272e-05j [19] -9.05326493e-03+9.38128822e-07j [20] -1.15064035e-04-1.25653909e-07j
2	Re -3.288322155533e-04 Im 6.550640975015e+00	[01] -2.56174055e-06+7.67301858e-03j [02] 6.57493952e-08+3.50591629e-03j [03] 5.13049445e-07+5.80103478e-03j [04] -6.92177560e-06-1.37888153e-01j [05] 1.16459566e-06+4.45465709e-02j [06] -4.43585600e-07+3.34776305e-02j [07] 5.51831316e-06+1.44631174e-02j [08] 4.68174762e-06+1.84170537e-02j [09] -7.38353225e-08+1.38204261e-03j [10] 2.00636775e-08+1.75653084e-05j [11] -5.02631891e-02-1.93041783e-05j [12] -2.29659989e-02-7.22157538e-07j [13] -3.80004963e-02+1.45323559e-06j [14] 9.03255788e-01+0.00000000e+00j [15] -2.91808593e-01-7.01949953e-06j [16] -2.19299938e-01-1.39142934e-05j [17] -9.47426914e-02+3.13925493e-05j [18] -1.20643508e-01+2.46123272e-05j [19] -9.05326493e-03-9.38128823e-07j [20] -1.15064035e-04+1.25653909e-07j
3	Re -3.246441690124e-04 Im 6.870225271878e+00	[01] 7.05380404e-07+1.44075147e-03j [02] -8.86682633e-08+3.16453283e-04j [03] -1.00752344e-07+4.69289979e-04j [04] -3.38923143e-06+1.22640385e-02j [05] 7.03501316e-07-2.24568935e-03j [06] 9.03419814e-06+8.04163731e-02j [07] -5.61547089e-06-1.18836418e-01j [08] -1.29060657e-06-2.87724951e-04j [09] -2.15500153e-08+2.71924041e-04j [10] -9.08537290e-09-5.17397546e-07j [11] -9.89828740e-03+4.37839071e-06j [12] -2.17410531e-03-7.11905655e-07j [13] -3.22412784e-03-8.44543559e-07j [14] -8.42567063e-02-2.72662320e-05j [15] 1.54283915e-02+5.56226247e-06j [16] -5.52478601e-01+3.59602697e-05j [17] 8.16432961e-01+0.00000000e+00j [18] 1.97673565e-03-8.77334964e-06j [19] -1.86817941e-03-2.36332014e-07j [20] 3.55464065e-06-6.22505882e-08j
4	Re -3.246441690075e-04 Im -6.870225271878e+00	[01] 7.05380405e-07-1.44075147e-03j [02] -8.86682634e-08-3.16453283e-04j [03] -1.00752344e-07-4.69289979e-04j [04] -3.38923143e-06-1.22640385e-02j [05] 7.03501317e-07+2.24568935e-03j [06] 9.03419814e-06-8.04163731e-02j [07] -5.61547089e-06+1.18836418e-01j [08] -1.29060657e-06+2.87724951e-04j [09] -2.15500151e-08-2.71924041e-04j [10] -9.08537290e-09+5.17397546e-07j [11] -9.89828740e-03-4.37839072e-06j [12] -2.17410531e-03+7.11905656e-07j [13] -3.22412784e-03+8.44543559e-07j [14] -8.42567063e-02+2.72662320e-05j [15] 1.54283915e-02-5.56226248e-06j [16] -5.52478601e-01-3.59602697e-05j [17] 8.16432961e-01+0.00000000e+00j [18] 1.97673565e-03+8.77334964e-06j [19] -1.86817941e-03+2.36332013e-07j [20] 3.55464065e-06+6.22505882e-08j
5	Re -3.164263569584e-04 Im 6.818223698330e+00	[01] 1.61673327e-05+8.11385885e-02j [02] -7.79097747e-07-6.10687861e-04j [03] -8.33920609e-07-5.17479686e-04j [04] -4.15360768e-06-7.99762638e-05j [05] 9.80175071e-07+5.93079554e-04j [06] -8.01061863e-07+1.40893228e-03j [07] 5.31679696e-07-1.27488340e-03j [08] -5.57992593e-06-1.20233926e-01j [09] -8.86355494e-07+3.64564816e-03j [10] -3.04264044e-07-5.53386933e-04j [11] -5.53221052e-01+8.45581029e-05j [12] 4.16380669e-03-5.11882499e-06j [13] 3.52829252e-03-5.52211304e-06j [14] 5.45297372e-04-2.82949197e-05j [15] -4.04374938e-03+6.49538689e-06j [16] -9.60641520e-03-5.90764229e-06j [17] 8.69244005e-03+4.02851781e-06j [18] 8.19781806e-01+0.00000000e+00j [19] -2.48568444e-02-7.19694920e-06j [20] 3.77311600e-03-1.89943411e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
6	Re -3.164263569532e-04 Im -6.818223698330e+00	[01] 1.61673327e-05-8.11385885e-02j [03] -8.33920609e-07+5.17479686e-04j [05] 9.80175071e-07-5.93079554e-04j [07] 5.31679696e-07+1.27488340e-03j [09] -8.86355494e-07-3.64564816e-03j [11] -5.53221052e-01-8.45581029e-05j [13] 3.52829252e-03+5.52211304e-06j [15] -4.04374938e-03-6.49538690e-06j [17] 8.69244005e-03-4.02851781e-06j [19] -2.48568444e-02+7.19694920e-06j	[02] -7.79097747e-07+6.10687861e-04j [04] -4.15360769e-06+7.99762638e-05j [06] -8.01061863e-07-1.40893228e-03j [08] -5.57992593e-06+1.20233926e-01j [10] -3.04264044e-07+5.53386933e-04j [12] 4.16380669e-03+5.11882499e-06j [14] 5.45297372e-04+2.82949198e-05j [16] -9.60641520e-03+5.90764229e-06j [18] 8.19781806e-01+0.00000000e+00j [20] 3.77311600e-03+1.89943411e-06j
7	Re -3.084791373026e-04 Im 5.105575755992e+00	[01] 9.04025120e-07-9.33538526e-03j [03] 5.19054647e-06+5.78766411e-02j [05] 7.24252659e-07+7.57239723e-02j [07] -1.00379999e-05-9.79649976e-02j [09] 2.60535295e-06+2.75822094e-02j [11] 4.76625164e-02+7.49534032e-06j [13] -2.95493577e-01+8.64699190e-06j [15] -3.86614477e-01-1.96615388e-05j [17] 5.00167720e-01-2.10296110e-05j [19] -1.40823060e-01+4.79329072e-06j	[02] 1.21017042e-05+7.50045152e-02j [04] -3.29080884e-06-1.38435046e-02j [06] -6.48349040e-06-1.07306937e-01j [08] 2.84160067e-07-9.15410541e-03j [10] -3.87586908e-07-8.35653625e-04j [12] -3.82941238e-01+3.86488396e-05j [14] 7.06790625e-02-1.25310415e-05j [16] 5.47863699e-01+0.00000000e+00j [18] 4.67369786e-02+4.27465129e-06j [20] 4.26649301e-03-1.72107261e-06j
8	Re -3.084791373026e-04 Im -5.105575755992e+00	[01] 9.04025119e-07+9.33538526e-03j [03] 5.19054647e-06-5.78766411e-02j [05] 7.24252659e-07-7.57239723e-02j [07] -1.00379999e-05+9.79649976e-02j [09] 2.60535295e-06-2.75822094e-02j [11] 4.76625164e-02-7.49534032e-06j [13] -2.95493577e-01-8.64699190e-06j [15] -3.86614477e-01+1.96615388e-05j [17] 5.00167720e-01+2.10296110e-05j [19] -1.40823060e-01-4.79329071e-06j	[02] 1.21017042e-05-7.50045152e-02j [04] -3.29080884e-06+1.38435046e-02j [06] -6.48349040e-06+1.07306937e-01j [08] 2.84160067e-07+9.15410541e-03j [10] -3.87586908e-07+8.35653625e-04j [12] -3.82941238e-01-3.86488396e-05j [14] 7.06790625e-02+1.25310415e-05j [16] 5.47863699e-01+0.00000000e+00j [18] 4.67369786e-02-4.27465129e-06j [20] 4.26649301e-03+1.72107261e-06j
9	Re -3.081160357837e-04 Im -4.289686990722e+00	[01] -4.58354636e-06+2.38404387e-02j [03] -8.18136123e-06+4.97849685e-03j [05] 2.26569818e-05-1.47579402e-01j [07] -3.66628072e-06-1.81991828e-02j [09] -1.13738688e-05+1.58350528e-01j [11] 1.02268021e-01+1.23163577e-05j [13] 2.13561957e-02+3.35615241e-05j [15] -6.33069449e-01-5.17197798e-05j [17] -7.80687965e-02+2.13346568e-05j [19] 6.79274202e-01+0.00000000e+00j	[02] -8.50188368e-06+6.73397499e-03j [04] 3.06753344e-06-4.58753338e-02j [06] -4.33701556e-06-1.90450692e-02j [08] -7.05603259e-06+3.52063827e-02j [10] 1.89551578e-06-3.40647639e-03j [12] 2.88866475e-02+3.43955741e-05j [14] -1.96790823e-01+9.76167672e-07j [16] -8.16973843e-02+2.44725304e-05j [18] 1.51024364e-01+1.94205201e-05j [20] -1.46127180e-02-7.08157938e-06j
10	Re -3.081160357826e-04 Im 4.289686990722e+00	[01] -4.58354636e-06-2.38404387e-02j [03] -8.18136124e-06-4.97849685e-03j [05] 2.26569818e-05+1.47579402e-01j [07] -3.66628072e-06+1.81991828e-02j [09] -1.13738688e-05-1.58350528e-01j [11] 1.02268021e-01-1.23163577e-05j [13] 2.13561957e-02-3.35615241e-05j [15] -6.33069449e-01+5.17197798e-05j [17] -7.80687965e-02-2.13346568e-05j [19] 6.79274202e-01+0.00000000e+00j	[02] -8.50188368e-06-6.73397499e-03j [04] 3.06753344e-06+4.58753338e-02j [06] -4.33701556e-06+1.90450692e-02j [08] -7.05603259e-06-3.52063827e-02j [10] 1.89551578e-06+3.40647639e-03j [12] 2.88866475e-02-3.43955741e-05j [14] -1.96790823e-01-9.76167672e-07j [16] -8.16973843e-02-2.44725304e-05j [18] 1.51024364e-01-1.94205201e-05j [20] -1.46127180e-02+7.08157938e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
11	Re -2.981449805238e-04 Im 4.636905704887e+00	[01] 8.94493029e-07-1.54359855e-02j [03] -2.88942962e-06-1.07545374e-01j [05] -1.68449253e-06+1.05041087e-01j [07] -7.58583008e-06-2.49948043e-02j [09] 1.52534270e-05+8.25368731e-02j [11] 7.15752089e-02+8.74984142e-06j [13] 4.98677757e-01+1.86661006e-05j [15] -4.87065617e-01-3.91283060e-05j [17] 1.15898553e-01-2.77227033e-05j [19] -3.82715702e-01+4.61207481e-05j	[02] -7.37339521e-06-1.14674875e-01j [04] -4.22190695e-06+1.34909429e-02j [06] -7.24644836e-06-2.66152048e-02j [08] 1.28439457e-06-9.05295451e-03j [10] 1.29836925e-06+6.71093718e-03j [12] 5.31736583e-01+0.00000000e+00j [14] -6.25562287e-02-2.35988413e-05j [16] 1.23412197e-01-2.56659080e-05j [18] 4.19776960e-02+8.65470943e-06j [20] -3.11179833e-02+4.01958353e-06j
12	Re -2.981449805223e-04 Im -4.636905704887e+00	[01] 8.94493029e-07+1.54359855e-02j [03] -2.88942962e-06+1.07545374e-01j [05] -1.68449253e-06-1.05041087e-01j [07] -7.58583008e-06+2.49948043e-02j [09] 1.52534270e-05-8.25368731e-02j [11] 7.15752089e-02-8.74984142e-06j [13] 4.98677757e-01-1.86661006e-05j [15] -4.87065617e-01+3.91283060e-05j [17] 1.15898553e-01+2.77227033e-05j [19] -3.82715702e-01-4.61207481e-05j	[02] -7.37339521e-06+1.14674875e-01j [04] -4.22190695e-06-1.34909429e-02j [06] -7.24644836e-06+2.66152048e-02j [08] 1.28439457e-06+9.05295451e-03j [10] 1.29836925e-06-6.71093718e-03j [12] 5.31736583e-01+0.00000000e+00j [14] -6.25562287e-02+2.35988413e-05j [16] 1.23412197e-01+2.56659080e-05j [18] 4.19776960e-02-8.65470943e-06j [20] -3.11179833e-02-4.01958353e-06j
13	Re -2.913215811404e-04 Im -5.737927004192e+00	[01] 9.16748197e-06+1.04536117e-01j [03] 4.78384840e-06-2.87979356e-02j [05] 1.62160540e-07-1.65706393e-03j [07] -3.91765321e-07-2.68582410e-02j [09] 4.81740933e-08-3.82515941e-02j [11] 5.99820604e-01-8.30559691e-05j [13] -1.65240454e-01-1.90599128e-05j [15] -9.50811191e-03-4.47726859e-07j [17] -1.54110626e-01+1.00723060e-05j [19] -2.19484855e-01+1.08670954e-05j	[02] -6.38257032e-06-1.46151183e-02j [04] -1.74887680e-07+3.94032656e-03j [06] -2.59599586e-06-3.18021102e-02j [08] -6.06703636e-06+1.19497538e-01j [10] 2.38910668e-07-3.63924037e-03j [12] -8.38604803e-02+4.08804220e-05j [14] 2.26093062e-02-1.44409419e-07j [16] -1.82478186e-01+2.41602758e-05j [18] 6.85668153e-01+0.00000000e+00j [20] -2.08816957e-02-3.10662715e-07j
14	Re -2.913215811374e-04 Im 5.737927004192e+00	[01] 9.16748197e-06-1.04536117e-01j [03] 4.78384840e-06+2.87979356e-02j [05] 1.62160540e-07+1.65706393e-03j [07] -3.91765321e-07+2.68582410e-02j [09] 4.81740933e-08+3.82515941e-02j [11] 5.99820604e-01+8.30559691e-05j [13] -1.65240454e-01+1.90599128e-05j [15] -9.50811191e-03+4.47726860e-07j [17] -1.54110626e-01-1.00723060e-05j [19] -2.19484855e-01-1.08670954e-05j	[02] -6.38257032e-06+1.46151183e-02j [04] -1.74887681e-07-3.94032656e-03j [06] -2.59599586e-06+3.18021102e-02j [08] -6.06703636e-06-1.19497538e-01j [10] 2.38910668e-07+3.63924037e-03j [12] -8.38604803e-02-4.08804220e-05j [14] 2.26093062e-02+1.44409414e-07j [16] -1.82478186e-01-2.41602758e-05j [18] 6.85668153e-01+0.00000000e+00j [20] -2.08816957e-02+3.10662715e-07j
15	Re -2.877535968805e-04 Im -5.635952357782e+00	[01] 6.10681683e-06-5.40059028e-03j [03] 9.93213757e-06-1.16152818e-01j [05] -5.17763978e-08+1.42776526e-04j [07] -3.74991654e-06+8.26491019e-03j [09] -2.78209883e-06+2.63405915e-03j [11] -3.04374713e-02-3.28636894e-05j [13] -6.54631750e-01-2.25536631e-05j [15] 8.04681711e-04+2.50724852e-07j [17] 4.65806411e-02+1.87560933e-05j [19] 1.48454327e-02+1.49218165e-05j	[02] -6.61662472e-06+1.29593451e-01j [04] 2.02254976e-07-3.69975051e-05j [06] -3.78349472e-06+9.64119185e-03j [08] 7.51978619e-06-6.16531804e-03j [10] -3.07174492e-07-5.69516323e-05j [12] 7.30382518e-01+0.00000000e+00j [14] -2.08516234e-04-1.12925324e-06j [16] 5.43372990e-02+1.85493084e-05j [18] -3.47474409e-02-4.06070643e-05j [20] -3.20976598e-04+1.74760884e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
16	Re -2.877535968799e-04 Im 5.635952357782e+00	[01] 6.10681683e-06+5.40059028e-03j [03] 9.93213758e-06+1.16152818e-01j [05] -5.17763979e-08-1.42776526e-04j [07] -3.74991654e-06-8.26491019e-03j [09] -2.78209883e-06-2.63405915e-03j [11] -3.04374713e-02+3.28636894e-05j [13] -6.54631750e-01+2.25536631e-05j [15] 8.04681711e-04-2.50724853e-07j [17] 4.65806411e-02-1.87560933e-05j [19] 1.48454327e-02-1.49218165e-05j	[02] -6.61662472e-06-1.29593451e-01j [04] 2.02254976e-07+3.69975051e-05j [06] -3.78349472e-06-9.64119185e-03j [08] 7.51978619e-06+6.16531804e-03j [10] -3.07174492e-07+5.69516323e-05j [12] 7.30382518e-01+0.00000000e+00j [14] -2.08516234e-04+1.12925324e-06j [16] 5.43372990e-02-1.85493084e-05j [18] -3.47474409e-02+4.06070643e-05j [20] -3.20976598e-04-1.74760884e-06j
17	Re -2.868555240877e-04 Im 0.000000000000e+00	[01] 3.16227752e-01+3.09449929e-16j [03] 3.16227753e-01+6.17321036e-16j [05] 3.16227755e-01+0.00000000e+00j [07] 3.16227754e-01+1.97353272e-16j [09] 3.16227754e-01+4.32598373e-16j [11] -9.07116776e-05+1.71767322e-15j [13] -9.07116778e-05-4.42161005e-16j [15] -9.07116785e-05-1.97046475e-15j [17] -9.07116782e-05-7.57926242e-16j [19] -9.07116781e-05-6.56881367e-16j	[02] 3.16227753e-01+6.94683496e-17j [04] 3.16227754e-01+4.10494796e-17j [06] 3.16227754e-01+9.94660482e-17j [08] 3.16227753e-01+4.32598373e-16j [10] 3.16227747e-01+2.44323344e-16j [12] -9.07116778e-05+1.17455701e-15j [14] -9.07116782e-05-1.11158331e-15j [16] -9.07116781e-05+2.07133028e-15j [18] -9.07116778e-05+5.55657144e-16j [20] -9.07116762e-05-1.51656987e-16j
18	Re -1.913021102221e-04 Im -2.834139802006e+00	[01] 7.67039110e-06+5.67349616e-02j [03] 6.46817360e-06+7.64706369e-02j [05] -1.11733584e-05+1.65533248e-01j [07] -1.13342254e-06+1.19760519e-01j [09] 2.67576588e-06+1.20557535e-01j [11] 1.60794811e-01-3.25924786e-05j [13] 2.16728474e-01-3.29607024e-05j [15] 4.69144370e-01+0.00000000e+00j [17] 3.39418054e-01-1.96981621e-05j [19] 3.41676909e-01-3.06464055e-05j	[02] 6.70826105e-06+6.70647190e-02j [04] -2.14666756e-06+1.22391022e-01j [06] 2.00479658e-07+1.21455787e-01j [08] 5.32962370e-06+6.78129890e-02j [10] 1.52763419e-05-8.08780389e-02j [12] 1.90070788e-01-3.18417719e-05j [14] 3.46873266e-01-1.73297047e-05j [16] 3.44222679e-01-2.38029356e-05j [18] 1.92191490e-01-2.80776666e-05j [20] -2.29219672e-01-2.78231492e-05j
19	Re -1.913021102213e-04 Im 2.834139802006e+00	[01] 7.67039110e-06-5.67349616e-02j [03] 6.46817360e-06-7.64706369e-02j [05] -1.11733584e-05-1.65533248e-01j [07] -1.13342254e-06-1.19760519e-01j [09] 2.67576588e-06-1.20557535e-01j [11] 1.60794811e-01+3.25924786e-05j [13] 2.16728474e-01+3.29607024e-05j [15] 4.69144370e-01+0.00000000e+00j [17] 3.39418054e-01+1.96981621e-05j [19] 3.41676909e-01+3.06464055e-05j	[02] 6.70826105e-06-6.70647190e-02j [04] -2.14666756e-06-1.22391022e-01j [06] 2.00479658e-07-1.21455787e-01j [08] 5.32962370e-06-6.78129890e-02j [10] 1.52763419e-05+8.08780389e-02j [12] 1.90070788e-01+3.18417719e-05j [14] 3.46873266e-01+1.73297047e-05j [16] 3.44222679e-01+2.38029356e-05j [18] 1.92191490e-01+2.80776666e-05j [20] -2.29219672e-01+2.78231492e-05j
20	Re -2.746330085749e-12 Im 0.000000000000e+00	[01] 3.16227766e-01+2.16348679e-16j [03] 3.16227766e-01+5.15631019e-16j [05] 3.16227766e-01-1.08174340e-16j [07] 3.16227766e-01+9.37510944e-17j [09] 3.16227766e-01+3.29931736e-16j [11] -8.70471700e-13+1.84617540e-15j [13] -8.68928413e-13-4.54332227e-16j [15] -8.66923581e-13-2.25002627e-15j [17] -8.68950047e-13-1.03847366e-15j [19] -8.68387541e-13-8.22124982e-16j	[02] 3.16227766e-01-3.33537547e-17j [04] 3.16227766e-01-5.22842642e-17j [06] 3.16227766e-01+0.00000000e+00j [08] 3.16227766e-01+3.33537547e-16j [10] 3.16227766e-01+1.42429547e-16j [12] -8.68055806e-13+1.24039910e-15j [14] -8.68394752e-13-1.03847366e-15j [16] -8.68712064e-13+1.75963593e-15j [18] -8.68063018e-13+8.07701736e-16j [20] -8.68538985e-13-1.58655698e-16j

Table S25 39-bus system Heavy load eig reference

Complete reference spectrum with 20 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S25. Classical reference eigenpairs (MATLAB eig) for the 39-bus system under the Heavy load scenario (Serving as the ground-truth benchmark for Table II in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -3.302836338590e-04 Im -6.539542147604e+00	[01] -2.60426811e-06-6.01707209e-03j [02] 7.83686992e-08-3.32710917e-03j [03] 5.31668236e-07-5.58115096e-03j [04] -6.95658058e-06+1.37738741e-01j [05] 1.35642653e-06-4.76572492e-02j [06] -6.21877871e-07-2.93702552e-02j [07] 5.17107276e-06-1.68623118e-02j [08] 4.36105612e-06-1.94809932e-02j [09] -7.11171100e-08-9.65005290e-04j [10] 2.87892295e-08-3.69532740e-05j [11] -3.93488957e-02+1.90180615e-05j [12] -2.17577707e-02+5.86394294e-07j [13] -3.64981721e-02-1.63350402e-06j [14] 9.00748305e-01+0.00000000e+00j [15] -3.11656590e-01+6.87000101e-06j [16] -1.92068021e-01+1.37673111e-05j [17] -1.10271800e-01-2.82471027e-05j [18] -1.27396777e-01-2.20850571e-05j [19] -6.31069274e-03+7.83798792e-07j [20] -2.41657502e-04-1.76063319e-07j
2	Re -3.302836338525e-04 Im 6.539542147604e+00	[01] -2.60426811e-06+6.01707209e-03j [02] 7.83686992e-08+3.32710917e-03j [03] 5.31668236e-07+5.58115096e-03j [04] -6.95658058e-06-1.37738741e-01j [05] 1.35642653e-06+4.76572492e-02j [06] -6.21877871e-07+2.93702552e-02j [07] 5.17107276e-06+1.68623118e-02j [08] 4.36105612e-06+1.94809932e-02j [09] -7.11171103e-08+9.65005290e-04j [10] 2.87892295e-08+3.69532740e-05j [11] -3.93488957e-02-1.90180615e-05j [12] -2.17577707e-02-5.86394294e-07j [13] -3.64981721e-02+1.63350402e-06j [14] 9.00748305e-01+0.00000000e+00j [15] -3.11656590e-01-6.87000101e-06j [16] -1.92068021e-01-1.37673111e-05j [17] -1.10271800e-01+2.82471027e-05j [18] -1.27396777e-01+2.20850571e-05j [19] -6.31069274e-03-7.83798793e-07j [20] -2.41657502e-04+1.76063319e-07j
3	Re -3.228148124302e-04 Im -6.866199362855e+00	[01] 1.12329950e-07-6.37116913e-04j [02] -8.29081107e-08-2.52797462e-04j [03] -1.02553160e-07-3.67371068e-04j [04] -3.40475901e-06-8.71397788e-03j [05] 7.48530718e-07+1.69684114e-03j [06] 8.97367531e-06-8.31341176e-02j [07] -5.51898674e-06+1.17387622e-01j [08] -5.66501697e-07-3.52535581e-04j [09] -3.32334186e-08-2.09063251e-04j [10] -4.21917593e-09-1.60991393e-07j [11] -4.37457178e-03-5.65609055e-07j [12] -1.73575774e-03+6.50870382e-07j [13] -2.52244296e-03+8.22743265e-07j [14] -5.98319083e-02+2.61907553e-05j [15] 1.16508493e-02-5.68732660e-06j [16] -5.70815428e-01-3.47781191e-05j [17] 8.06006819e-01+0.00000000e+00j [18] -2.42057940e-03+4.00351730e-06j [19] -1.43546995e-03+2.95675992e-07j [20] -1.10539764e-06+2.90216734e-08j
4	Re -3.228148124249e-04 Im 6.866199362855e+00	[01] 1.12329952e-07+6.37116913e-04j [02] -8.29081106e-08+2.52797462e-04j [03] -1.02553160e-07+3.67371068e-04j [04] -3.40475901e-06+8.71397788e-03j [05] 7.48530719e-07-1.69684114e-03j [06] 8.97367531e-06+8.31341176e-02j [07] -5.51898674e-06-1.17387622e-01j [08] -5.66501699e-07+3.52535581e-04j [09] -3.32334182e-08+2.09063251e-04j [10] -4.21917595e-09+1.60991393e-07j [11] -4.37457178e-03+5.65609065e-07j [12] -1.73575774e-03-6.50870381e-07j [13] -2.52244296e-03-8.22743264e-07j [14] -5.98319083e-02-2.61907553e-05j [15] 1.16508493e-02+5.68732660e-06j [16] -5.70815428e-01+3.47781191e-05j [17] 8.06006819e-01+0.00000000e+00j [18] -2.42057940e-03-4.00351731e-06j [19] -1.43546995e-03-2.95675990e-07j [20] -1.10539764e-06-2.90216735e-08j
5	Re -3.170621387242e-04 Im 0.000000000000e+00	[01] 3.16227749e-01+5.38372766e-17j [02] 3.16227750e-01-4.49424211e-16j [03] 3.16227750e-01-2.24712105e-16j [04] 3.16227752e-01+3.74520181e-17j [05] 3.16227753e-01+0.00000000e+00j [06] 3.16227751e-01-9.36300391e-18j [07] 3.16227752e-01-2.43438114e-16j [08] 3.16227750e-01-8.42670389e-17j [09] 3.16227751e-01-3.08979145e-16j [10] 3.16227743e-01+1.02993052e-16j [11] -1.00263846e-04+1.49837757e-16j [12] -1.00263847e-04-1.87257120e-15j [13] -1.00263847e-04+5.99261969e-16j [14] -1.00263847e-04-7.49010667e-16j [15] -1.00263848e-04+3.59542338e-15j [16] -1.00263847e-04-1.49805102e-15j [17] -1.00263847e-04-7.48743488e-17j [18] -1.00263847e-04+1.49837757e-16j [19] -1.00263847e-04-1.87257120e-15j [20] -1.00263845e-04+5.43083943e-16j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
6	Re -3.137199659973e-04 Im 6.801734936276e+00	[01] 1.63421280e-05+8.35727295e-02j [03] -7.24690656e-07-1.83945468e-04j [05] 1.05209532e-06+5.96342963e-04j [07] 2.72656847e-08-1.71915802e-03j [09] -6.87383255e-07+4.09357225e-03j [11] -5.68439559e-01+8.49363889e-05j [13] 1.25114854e-03-4.87144639e-06j [15] -4.05616710e-03+6.96898879e-06j [17] 1.16932572e-02+7.24788157e-07j [19] -2.78433932e-02-5.95963405e-06j	[02] -7.14764377e-07-3.20456866e-04j [04] -3.78758312e-06+1.05901785e-03j [06] -2.50838661e-07+2.25906192e-03j [08] -5.48592587e-06-1.18939875e-01j [10] -3.22434563e-07-6.37788871e-04j [12] 2.17966288e-03-4.76110412e-06j [14] -7.20315750e-03-2.60943715e-05j [16] -1.53655403e-02-2.41485091e-06j [18] 8.08997508e-01+0.00000000e+00j [20] 4.33807095e-03-1.99302733e-06j
7	Re -3.137199659889e-04 Im -6.801734936276e+00	[01] 1.63421280e-05-8.35727295e-02j [03] -7.24690656e-07+1.83945468e-04j [05] 1.05209532e-06-5.96342963e-04j [07] 2.72656843e-08+1.71915802e-03j [09] -6.87383255e-07-4.09357225e-03j [11] -5.68439559e-01-8.49363889e-05j [13] 1.25114854e-03+4.87144639e-06j [15] -4.05616710e-03-6.96898879e-06j [17] 1.16932572e-02-7.24788155e-07j [19] -2.78433932e-02+5.95963405e-06j	[02] -7.14764377e-07+3.20456866e-04j [04] -3.78758312e-06-1.05901785e-03j [06] -2.50838661e-07-2.25906192e-03j [08] -5.48592587e-06+1.18939875e-01j [10] -3.22434563e-07+6.37788871e-04j [12] 2.17966288e-03+4.76110412e-06j [14] -7.20315750e-03+2.60943715e-05j [16] -1.53655403e-02+2.41485091e-06j [18] 8.08997508e-01+0.00000000e+00j [20] 4.33807095e-03+1.99302733e-06j
8	Re -3.104300747913e-04 Im -5.009898754153e+00	[01] 9.24367664e-07+8.66151786e-03j [03] 6.34069560e-06-5.93775610e-02j [05] 1.05197946e-06-8.09701003e-02j [07] -1.04023968e-05+1.02536201e-01j [09] 1.93107892e-06-2.20369395e-02j [11] 4.33933273e-02-7.31978405e-06j [13] -2.97475571e-01-1.33336623e-05j [15] -4.05652005e-01+1.98652437e-05j [17] 5.13695987e-01+2.02846344e-05j [19] -1.10402836e-01-2.83358112e-06j	[02] 1.19950374e-05-7.19447081e-02j [04] -3.06393738e-06+8.86219440e-03j [06] -6.73896411e-06+1.08757271e-01j [08] 3.64058041e-07+8.69786939e-03j [10] -4.61337273e-07+9.84262343e-04j [12] -3.60435707e-01-3.77601217e-05j [14] 4.43986976e-02+1.25989244e-05j [16] 5.44862921e-01+0.00000000e+00j [18] 4.35754449e-02-4.52397417e-06j [20] 4.93105483e-03+2.00570839e-06j
9	Re -3.104300747905e-04 Im 5.009898754153e+00	[01] 9.24367664e-07-8.66151786e-03j [03] 6.34069560e-06+5.93775610e-02j [05] 1.05197946e-06+8.09701003e-02j [07] -1.04023968e-05-1.02536201e-01j [09] 1.93107892e-06+2.20369395e-02j [11] 4.33933273e-02+7.31978405e-06j [13] -2.97475571e-01+1.33336623e-05j [15] -4.05652005e-01-1.98652437e-05j [17] 5.13695987e-01-2.02846344e-05j [19] -1.10402836e-01+2.83358112e-06j	[02] 1.19950374e-05+7.19447081e-02j [04] -3.06393738e-06-8.86219440e-03j [06] -6.73896411e-06-1.08757271e-01j [08] 3.64058041e-07-8.69786939e-03j [10] -4.61337273e-07-9.84262343e-04j [12] -3.60435707e-01+3.77601217e-05j [14] 4.43986976e-02-1.25989244e-05j [16] 5.44862921e-01+0.00000000e+00j [18] 4.35754449e-02+4.52397417e-06j [20] 4.93105483e-03-2.00570839e-06j
10	Re -3.018913359822e-04 Im -4.524290517924e+00	[01] 3.45392656e-08+2.01113682e-02j [03] -3.88634166e-06+1.13218183e-01j [05] 2.63163977e-06-1.16533729e-01j [07] -8.49546857e-06+2.36136493e-02j [09] 1.22242719e-05-6.25791101e-02j [11] 9.09896726e-02-6.22771350e-06j [13] 5.12231954e-01-1.65966499e-05j [15] -5.27232447e-01+2.32742204e-05j [17] 1.06835012e-01+3.13072118e-05j [19] -2.83126078e-01-3.64140663e-05j	[02] -7.88851527e-06+1.18221130e-01j [04] -3.36223835e-06-2.06039829e-02j [06] -8.11774758e-06+2.46539761e-02j [08] -4.98922881e-08+1.50824704e-02j [10] 1.75777021e-06-8.09941710e-03j [12] 5.34866739e-01+0.00000000e+00j [14] -9.32184036e-02+2.14319070e-05j [16] 1.11541753e-01+2.92842266e-05j [18] 6.82374780e-02-4.32753995e-06j [20] -3.66441165e-02-5.50751923e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
11	Re -3.018913359814e-04 Im 4.524290517924e+00	[01] 3.45392658e-08-2.01113682e-02j [02] -7.88851527e-06-1.18221130e-01j [03] -3.88634166e-06-1.13218183e-01j [04] -3.36223835e-06+2.06039829e-02j [05] 2.63163977e-06+1.16533729e-01j [06] -8.11774758e-06-2.46539761e-02j [07] -8.49546857e-06-2.36136493e-02j [08] -4.98922879e-08-1.50824704e-02j [09] 1.22242719e-05+6.25791101e-02j [10] 1.75777021e-06+8.09941710e-03j [11] 9.09896726e-02+6.22771350e-06j [12] 5.34866739e-01+0.00000000e+00j [13] 5.12231954e-01+1.65966499e-05j [14] -9.32184036e-02-2.14319070e-05j [15] -5.27232447e-01-2.32742204e-05j [16] 1.11541753e-01-2.92842266e-05j [17] 1.06835012e-01-3.13072118e-05j [18] 6.82374780e-02+4.32753995e-06j [19] -2.83126078e-01+3.64140663e-05j [20] -3.66441165e-02+5.50751923e-06j
12	Re -3.017063793214e-04 Im -4.095220235418e+00	[01] -4.10001531e-06+2.03865549e-02j [02] -6.26953566e-06-5.13424677e-03j [03] -6.38606107e-06-6.92527266e-03j [04] 3.45843216e-06-5.23626272e-02j [05] 1.88061559e-05-1.37934169e-01j [06] -2.89050455e-06-2.94751438e-02j [07] -2.13982616e-06-2.87186649e-02j [08] -6.57233589e-06+3.41729471e-02j [09] -1.30022135e-05+1.76485919e-01j [10] 1.68613178e-06-2.40160388e-03j [11] 8.34874335e-02+1.06397120e-05j [12] -2.10258694e-02+2.72241643e-05j [13] -2.83605148e-02+2.82417255e-05j [14] -2.14436492e-01+1.63509731e-06j [15] -5.64870805e-01-3.53997316e-05j [16] -1.20707204e-01+2.07300916e-05j [17] -1.17609257e-01+1.74276638e-05j [18] 1.39945746e-01+1.66049668e-05j [19] 7.22748712e-01+0.00000000e+00j [20] -9.83509734e-03-6.18050177e-06j
13	Re -3.017063793170e-04 Im 4.095220235418e+00	[01] -4.10001531e-06-2.03865549e-02j [02] -6.26953565e-06+5.13424677e-03j [03] -6.38606107e-06+6.92527266e-03j [04] 3.45843216e-06+5.23626272e-02j [05] 1.88061559e-05+1.37934169e-01j [06] -2.89050455e-06+2.94751438e-02j [07] -2.13982616e-06+2.87186649e-02j [08] -6.57233589e-06-3.41729471e-02j [09] -1.30022135e-05-1.76485919e-01j [10] 1.68613178e-06+2.40160388e-03j [11] 8.34874335e-02-1.06397120e-05j [12] -2.10258694e-02-2.72241643e-05j [13] -2.83605148e-02-2.82417255e-05j [14] -2.14436492e-01-1.63509732e-06j [15] -5.64870805e-01+3.53997316e-05j [16] -1.20707204e-01-2.07300916e-05j [17] -1.17609257e-01-1.74276638e-05j [18] 1.39945746e-01-1.66049668e-05j [19] 7.22748712e-01+0.00000000e+00j [20] -9.83509734e-03+6.18050177e-06j
14	Re -2.931845829688e-04 Im -5.682189721349e+00	[01] 8.58502135e-06+1.05202514e-01j [02] -1.57169137e-05-9.98998444e-03j [03] 1.34357107e-05-3.25633715e-02j [04] -8.97378219e-08+4.69797181e-03j [05] 1.08219257e-07-1.69174132e-03j [06] -2.45151393e-06-2.70014975e-02j [07] -6.36547652e-07-2.43065592e-02j [08] -6.37454870e-06+1.23544679e-01j [09] 2.75346194e-07-3.45050957e-02j [10] 3.02712701e-07-3.87954825e-03j [11] 5.97780642e-01-7.96254753e-05j [12] -5.67649823e-02+9.22353946e-05j [13] -1.85031259e-01-6.67971786e-05j [14] 2.66947672e-02-8.67465577e-07j [15] -9.61279515e-03-1.18929876e-07j [16] -1.53427631e-01+2.18463900e-05j [17] -1.38114481e-01+1.07432930e-05j [18] 7.02004305e-01+0.00000000e+00j [19] -1.96064500e-01+8.55179277e-06j [20] -2.20443293e-02-5.82647262e-07j
15	Re -2.931845829686e-04 Im 5.682189721349e+00	[01] 8.58502135e-06-1.05202514e-01j [02] -1.57169137e-05+9.98998444e-03j [03] 1.34357107e-05+3.25633715e-02j [04] -8.97378221e-08-4.69797181e-03j [05] 1.08219257e-07+1.69174132e-03j [06] -2.45151393e-06+2.70014975e-02j [07] -6.36547652e-07+2.43065592e-02j [08] -6.37454870e-06-1.23544679e-01j [09] 2.75346194e-07+3.45050957e-02j [10] 3.02712701e-07+3.87954825e-03j [11] 5.97780642e-01+7.96254753e-05j [12] -5.67649823e-02-9.22353946e-05j [13] -1.85031259e-01+6.67971786e-05j [14] 2.66947672e-02+8.67465576e-07j [15] -9.61279515e-03+1.18929877e-07j [16] -1.53427631e-01-2.18463900e-05j [17] -1.38114481e-01-1.07432930e-05j [18] 7.02004305e-01+0.00000000e+00j [19] -1.96064500e-01-8.55179277e-06j [20] -2.20443293e-02+5.82647262e-07j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
16	Re -2.888538881662e-04 Im -5.640131056641e+00	[01] 1.56432975e-05-7.94893842e-03j [03] 5.21606748e-06-1.13205611e-01j [05] -3.19229121e-07+8.10066785e-04j [07] -5.42169630e-06+7.49396666e-03j [09] -5.68163805e-06+3.27555944e-03j [11] -4.48330590e-02-8.59341663e-05j [13] -6.38494482e-01+3.28057659e-06j [15] 4.56888293e-03+1.56650314e-06j [17] 4.22669556e-02+2.84144163e-05j [19] 1.84745862e-02+3.10990251e-05j	[02] -6.74979251e-06+1.31795748e-01j [04] 8.18887658e-07-5.74599647e-04j [06] -5.49599226e-06+8.36774545e-03j [08] 1.93329583e-05-9.52883721e-03j [10] -6.68044677e-07-4.28382501e-05j [12] 7.43345295e-01+0.00000000e+00j [14] -3.24081755e-03-4.45265837e-06j [16] 4.71951826e-02+2.85810608e-05j [18] -5.37438963e-02-1.06287977e-04j [20] -2.41613152e-04+3.78023353e-06j
17	Re -2.888538881645e-04 Im 5.640131056641e+00	[01] 1.56432975e-05+7.94893842e-03j [03] 5.21606748e-06+1.13205611e-01j [05] -3.19229121e-07-8.10066785e-04j [07] -5.42169630e-06-7.49396666e-03j [09] -5.68163805e-06-3.27555944e-03j [11] -4.48330590e-02+8.59341663e-05j [13] -6.38494482e-01-3.28057659e-06j [15] 4.56888293e-03-1.56650314e-06j [17] 4.22669556e-02-2.84144163e-05j [19] 1.84745862e-02-3.10990251e-05j	[02] -6.74979251e-06-1.31795748e-01j [04] 8.18887658e-07+5.74599647e-04j [06] -5.49599226e-06-8.36774545e-03j [08] 1.93329583e-05+9.52883721e-03j [10] -6.68044677e-07+4.28382501e-05j [12] 7.43345295e-01+0.00000000e+00j [14] -3.24081755e-03+4.45265837e-06j [16] 4.71951826e-02-2.85810608e-05j [18] -5.37438963e-02+1.06287977e-04j [20] -2.41613152e-04-3.78023353e-06j
18	Re -1.770322067688e-04 Im 2.663607833865e+00	[01] 1.01429296e-05-5.12306823e-02j [03] 8.51696388e-06-7.48050205e-02j [05] -1.15194132e-05-1.73319870e-01j [07] -8.94325083e-07-1.27253677e-01j [09] 3.19496718e-06-1.28545749e-01j [11] 1.36458445e-01+3.60862676e-05j [13] 1.99251237e-01+3.59287496e-05j [15] 4.61656167e-01+0.00000000e+00j [17] 3.38953891e-01+2.01458680e-05j [19] 3.42395464e-01+3.12668772e-05j	[02] 9.15525489e-06-6.37108729e-02j [04] -2.09649594e-06-1.30218247e-01j [06] 4.74311186e-07-1.28082289e-01j [08] 7.63891528e-06-6.51883914e-02j [10] 2.05027338e-05+1.04344949e-01j [12] 1.69700779e-01+3.56648851e-05j [14] 3.46850342e-01+1.74685805e-05j [16] 3.41160988e-01+2.39380692e-05j [18] 1.73636309e-01+3.18875194e-05j [20] -2.77934026e-01+3.61388258e-05j
19	Re -1.770322067663e-04 Im -2.663607833865e+00	[01] 1.01429296e-05+5.12306823e-02j [03] 8.51696388e-06+7.48050205e-02j [05] -1.15194132e-05+1.73319870e-01j [07] -8.94325082e-07+1.27253677e-01j [09] 3.19496718e-06+1.28545749e-01j [11] 1.36458445e-01-3.60862676e-05j [13] 1.99251237e-01-3.59287496e-05j [15] 4.61656167e-01+0.00000000e+00j [17] 3.38953891e-01-2.01458680e-05j [19] 3.42395464e-01-3.12668772e-05j	[02] 9.15525489e-06+6.37108729e-02j [04] -2.09649594e-06+1.30218247e-01j [06] 4.74311186e-07+1.28082289e-01j [08] 7.63891528e-06+6.51883914e-02j [10] 2.05027338e-05-1.04344949e-01j [12] 1.69700779e-01-3.56648851e-05j [14] 3.46850342e-01-1.74685805e-05j [16] 3.41160988e-01-2.39380692e-05j [18] 1.73636309e-01-3.18875194e-05j [20] -2.77934026e-01-3.61388258e-05j
20	Re 5.714733604915e-12 Im 0.000000000000e+00	[01] 3.16227766e-01+9.16116595e-17j [03] 3.16227766e-01-1.49909988e-16j [05] 3.16227766e-01+0.00000000e+00j [07] 3.16227766e-01-2.20700816e-16j [09] 3.16227766e-01-2.99819977e-16j [11] 1.80654861e-12+5.33013292e-16j [13] 1.80838084e-12+3.99759969e-16j [15] 1.80721488e-12+3.86434636e-15j [17] 1.80651530e-12-1.99879984e-16j [19] 1.80595001e-12-1.93217318e-15j	[02] 3.16227766e-01-4.24744967e-16j [04] 3.16227766e-01+3.33133307e-17j [06] 3.16227766e-01+6.16297582e-33j [08] 3.16227766e-01-4.58058297e-17j [10] 3.16227766e-01+1.91551652e-16j [12] 1.80857240e-12-1.66566654e-15j [14] 1.80624879e-12-7.32893276e-16j [16] 1.80748139e-12-2.13205317e-15j [18] 1.80696503e-12+2.66506646e-16j [20] 1.80676515e-12+5.57998290e-16j

Table S26 39-bus system Reactive stress eig reference

Complete reference spectrum with 20 right eigenvectors, ordered by eigenvalue real part and then imaginary part.

Table S26. Classical reference eigenpairs (MATLAB eig) for the 39-bus system under the Reactive stress scenario (Serving as the ground-truth benchmark for Table II in the main paper).

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value
1	Re -3.298832963763e-04 Im -6.564343772335e+00	[01] -2.53529146e-06-6.73039721e-03j [02] 7.10187756e-08-3.42466613e-03j [03] 5.27109040e-07-5.67937363e-03j [04] -6.89854765e-06+1.37274117e-01j [05] 1.29602311e-06-4.71400640e-02j [06] -6.22785109e-07-3.04442712e-02j [07] 5.30696717e-06-1.61983546e-02j [08] 4.54201608e-06-1.82814524e-02j [09] -7.37999410e-08-1.20964858e-03j [10] 2.29021421e-08-2.51653041e-05j [11] -4.41806402e-02+1.88627704e-05j [12] -2.24806858e-02+6.63548495e-07j [13] -3.72813611e-02-1.58659445e-06j [14] 9.01114499e-01+0.00000000e+00j [15] -3.09443586e-01+7.04317852e-06j [16] -1.99846662e-01+1.41312321e-05j [17] -1.06331570e-01-2.94931903e-05j [18] -1.20005740e-01-2.37846092e-05j [19] -7.94054907e-03+8.83491042e-07j [20] -1.65193715e-04-1.42035920e-07j
2	Re -3.298832963713e-04 Im 6.564343772335e+00	[01] -2.53529146e-06+6.73039721e-03j [02] 7.10187756e-08+3.42466613e-03j [03] 5.27109040e-07+5.67937363e-03j [04] -6.89854765e-06-1.37274117e-01j [05] 1.29602311e-06+4.71400640e-02j [06] -6.22785109e-07+3.04442712e-02j [07] 5.30696717e-06+1.61983546e-02j [08] 4.54201608e-06+1.82814524e-02j [09] -7.37999410e-08+1.20964858e-03j [10] 2.29021421e-08+2.51653041e-05j [11] -4.41806402e-02-1.88627704e-05j [12] -2.24806858e-02-6.63548495e-07j [13] -3.72813611e-02+1.58659445e-06j [14] 9.01114499e-01+0.00000000e+00j [15] -3.09443586e-01-7.04317852e-06j [16] -1.99846662e-01-1.41312321e-05j [17] -1.06331570e-01+2.94931903e-05j [18] -1.20005740e-01+2.37846092e-05j [19] -7.94054907e-03-8.83491042e-07j [20] -1.65193715e-04+1.42035920e-07j
3	Re -3.233130477534e-04 Im 6.884868287455e+00	[01] 2.32744892e-07+7.50366459e-04j [02] -8.33213818e-08+2.64437184e-04j [03] -1.00623384e-07+3.86234807e-04j [04] -3.44217331e-06+9.59435945e-03j [05] 7.53746218e-07-1.86752925e-03j [06] 8.99316769e-06+8.21944211e-02j [07] -5.51818030e-06-1.17508232e-01j [08] -7.19330139e-07+2.78731540e-04j [09] -3.05956964e-08+2.18573937e-04j [10] -4.91406822e-09+4.80689559e-07j [11] -5.16617431e-03+1.35981466e-06j [12] -1.82061515e-03-6.59152731e-07j [13] -2.65917574e-03-8.17653499e-07j [14] -6.60559000e-02-2.68008915e-05j [15] 1.28576927e-02+5.79324001e-06j [16] -5.65897766e-01+3.53422462e-05j [17] 8.09028700e-01+0.00000000e+00j [18] -1.91902971e-03-5.04261080e-06j [19] -1.50485276e-03-2.81315146e-07j [20] -3.30948271e-06-3.39881258e-08j
4	Re -3.233130477477e-04 Im -6.884868287455e+00	[01] 2.32744893e-07-7.50366459e-04j [02] -8.33213820e-08-2.64437184e-04j [03] -1.00623384e-07-3.86234807e-04j [04] -3.44217331e-06-9.59435945e-03j [05] 7.53746219e-07+1.86752925e-03j [06] 8.99316769e-06-8.21944211e-02j [07] -5.51818030e-06+1.17508232e-01j [08] -7.19330140e-07-2.78731540e-04j [09] -3.05956961e-08-2.18573937e-04j [10] -4.91406825e-09-4.80689559e-07j [11] -5.16617431e-03-1.35981467e-06j [12] -1.82061515e-03+6.59152731e-07j [13] -2.65917574e-03+8.17653500e-07j [14] -6.60559000e-02+2.68008915e-05j [15] 1.28576927e-02-5.79324001e-06j [16] -5.65897766e-01-3.53422462e-05j [17] 8.09028700e-01+0.00000000e+00j [18] -1.91902971e-03+5.04261081e-06j [19] -1.50485276e-03+2.81315145e-07j [20] -3.30948271e-06+3.39881259e-08j
5	Re -3.158410793816e-04 Im -6.823011459430e+00	[01] 1.59968936e-05-8.16369625e-02j [02] -7.43710456e-07+4.50352452e-04j [03] -7.62499753e-07+3.19994154e-04j [04] -3.92668455e-06-7.03814522e-04j [05] 1.03652864e-06-5.24532384e-04j [06] -4.32844356e-07-1.81559931e-03j [07] 1.83336894e-07+1.42906325e-03j [08] -5.54367468e-06+1.19758190e-01j [09] -7.67522528e-07-3.94814797e-03j [10] -3.05258710e-07+5.76728337e-04j [11] -5.57009936e-01-8.33626818e-05j [12] 3.07276018e-03+4.93210516e-06j [13] 2.18332402e-03+5.10147725e-06j [14] -4.80213331e-03+2.70141072e-05j [15] -3.57889079e-03-6.90657793e-06j [16] -1.23878548e-02+3.52674285e-06j [17] 9.75051486e-03-1.70226661e-06j [18] 8.17111502e-01+0.00000000e+00j [19] -2.69382586e-02+6.48380232e-06j [20] 3.93502415e-03+1.90062918e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
6	Re -3.158410793773e-04 Im 6.823011459430e+00	[01] 1.59968936e-05+8.16369625e-02j [03] -7.62499753e-07-3.19994154e-04j [05] 1.03652864e-06+5.24532384e-04j [07] 1.83336894e-07-1.42906325e-03j [09] -7.67522528e-07+3.94814797e-03j [11] -5.57009936e-01+8.33626818e-05j [13] 2.18332402e-03-5.10147725e-06j [15] -3.57889079e-03+6.90657793e-06j [17] 9.75051486e-03+1.70226660e-06j [19] -2.69382586e-02-6.48380232e-06j	[02] -7.43710456e-07-4.50352452e-04j [04] -3.92668455e-06+7.03814522e-04j [06] -4.32844356e-07+1.81559931e-03j [08] -5.54367468e-06-1.19758190e-01j [10] -3.05258710e-07-5.76728337e-04j [12] 3.07276018e-03-4.93210516e-06j [14] -4.80213331e-03-2.70141072e-05j [16] -1.23878548e-02-3.52674284e-06j [18] 8.17111502e-01+0.00000000e+00j [20] 3.93502415e-03-1.90062918e-06j
7	Re -3.099940826694e-04 Im -5.068979941778e+00	[01] 9.90212344e-07+8.48342986e-03j [03] 6.07243122e-06-5.86864499e-02j [05] 8.25733545e-07-8.01162986e-02j [07] -1.02278335e-05+1.00518380e-01j [09] 2.10921212e-06-2.38526131e-02j [11] 4.30023355e-02-7.64917957e-06j [13] -2.97480439e-01-1.25885798e-05j [15] -4.06107911e-01+2.06499517e-05j [17] 5.09525657e-01+2.06845800e-05j [19] -1.20908418e-01-3.29738502e-06j	[02] 1.19103930e-05-7.15379075e-02j [04] -3.12679606e-06+9.58484989e-03j [06] -6.57867189e-06+1.07573524e-01j [08] 4.31081347e-07+8.40923605e-03j [10] -4.41878621e-07+9.57988545e-04j [12] -3.62624222e-01-3.81972154e-05j [14] 4.85854128e-02+1.28784198e-05j [16] 5.45288037e-01+0.00000000e+00j [18] 4.26262488e-02-4.79195612e-06j [20] 4.85602485e-03+1.94290309e-06j
8	Re -3.099940826660e-04 Im 5.068979941778e+00	[01] 9.90212345e-07-8.48342986e-03j [03] 6.07243122e-06+5.86864499e-02j [05] 8.25733545e-07+8.01162986e-02j [07] -1.02278335e-05-1.00518380e-01j [09] 2.10921212e-06+2.38526131e-02j [11] 4.30023355e-02+7.64917957e-06j [13] -2.97480439e-01+1.25885798e-05j [15] -4.06107911e-01-2.06499517e-05j [17] 5.09525657e-01-2.06845800e-05j [19] -1.20908418e-01+3.29738502e-06j	[02] 1.19103930e-05+7.15379075e-02j [04] -3.12679606e-06-9.58484989e-03j [06] -6.57867189e-06-1.07573524e-01j [08] 4.31081347e-07-8.40923605e-03j [10] -4.41878621e-07-9.57988545e-04j [12] -3.62624222e-01+3.81972154e-05j [14] 4.85854128e-02-1.28784198e-05j [16] 5.45288037e-01+0.00000000e+00j [18] 4.26262488e-02+4.79195612e-06j [20] 4.85602485e-03-1.94290309e-06j
9	Re -3.034620911457e-04 Im -4.208214869649e+00	[01] -4.27450325e-06+2.21024270e-02j [03] -7.03602309e-06-2.75832517e-03j [05] 2.00479374e-05-1.38812664e-01j [07] -2.58101831e-06-2.51489553e-02j [09] -1.22128931e-05+1.69360457e-01j [11] 9.30117632e-02+1.12807794e-05j [13] -1.16076229e-02+3.04461441e-05j [15] -5.84153523e-01-4.22416469e-05j [17] -1.05832207e-01+1.84932342e-05j [19] 7.12705197e-01+0.00000000e+00j	[02] -7.03454216e-06-1.01843678e-03j [04] 3.43203114e-06-4.97548337e-02j [06] -3.30871580e-06-2.59674437e-02j [08] -6.73898701e-06+3.47990935e-02j [10] 1.70730256e-06-2.76077148e-03j [12] -4.28579867e-03+2.99119219e-05j [14] -2.09379032e-01+6.55981391e-07j [16] -1.09276582e-01+2.18039218e-05j [18] 1.46442065e-01+1.77988996e-05j [20] -1.16179201e-02-6.34690652e-06j
10	Re -3.034620911457e-04 Im 4.208214869649e+00	[01] -4.27450325e-06-2.21024270e-02j [03] -7.03602309e-06+2.75832517e-03j [05] 2.00479374e-05+1.38812664e-01j [07] -2.58101831e-06+2.51489553e-02j [09] -1.22128931e-05-1.69360457e-01j [11] 9.30117632e-02-1.12807794e-05j [13] -1.16076229e-02-3.04461441e-05j [15] -5.84153523e-01+4.22416469e-05j [17] -1.05832207e-01-1.84932342e-05j [19] 7.12705197e-01+0.00000000e+00j	[02] -7.03454216e-06+1.01843678e-03j [04] 3.43203114e-06+4.97548337e-02j [06] -3.30871580e-06+2.59674437e-02j [08] -6.73898701e-06-3.47990935e-02j [10] 1.70730256e-06+2.76077148e-03j [12] -4.28579867e-03-2.99119219e-05j [14] -2.09379032e-01-6.55981391e-07j [16] -1.09276582e-01-2.18039218e-05j [18] 1.46442065e-01-1.77988996e-05j [20] -1.16179201e-02+6.34690652e-06j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
11	Re -3.008362121401e-04 Im 4.597391141537e+00	[01] 3.79419175e-07-1.82590449e-02j [03] -3.47546015e-06-1.10771328e-01j [05] 1.24384809e-06+1.12672761e-01j [07] -8.26474928e-06-2.30370774e-02j [09] 1.34746975e-05+6.89716613e-02j [11] 8.39439712e-02+7.23732026e-06j [13] 5.09259125e-01+1.73459772e-05j [15] -5.18000752e-01-2.81775903e-05j [17] 1.05910458e-01-3.10658980e-05j [19] -3.17089709e-01+4.11992817e-05j	[02] -7.58262518e-06-1.15877984e-01j [04] -3.66417935e-06+1.89959643e-02j [06] -7.92579945e-06-2.41857633e-02j [08] 4.79799872e-07-1.26984695e-02j [10] 1.51651634e-06+7.34061398e-03j [12] 5.32736420e-01+0.00000000e+00j [14] -8.73318770e-02-2.25603396e-05j [16] 1.11191416e-01-2.91620468e-05j [18] 5.83798311e-02+6.02598715e-06j [20] -3.37476741e-02+4.76369627e-06j
12	Re -3.008362121376e-04 Im -4.597391141537e+00	[01] 3.79419175e-07+1.82590449e-02j [03] -3.47546015e-06+1.10771328e-01j [05] 1.24384809e-06-1.12672761e-01j [07] -8.26474928e-06+2.30370774e-02j [09] 1.34746975e-05-6.89716613e-02j [11] 8.39439712e-02-7.23732026e-06j [13] 5.09259125e-01-1.73459772e-05j [15] -5.18000752e-01+2.81775903e-05j [17] 1.05910458e-01+3.10658980e-05j [19] -3.17089709e-01-4.11992817e-05j	[02] -7.58262518e-06+1.15877984e-01j [04] -3.66417935e-06-1.89959643e-02j [06] -7.92579945e-06+2.41857633e-02j [08] 4.79799872e-07+1.26984695e-02j [10] 1.51651634e-06-7.34061398e-03j [12] 5.32736420e-01+0.00000000e+00j [14] -8.73318770e-02+2.25603396e-05j [16] 1.11191416e-01+2.91620468e-05j [18] 5.83798311e-02-6.02598715e-06j [20] -3.37476741e-02-4.76369627e-06j
13	Re -2.999329549577e-04 Im 0.000000000000e+00	[01] 3.16227751e-01+3.59483049e-16j [03] 3.16227752e-01+1.96081662e-16j [05] 3.16227754e-01+0.00000000e+00j [07] 3.16227753e-01+1.43793220e-16j [09] 3.16227753e-01+2.61442209e-17j [11] -9.48471237e-05-2.50990019e-15j [13] -9.48471240e-05+1.04521997e-16j [15] -9.48471248e-05+1.83004064e-15j [17] -9.48471245e-05-1.62099665e-15j [19] -9.48471243e-05+9.93425541e-16j	[02] 3.16227752e-01+4.70595992e-16j [04] 3.16227753e-01+4.24843605e-16j [06] 3.16227753e-01+2.22225885e-16j [08] 3.16227752e-01+2.94122495e-16j [10] 3.16227746e-01+1.56865326e-16j [12] -9.48471240e-05+9.14992875e-16j [14] -9.48471245e-05-1.25497754e-15j [16] -9.48471244e-05+2.30063664e-15j [18] -9.48471240e-05-6.27516216e-16j [20] -9.48471222e-05+5.22829547e-16j
14	Re -2.913079366756e-04 Im 5.712180903185e+00	[01] 8.76422065e-06-1.05270958e-01j [03] 1.39987020e-05+3.37239205e-02j [05] 1.27363667e-07+1.70896081e-03j [07] -8.05464299e-07+2.52239002e-02j [09] 1.31192054e-07+3.67933536e-02j [11] 6.01326752e-01+8.07290794e-05j [13] -1.92637139e-01+7.01390726e-05j [15] -9.76189333e-03+2.29690460e-07j [17] -1.44083481e-01-1.19488801e-05j [19] -2.10170292e-01-9.96880316e-06j	[02] -1.64832820e-05+9.70337734e-03j [04] -1.35536719e-07-4.12063425e-03j [06] -2.76026617e-06+2.84875635e-02j [08] -6.15939041e-06-1.20777871e-01j [10] 2.83242547e-07+3.75034084e-03j [12] -5.54274419e-02-9.69821595e-05j [14] 2.35378083e-02+4.26163201e-07j [16] -1.62726115e-01-2.40657930e-05j [18] 6.89905048e-01+0.00000000e+00j [20] -2.14226254e-02+5.25428616e-07j
15	Re -2.913079366745e-04 Im -5.712180903185e+00	[01] 8.76422065e-06+1.05270958e-01j [03] 1.39987020e-05-3.37239205e-02j [05] 1.27363667e-07-1.70896081e-03j [07] -8.05464299e-07-2.52239002e-02j [09] 1.31192054e-07-3.67933536e-02j [11] 6.01326752e-01-8.07290794e-05j [13] -1.92637139e-01-7.01390726e-05j [15] -9.76189333e-03-2.29690461e-07j [17] -1.44083481e-01+1.19488801e-05j [19] -2.10170292e-01+9.96880316e-06j	[02] -1.64832820e-05-9.70337734e-03j [04] -1.35536720e-07+4.12063425e-03j [06] -2.76026617e-06-2.84875635e-02j [08] -6.15939041e-06+1.20777871e-01j [10] 2.83242547e-07-3.75034084e-03j [12] -5.54274419e-02+9.69821595e-05j [14] 2.35378083e-02-4.26163200e-07j [16] -1.62726115e-01+2.40657930e-05j [18] 6.89905048e-01+0.00000000e+00j [20] -2.14226254e-02-5.25428616e-07j

ID	Eigenvalue real and imaginary parts	Complete right eigenvector component index followed by complex value	
16	Re -2.888449669960e-04 Im 5.672456898736e+00	[01] 1.60180471e-05+8.77900062e-03j [03] 5.02979074e-06+1.12499355e-01j [05] -3.15788975e-07-7.16957239e-04j [07] -5.72311782e-06-7.80689308e-03j [09] -6.21082889e-06-3.71150027e-03j [11] -4.97985072e-02+8.83259115e-05j [13] -6.38147746e-01-3.96360142e-06j [15] 4.06690913e-03-1.58420986e-06j [17] 4.42842661e-02-3.02091574e-05j [19] 2.10533271e-02-3.41586110e-05j	[02] -6.66856534e-06-1.30960044e-01j [04] 7.28608461e-07+5.17319779e-04j [06] -5.89306048e-06-8.83194600e-03j [08] 1.94574660e-05+1.02155253e-02j [10] -6.66559780e-07+6.26887826e-06j [12] 7.42865204e-01+0.00000000e+00j [14] -2.93447436e-03+3.98357488e-06j [16] 5.00988347e-02-3.08770684e-05j [18] -5.79471327e-02+1.07420934e-04j [20] -3.55597492e-05-3.78284236e-06j
17	Re -2.888449669940e-04 Im -5.672456898736e+00	[01] 1.60180471e-05-8.77900062e-03j [03] 5.02979074e-06-1.12499355e-01j [05] -3.15788975e-07+7.16957239e-04j [07] -5.72311782e-06+7.80689308e-03j [09] -6.21082889e-06+3.71150027e-03j [11] -4.97985072e-02-8.83259115e-05j [13] -6.38147746e-01+3.96360143e-06j [15] 4.06690913e-03+1.58420986e-06j [17] 4.42842661e-02+3.02091574e-05j [19] 2.10533271e-02+3.41586110e-05j	[02] -6.66856534e-06+1.30960044e-01j [04] 7.28608461e-07-5.17319779e-04j [06] -5.89306048e-06+8.83194600e-03j [08] 1.94574660e-05-1.02155253e-02j [10] -6.66559780e-07-6.26887826e-06j [12] 7.42865204e-01+0.00000000e+00j [14] -2.93447436e-03-3.98357488e-06j [16] 5.00988347e-02+3.08770684e-05j [18] -5.79471327e-02-1.07420934e-04j [20] -3.55597492e-05+3.78284236e-06j
18	Re -1.849987583238e-04 Im 2.748388587592e+00	[01] 8.58158503e-06-5.48461360e-02j [03] 7.14423633e-06-7.61156918e-02j [05] -1.13249171e-05-1.68245847e-01j [07] -1.40276983e-06-1.24299203e-01j [09] 2.63183871e-06-1.24392991e-01j [11] 1.50738493e-01+3.37319974e-05j [13] 2.09195497e-01+3.37164461e-05j [15] 4.62404968e-01+0.00000000e+00j [17] 3.41622510e-01+1.91398416e-05j [19] 3.41880277e-01+3.02458644e-05j	[02] 7.59542707e-06-6.57714005e-02j [04] -2.43883998e-06-1.26790354e-01j [06] -4.14977461e-08-1.25421170e-01j [08] 6.15813859e-06-6.71235086e-02j [10] 1.76904439e-05+9.07706843e-02j [12] 1.80765365e-01+3.30428125e-05j [14] 3.48469162e-01+1.67531780e-05j [16] 3.44706112e-01+2.30887087e-05j [18] 1.84481484e-01+2.93427236e-05j [20] -2.49473116e-01+3.18277502e-05j
19	Re -1.849987583225e-04 Im -2.748388587592e+00	[01] 8.58158503e-06+5.48461360e-02j [03] 7.14423633e-06+7.61156918e-02j [05] -1.13249171e-05+1.68245847e-01j [07] -1.40276983e-06+1.24299203e-01j [09] 2.63183871e-06+1.24392991e-01j [11] 1.50738493e-01-3.37319974e-05j [13] 2.09195497e-01-3.37164461e-05j [15] 4.62404968e-01+0.00000000e+00j [17] 3.41622510e-01-1.91398416e-05j [19] 3.41880277e-01-3.02458644e-05j	[02] 7.59542707e-06+6.57714005e-02j [04] -2.43883998e-06+1.26790354e-01j [06] -4.14977464e-08+1.25421170e-01j [08] 6.15813859e-06+6.71235086e-02j [10] 1.76904439e-05-9.07706843e-02j [12] 1.80765365e-01-3.30428125e-05j [14] 3.48469162e-01-1.67531780e-05j [16] 3.44706112e-01-2.30887087e-05j [18] 1.84481484e-01-2.93427236e-05j [20] -2.49473116e-01-3.18277502e-05j
20	Re 4.284918513622e-12 Im 0.000000000000e+00	[01] 3.16227766e-01-1.25595209e-16j [03] 3.16227766e-01-3.25069953e-16j [05] 3.16227766e-01-4.72829023e-16j [07] 3.16227766e-01-3.10294046e-16j [09] 3.16227766e-01-4.43277209e-16j [11] 1.35514276e-12-2.48235237e-15j [13] 1.35471425e-12+5.91036279e-17j [15] 1.35429314e-12+1.83221246e-15j [17] 1.35604409e-12-1.71400521e-15j [19] 1.35464038e-12+1.09341712e-15j	[02] 3.16227766e-01+0.00000000e+00j [04] 3.16227766e-01-5.91036279e-17j [06] 3.16227766e-01-2.58578372e-16j [08] 3.16227766e-01-1.92086791e-16j [10] 3.16227766e-01-3.39845860e-16j [12] 1.35367994e-12+1.03431349e-15j [14] 1.35400501e-12-1.53669433e-15j [16] 1.35660557e-12+2.12773060e-15j [18] 1.35554171e-12-7.68347163e-16j [20] 1.35613274e-12+5.31932651e-16j